PROBABILITY & STATISTICS · COMMON CORE HIGH SCHOOL MATH

# Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

54 question sets • 1435 questions

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# Represent data with dot plots

| | | |
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| **BACKWARD 2** | Locate values on a number-line scale | **6.NS.C.6.c** |
| **BACKWARD 3** | Read counts from a dot plot | **6.SP.B.4** |
| **FORWARD 1** | Find the median and mean from a dot plot | **HSS-ID.A.2** |
| **FORWARD 2** | Choose the display that keeps the information you need | **HSS-ID.A.1** |

SET 1 • HSS-ID.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

The numbers of books 15 students read over the summer are 2, 4, 3, 2, 5, 3, 2, 6, 3, 4, 2, 3, 1, 3, 4. A dot plot shows one dot for each student above the number of books. How many dots are above 2, above 3, and above 4?

Dots above 2: _____

Dots above 3: _____

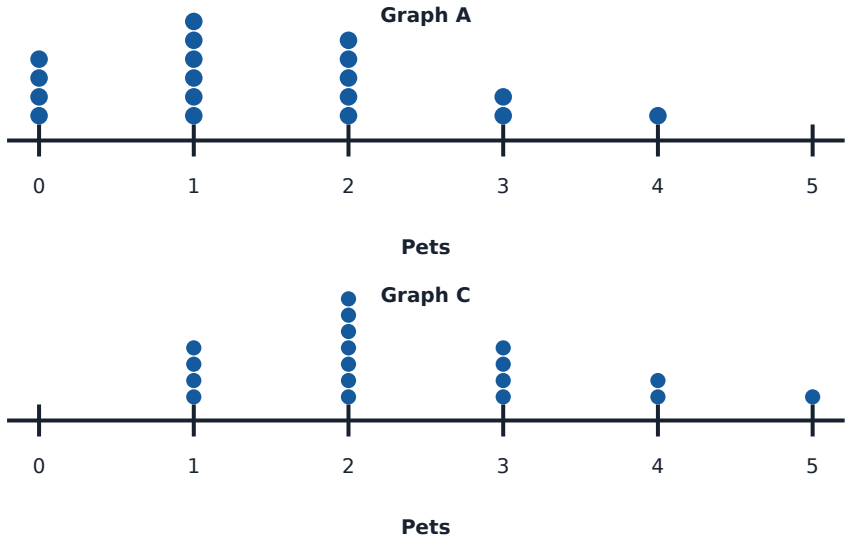
Dots above 4: _____

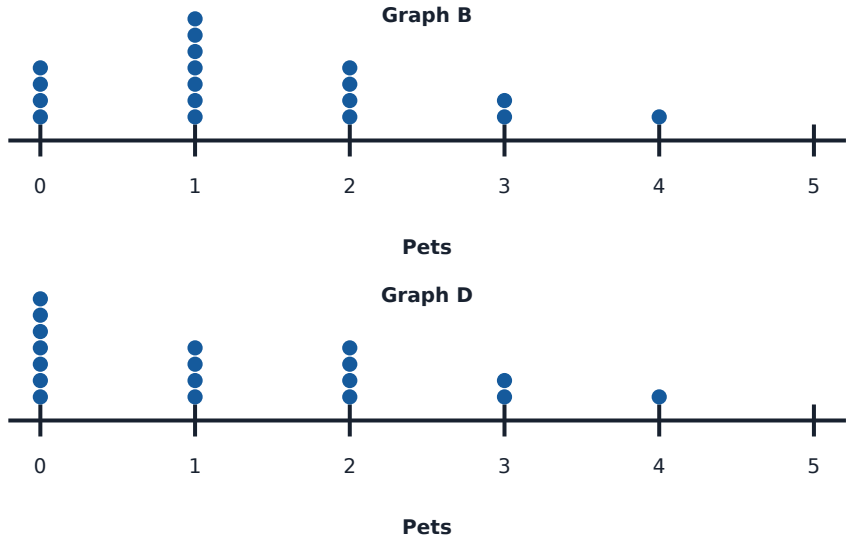
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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q1/5**

The numbers of pets in 18 households are 0, 1, 1, 2, 0, 3, 1, 2, 1, 0, 4, 1, 2, 1, 0, 2, 1, 3. Which dot plot shows the data?










**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 1 • MAIN • Q2/5**

Fourteen quiz scores (out of 10) are 7, 9, 8, 10, 7, 8, 8, 6, 9, 8, 7, 10, 8, 9. In a dot plot of the scores, which score has the tallest stack of dots, and how many dots are in that stack?

Score: _____

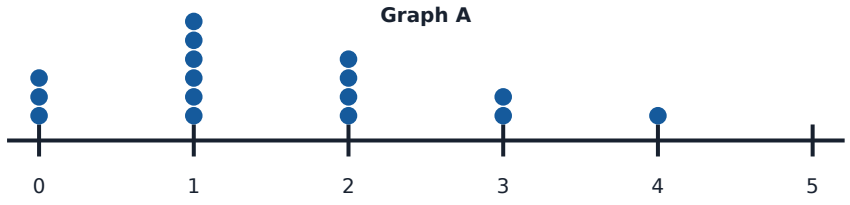
Dots in that stack: _____

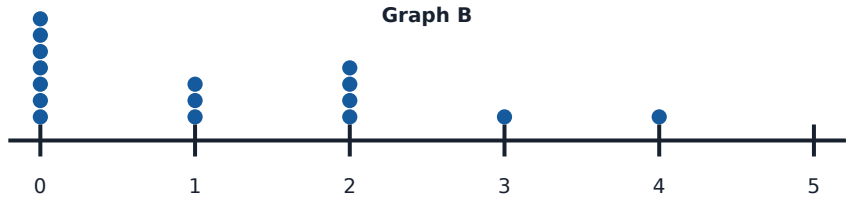
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

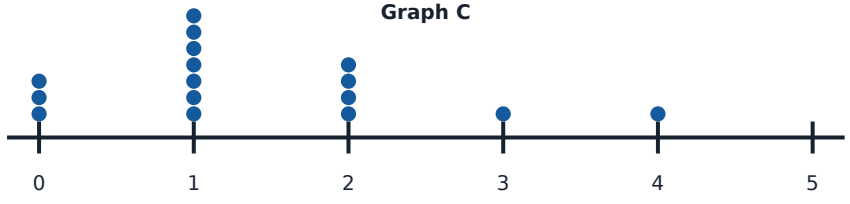
ID S1-M-Q3

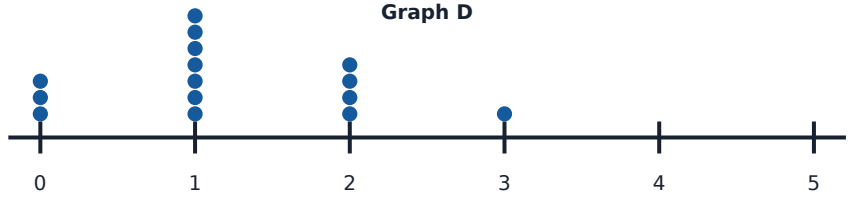
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q3/5

A team scored these numbers of goals in 16 games: 1, 0, 2, 1, 3, 1, 0, 2, 2, 1, 4, 1, 0, 1, 2, 1. Which dot plot shows the data?




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q4/5**

Twenty students reported their numbers of siblings: 1, 2, 0, 1, 3, 2, 1, 1, 0, 2, 5, 1, 2, 0, 1, 2, 3, 1, 1, 2. In a dot plot of the data, how many dots are at values of 2 or more, and how many dots are there in all?

Dots at 2 or more: _____

Dots in all: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSS-ID.A.1

# BACKWARD 1

Count how often each value occurs • Grade 6 • Statistics & Probability • 6.SP.B.5.a

In the list 3, 5, 3, 4, 3, 5, how many times does 3 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q1/5**

In the list 1, 2, 2, 4, 2, 1, 2, how many times does 2 appear?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q2/5**

## True or false?

The list 6, 6, 7, 8, 6 has three values equal to 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q3/5**

How many values are in the list 2, 4, 4, 5, 2, 4, 4?

**A.** 3

**B.** 5

**C.** 4

**D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q4/5**

In the list 0, 1, 0, 2, 1, 0, how many values are 0 or 1?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 1 • BACKWARD 1 • Q5/5**

SET 1 • HSS-ID.A.1

# BACKWARD 2

Locate values on a number-line scale • Grade 6 • The Number System • 6.NS.C.6.c

A number line has marks labeled 0, 1, 2, 3, 4, 5, 6. Where does a dot for the value 4 go?

**A.** at the right end of the number line

**B.** above the fourth mark from the left, labeled 3

**C.** above the mark labeled 4

**D.** above the mark labeled 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q1/5**

## True or false?

On a number line with marks at 0, 2, 4, 6, 8, the value 5 is halfway between the marks 4 and 6.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q2/5**

A number line has marks at 10, 15, 20, 25, 30. What value is halfway between the marks 15 and 20?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q3/5**

A number line runs from 0 to 10 with a mark at every whole number. How many marks are there?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q4/5**

## True or false?

On a horizontal number line, a larger value is placed to the left of a smaller value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

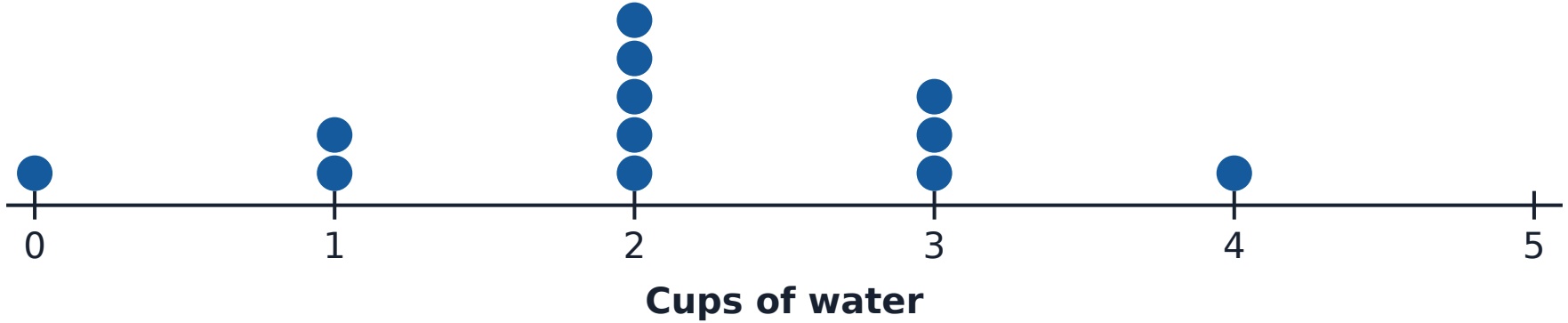
**Grade 6 • The Number System • 6.NS.C.6.c • Set 1 • BACKWARD 2 • Q5/5**

SET 1 • HSS-ID.A.1

# BACKWARD 3

Read counts from a dot plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank 3 cups?



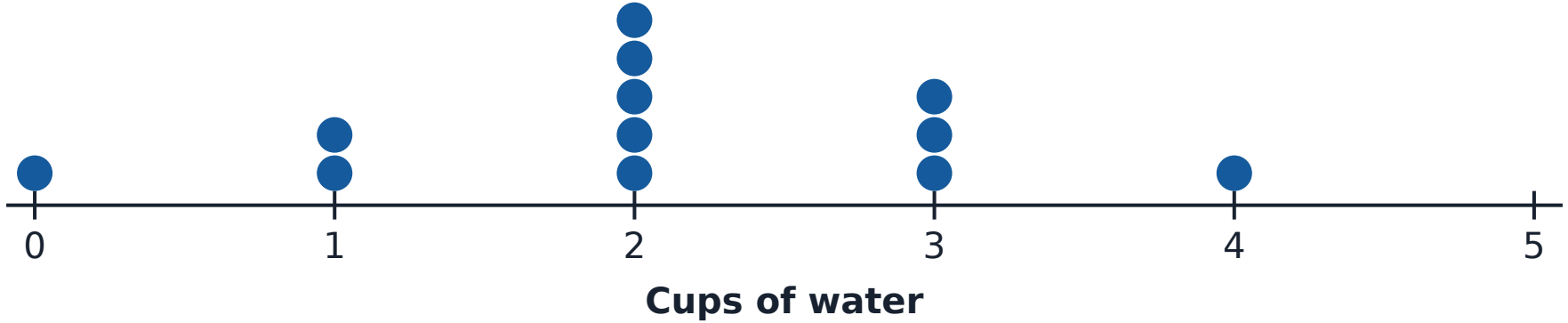

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q1/5**

The dot plot shows the cups of water some workers drank before noon. How many workers are shown in all?




Answer: _____

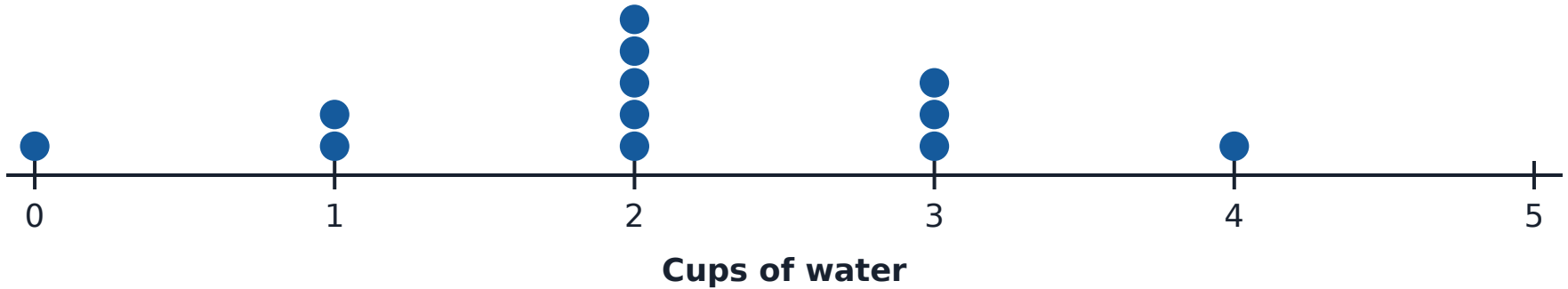
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q2/5**

## True or false?

The dot plot shows the cups of water 12 workers drank before noon. No worker drank 5 cups.




**A.** True

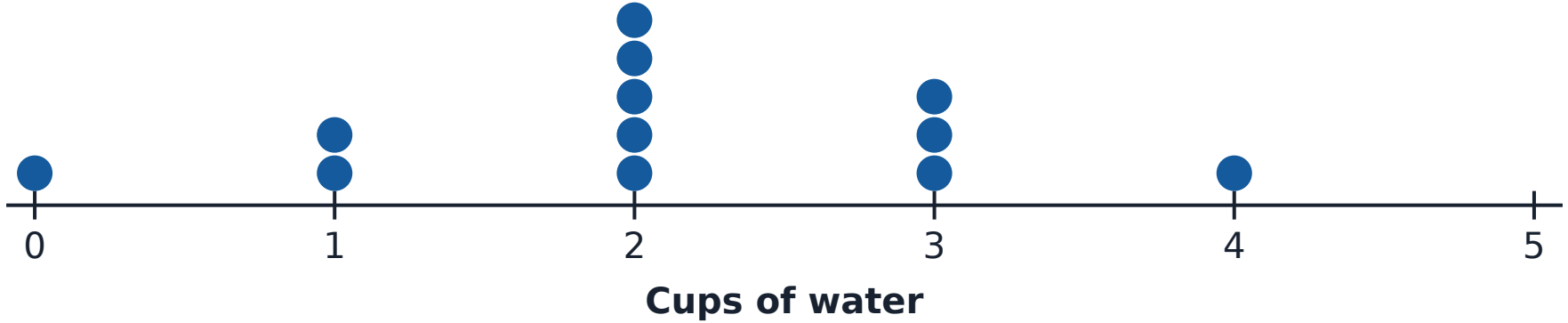
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q3/5**

The dot plot shows the cups of water 12 workers drank before noon. Which number of cups occurs most often?




**A.** 3 **B.** 12 **C.** 2 **D.** 5

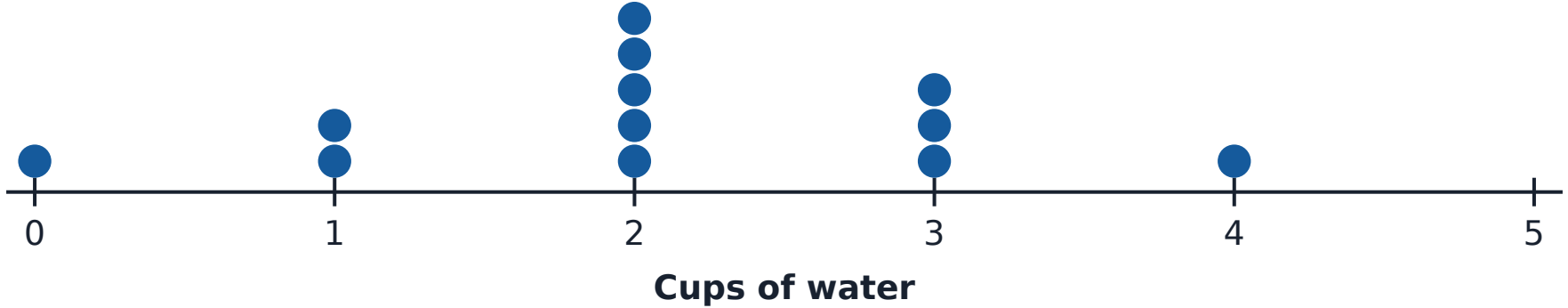
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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q4/5**

The dot plot shows the cups of water 12 workers drank before noon. How many workers drank fewer than 2 cups?




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 1 • BACKWARD 3 • Q5/5**

SET 1 • HSS-ID.A.1

# FORWARD 1

Find the median and mean from a dot plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The dot plot shows rainy days in 10 two-week periods. Find the median number of rainy days.




Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 1 • FORWARD 1 • Q1/5

The dot plot shows rainy days in 10 two-week periods. Find the mean number of rainy days. Give a decimal.




Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 1 • FORWARD 1 • Q2/5

## True or false?

A dot plot shows the values 3, 3, 4, 4, 4, 4, 5, 5, 6, 7. Adding one more dot at 20 changes the median more than it changes the mean.

**A.** True

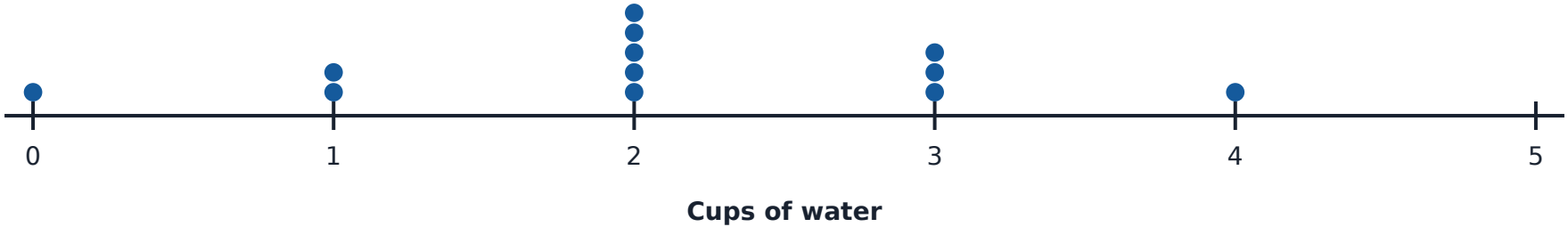
**B.** False

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 1 • FORWARD 1 • Q3/5**

The dot plot shows the cups of water 12 workers drank before noon. Which statement is correct?




**A.** The mean is 12 cups, because 12 workers are shown in the plot.

**B.** The median is 2.5 cups, halfway between the smallest and largest values.

**C.** The median is 5 cups, the height of the tallest stack in the plot.

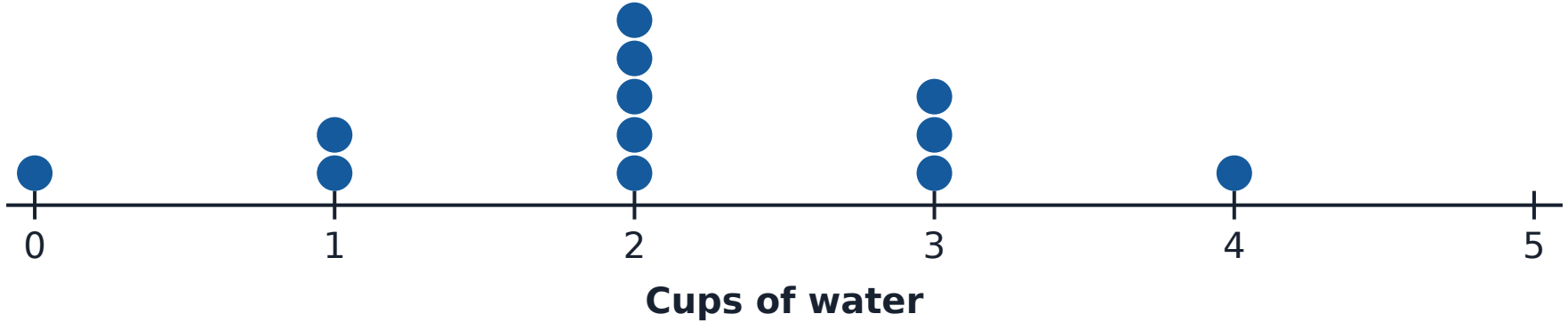
**D.** The median is 2 cups and the mean is 2.08 cups (to the nearest hundredth).

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 1 • FORWARD 1 • Q4/5**

The dot plot shows the cups of water 12 workers drank before noon. Find the total number of cups drunk.




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSS-ID.A.1

# FORWARD 2

Choose the display that keeps the information you need • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Which display always shows every individual value of a small data set?

**A.** a box plot

**B.** a histogram with bins of width 10

**C.** a dot plot

**D.** a box plot with outliers marked

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 1 • FORWARD 2 • Q1/5**

From a box plot alone, which of these can always be found?

**A.** the median

**B.** the number of values

**C.** the most common value

**D.** the mean

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 1 • FORWARD 2 • Q2/5**

## True or false?

A histogram with bins 0-10, 10-20, 20-30 shows how many values are exactly 15.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • FORWARD 2 • Q3/5

A school records the number of people in each of 400 students' households (whole numbers from 1 to 9). Which display shows how many households have each size and stays readable?

**A.** a histogram with one bin for each whole number

**B.** a list of all 400 household sizes

**C.** a box plot of the household sizes

**D.** a dot plot with one dot per household

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S1-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • FORWARD 2 • Q4/5**

## True or false?

Two different data sets can have the same box plot but different dot plots.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 1 • FORWARD 2 • Q5/5

# Represent data with histograms

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.1** |
| **BACKWARD 1** | Assign values to bins with an endpoint rule | **6.SP.B.4** |
| **BACKWARD 2** | Count the values in an interval | **6.SP.B.5.a** |
| **BACKWARD 3** | Read a histogram | **6.SP.B.4** |
| **FORWARD 1** | Regroup a histogram with wider bins | **HSS-ID.A.1** |
| **FORWARD 2** | Locate the median in a histogram | **HSS-ID.A.2** |

SET 2 • HSS-ID.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Commute times (minutes) of 20 employees are 12, 25, 8, 31, 17, 22, 15, 40, 19, 27, 10, 33, 21, 14, 26, 18, 35, 23, 29, 16. A histogram uses the bins 0-10, 10-20, 20-30, 30-40 and 40-50. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 10-20, 30-40 and 40-50?

10-20: _____

30-40: _____

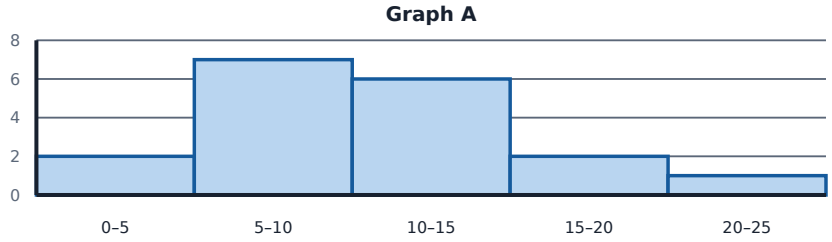
40-50: _____

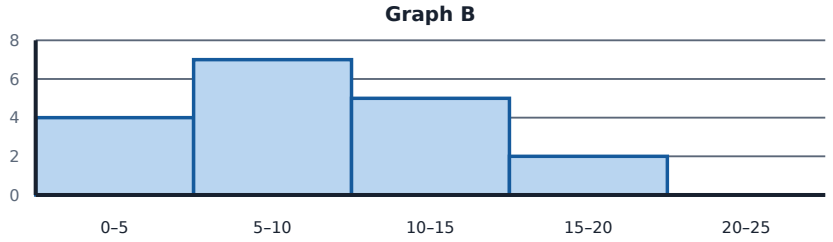
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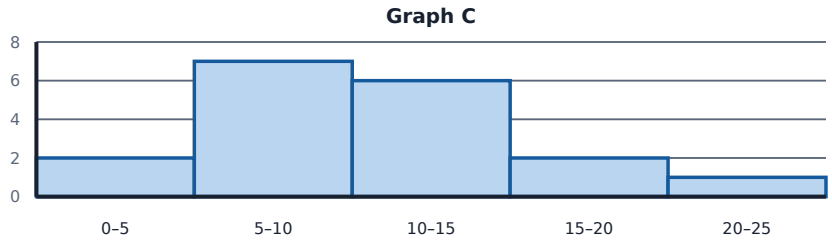
**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q1**

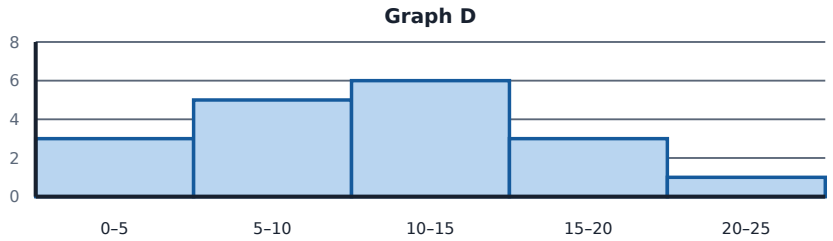
**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • MAIN • Q1/5**

Waiting times (minutes) of 18 patients are 4, 11, 7, 15, 9, 5, 12, 20, 10, 6, 14, 8, 17, 10, 3, 13, 5, 9. Which histogram shows the data with bins 0-5, 5-10, 10-15, 15-20, 20-25? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows minutes and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q2/5**

Standing long-jump distances (cm) of 16 students are 152, 168, 145, 175, 160, 158, 171, 149, 163, 180, 155, 166, 150, 172, 161, 157. A histogram uses the bins 140-150, 150-160, 160-170, 170-180 and 180-190. Each bin includes its left endpoint and excludes its right endpoint. How many values are in the bins 150-160 and 170-180?

150-160: _____

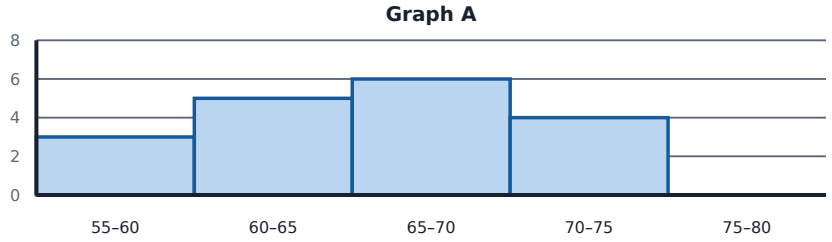
170-180: _____

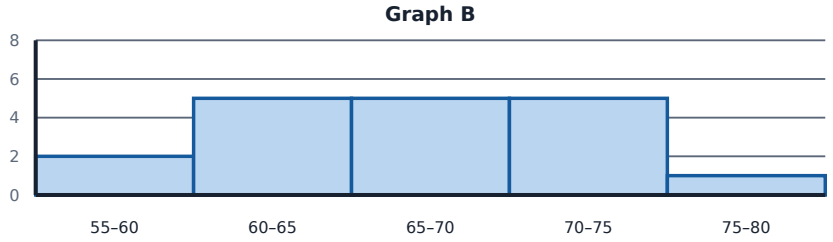
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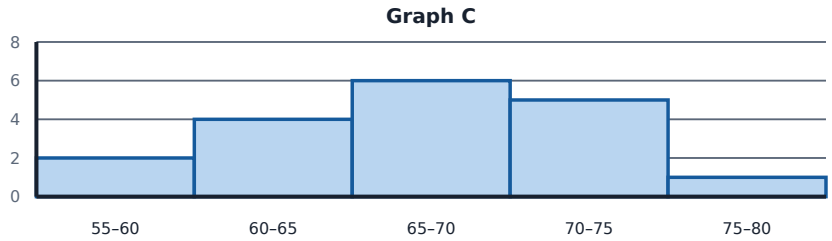
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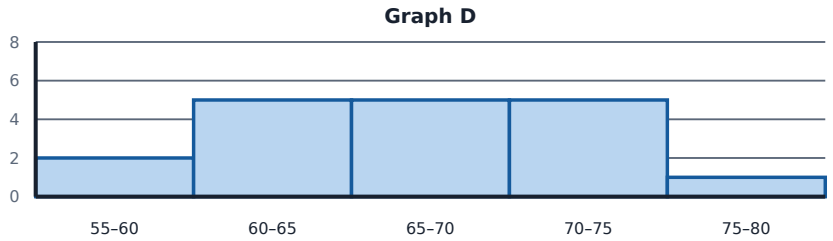
**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q3/5**

Daily high temperatures (°F) over 18 days are 61, 58, 70, 66, 72, 59, 64, 75, 68, 63, 71, 60, 67, 65, 74, 69, 62, 70. Which histogram shows the data with bins 55-60, 60-65, 65-70, 70-75, 75-80? Each bin includes its left endpoint and excludes its right endpoint. In each histogram, the horizontal axis shows °F and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q4/5**

Page counts of 14 novels are 120, 255, 188, 310, 96, 205, 150, 275, 230, 182, 140, 200, 260, 175. A histogram uses bins of width 50: 50-100, 100-150, ..., 300-350. Each bin includes its left endpoint and excludes its right endpoint. Which bin has the most values (give its left endpoint), and how many values does it have?

Left endpoint of the tallest bin: _____

Values in that bin: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • MAIN • Q5/5**

SET 2 • HSS-ID.A.1

# BACKWARD 1

Assign values to bins with an endpoint rule • Grade 6 • Statistics & Probability • 6.SP.B.4

Bins 10-20 and 20-30 each include their left endpoint and exclude their right endpoint. Which bin holds 20?

**A.** 20-30

**B.** 10-20

**C.** neither bin

**D.** both bins

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q1/5**

## True or false?

With bins that include the left endpoint and exclude the right endpoint, the value 30 goes in the bin 20-30.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q2/5**

Bins 0-5, 5-10 and 10-15 include their left endpoint and exclude their right endpoint. Which bin holds 9.9? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q3/5**

How many of the values 4, 5, 9, 10, 12 fall in the bin 5-10, which includes 5 and excludes 10?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q4/5**

**True or false?**

If every bin included both of its endpoints, a value equal to an endpoint shared by two bins would be counted twice.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 1 • Q5/5**

SET 2 • HSS-ID.A.1

# BACKWARD 2

Count the values in an interval • Grade 6 • Statistics & Probability • 6.SP.B.5.a

How many of the values 12, 18, 25, 31, 19, 22 are at least 15 and less than 25?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q1/5**

How many of the values 3.5, 4, 6.2, 7, 8 are at least 4 and less than 7?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q2/5**

## True or false?

Exactly two of the values 40, 45, 50, 55 are at least 45 and less than 55.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q3/5**

How many of the values 2, 9, 11, 14, 20, 21 are at least 10 and less than 20?

**A.** 3

**B.** 4

**C.** 1

**D.** 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q4/5**

How many of the values 100, 150, 199, 200, 250 are at least 100 and less than 200?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 2 • BACKWARD 2 • Q5/5**

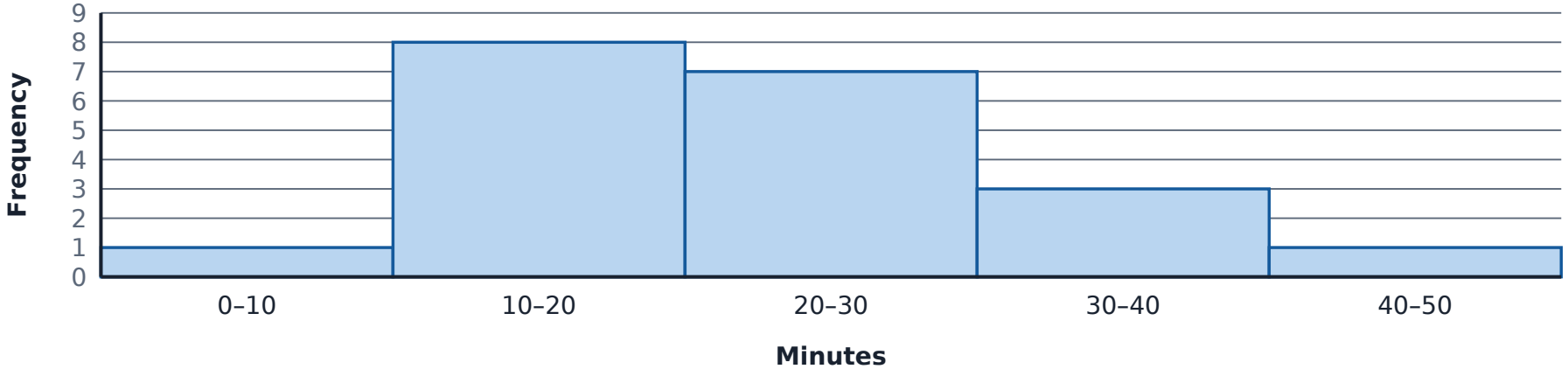
SET 2 • HSS-ID.A.1

# BACKWARD 3

Read a histogram • Grade 6 • Statistics & Probability • 6.SP.B.4

The histogram shows the commute times of some employees.

How many employees have commutes in the bin 20-30?




Answer: _____

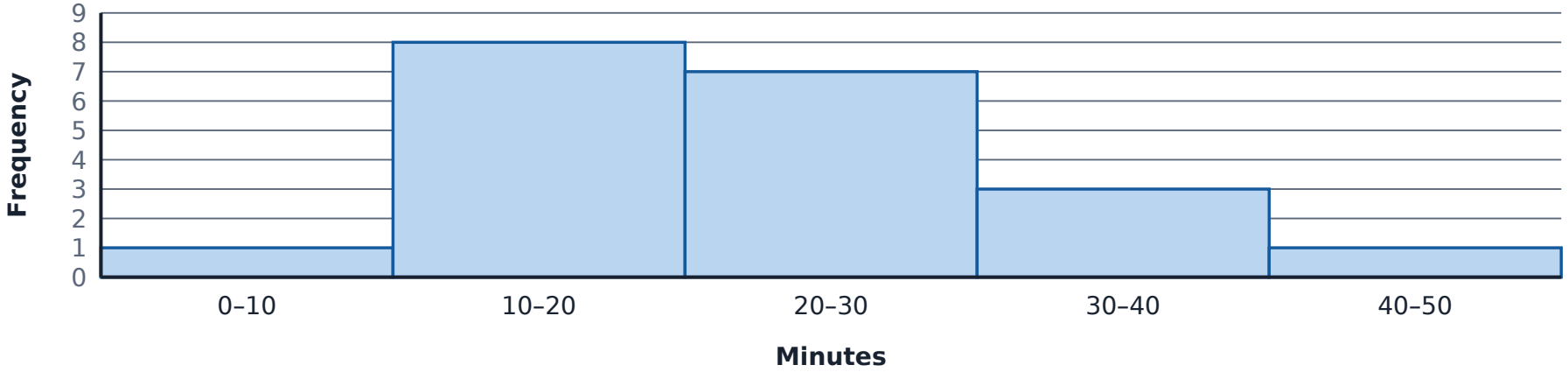
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q1/5**

The histogram shows the commute times of some employees.

How many employees are shown in all?




Answer: _____

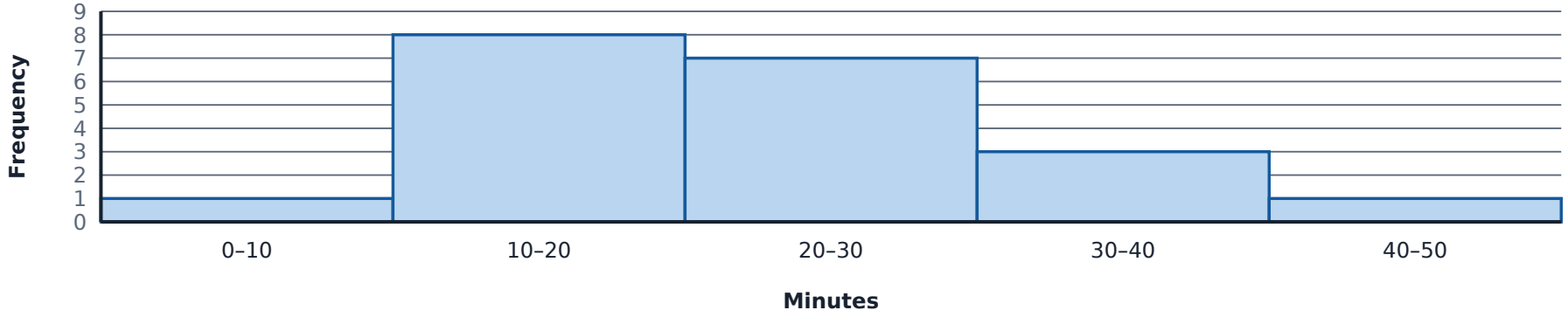
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q2/5**

## True or false?

The histogram shows the commute times of some employees. It shows that one employee's commute was exactly 15 minutes.




**A.** True

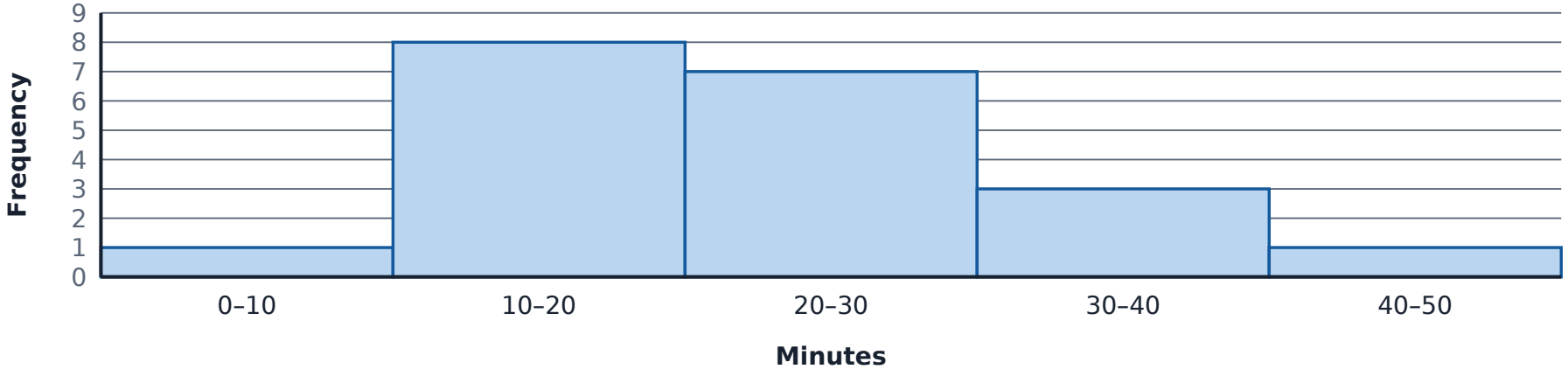
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q3/5**

The histogram shows the commute times of some employees. Which bin has the most values?




**A.** 10-20 **B.** 0-10 **C.** 20-30 **D.** 30-40

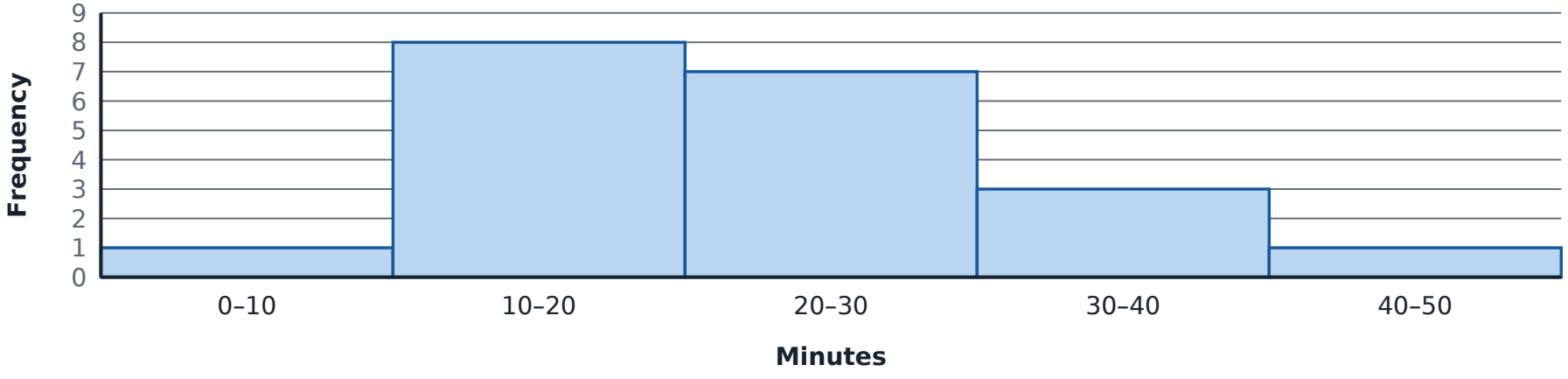
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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q4/5**

The histogram shows the commute times of some employees.
How many employees have commutes of at least 30 minutes?




Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 2 • BACKWARD 3 • Q5/5**

SET 2 • HSS-ID.A.1

# FORWARD 1

Regroup a histogram with wider bins • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

A histogram with bins 0-5, 5-10, 10-15, 15-20 has counts 3, 6, 8, 2. What are the counts for the wider bins 0-10 and 10-20?

0-10: _____

10-20: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q1/5

A histogram with bins 60-65, 65-70, 70-75, 75-80 has counts 2, 5, 6, 4. What are the counts for 60-70 and 70-80?

60-70: _____

70-80: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q2/5

## True or false?

Combining each pair of neighboring bins into one bin never changes the total number of values shown.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q3/5

A histogram with bins of width 1 has counts 1, 0, 2, 0, 1, 0, 3, 0, 2, so it looks jagged. Why might bins of width 2 help?

**A.** Wider bins always make a distribution look symmetric.

**B.** Wider bins change the data values, so the shape becomes smoother.

**C.** Wider bins add values to the empty bins, so no bin is 0.

**D.** Wider bins group nearby values, so the overall shape is easier to see.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 2 • FORWARD 1 • Q4/5**

A histogram with bins of width 10 has counts 4, 9, 5. Can the counts for bins of width 5 be found from it?

**A.** Yes: halve each count, because each narrower bin is half as wide.

**B.** Yes: keep the same counts, because the data have not changed.

**C.** No: each count could be split between the two narrower bins in many ways.

**D.** No: a histogram with bins of width 10 cannot be redrawn at all.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 2 • FORWARD 1 • Q5/5**

SET 2 • HSS-ID.A.1

# FORWARD 2

Locate the median in a histogram • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

A histogram of 21 values has counts 0-10: 4, 10-20: 6, 20-30: 7, 30-40: 4. The median is the 11th value in order. Which bin holds it? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q1/5

A histogram of 20 values has counts 50-60: 3, 60-70: 5, 70-80: 8, 80-90: 4. The median is the mean of the 10th and 11th values. Which bin holds both? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q2/5

## True or false?

The bin with the tallest bar always holds the median.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q3/5

A histogram of 15 values has counts 0-5: 10, 5-10: 2, 10-15: 2, 15-20: 1. Which bin holds the median (the 8th value)?

**A.** 15-20

**B.** 10-15

**C.** 5-10

**D.** 0-5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S2-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 2 • FORWARD 2 • Q4/5**

A histogram of 15 values has counts 0-10: 2, 10-20: 3, 20-30: 3, 30-40: 7. Which bin holds the median (the 8th value)? Give its left endpoint.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 2 • FORWARD 2 • Q5/5

# Represent data with box plots

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.1** |
| **BACKWARD 1** | Order data and find the median | **6.SP.B.5.c** |
| **BACKWARD 2** | Find quartiles as medians of the halves | **6.SP.B.5.c** |
| **BACKWARD 3** | Read a box plot | **6.SP.B.4** |
| **FORWARD 1** | Mark outliers on a box plot | **HSS-ID.A.3** |
| **FORWARD 2** | Compare two box plots | **HSS-ID.A.2** |

SET 3 • HSS-ID.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.1

Laps swum by 11 swimmers: 12, 15, 9, 18, 14, 11, 16, 20, 13, 10, 17. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

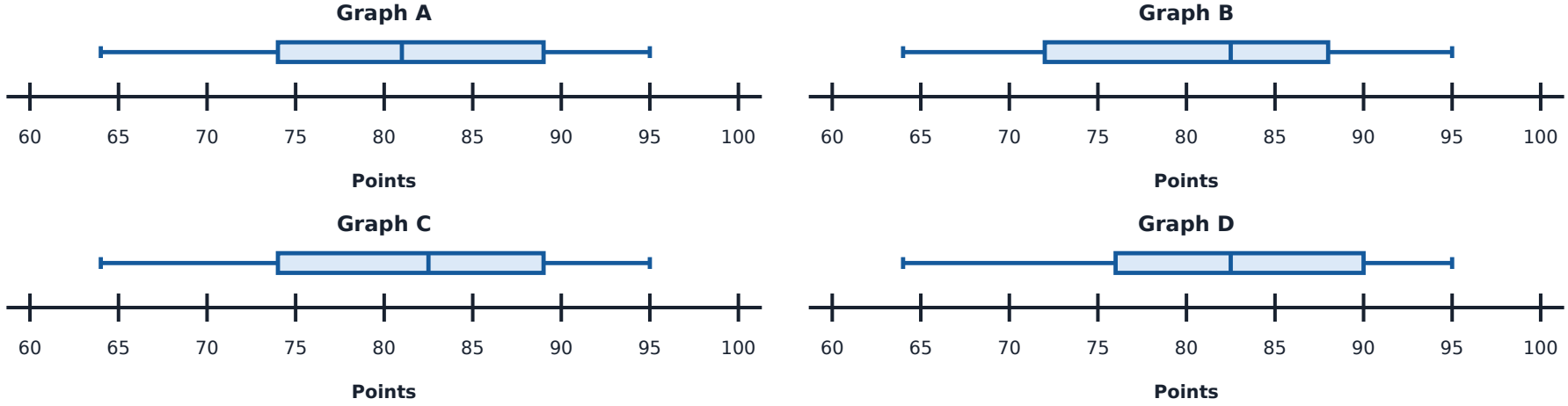
Maximum: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q1/5**

Test scores of 12 students: 72, 85, 90, 64, 78, 88, 95, 70, 81, 76, 84, 92. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q2/5**

Hours of sleep of 14 teenagers: 7.5, 6, 8, 6.5, 9, 7, 5.5, 8.5, 7, 6.5, 7.5, 8, 6, 7. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

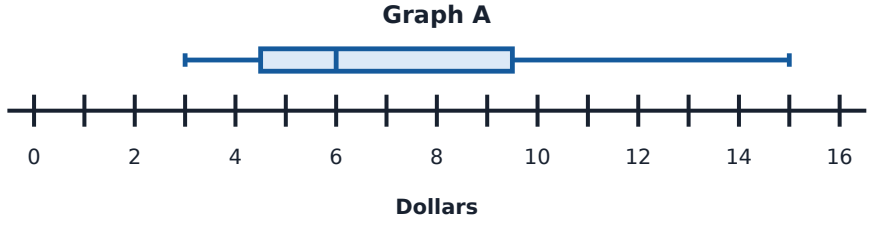
Maximum: _____

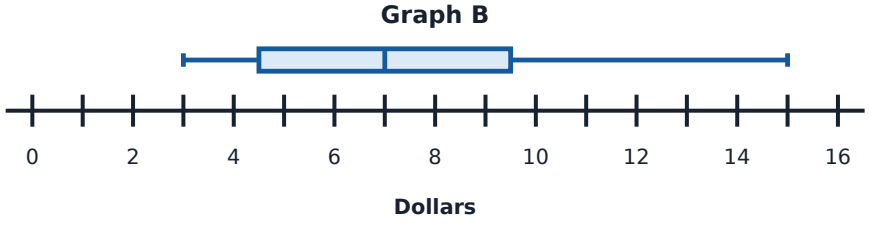
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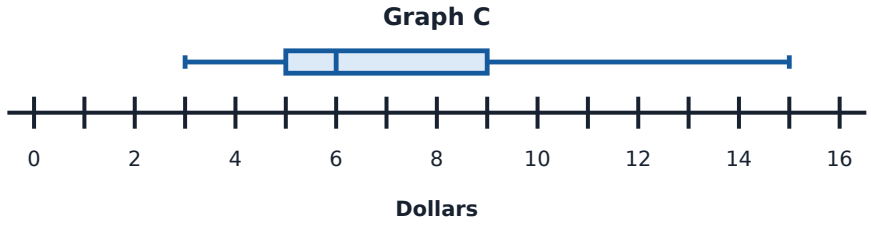
**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q3**

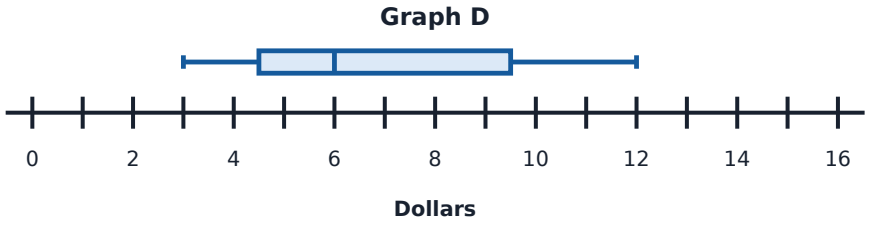
**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.1 • Set 3 • MAIN • Q3/5**

Prices (dollars) of 13 menu items: 3, 8, 5, 12, 6, 4, 9, 7, 10, 5, 6, 15, 4. Which box plot shows the data? Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.




**A.** Graph A **B.** Graph B **C.** Graph C **D.** Graph D

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q4/5**

Distances (km) 10 students live from school: 2.1, 3.4, 1.8, 5, 2.7, 4.2, 3, 2.4, 3.9, 1.5. Give the five-number summary for a box plot. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Minimum: _____

Q1: _____

Median: _____

Q3: _____

Maximum: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.1 • Set 3 • MAIN • Q5/5**

SET 3 • HSS-ID.A.1

# BACKWARD 1

Order data and find the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 9, 3, 7, 5, 11.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q1/5**

Find the median of 4, 10, 6, 8.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q2/5**

## True or false?

The median of 2, 9, 4 is 9, the middle value of the list as written.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q3/5**

What is the median of 1, 3, 3, 6, 7, 8?

**A.** 3

**B.** 6

**C.** 4.67

**D.** 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q4/5**

Find the median of 12, 15, 11, 14, 13, 16, 10.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 1 • Q5/5**

SET 3 • HSS-ID.A.1

# BACKWARD 2

Find quartiles as medians of the halves • Grade 6 • Statistics & Probability • 6.SP.B.5.c

For 2, 4, 6, 8, 10, 12, 14, 16, find Q1 and Q3 (each the median of a half).

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q1/5**

For 1, 3, 5, 7, 9, 11, 13 (an odd count), leave out the median and find Q1 and Q3.

Q1: _____

Q3: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q2/5**

## True or false?

For 1, 2, 3, 4, 5, 6, 7, 8, 9, Q1 is 2.5 when the median 5 is left out of both halves.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q3/5**

For 10, 20, 30, 40, 50, 60, what is the interquartile range Q3 − Q1 (Q1 and Q3 are the medians of the halves)?

**A.** 30

**B.** 50

**C.** 20

**D.** 35

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q4/5**

For 5, 6, 8, 9, 12, 15, find Q1 (the median of the lower half).

Q1: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

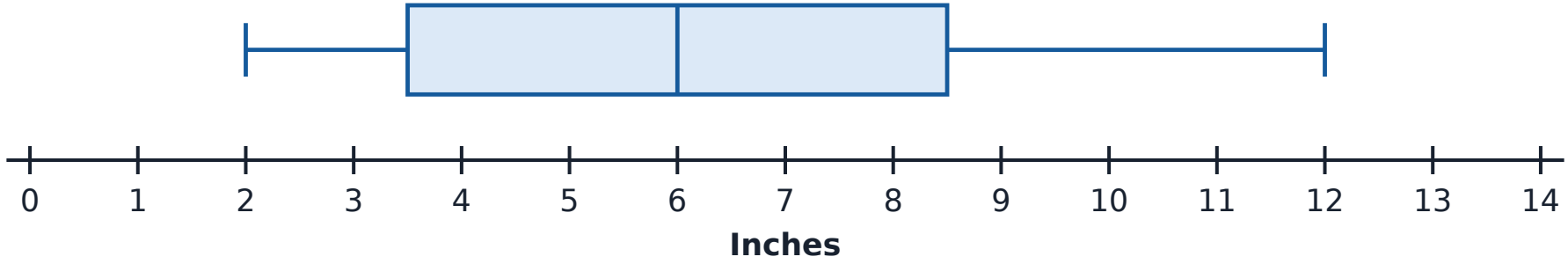
**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 3 • BACKWARD 2 • Q5/5**

SET 3 • HSS-ID.A.1

# BACKWARD 3

Read a box plot • Grade 6 • Statistics & Probability • 6.SP.B.4

The box plot shows the snowfall in some storms. What is the median snowfall?



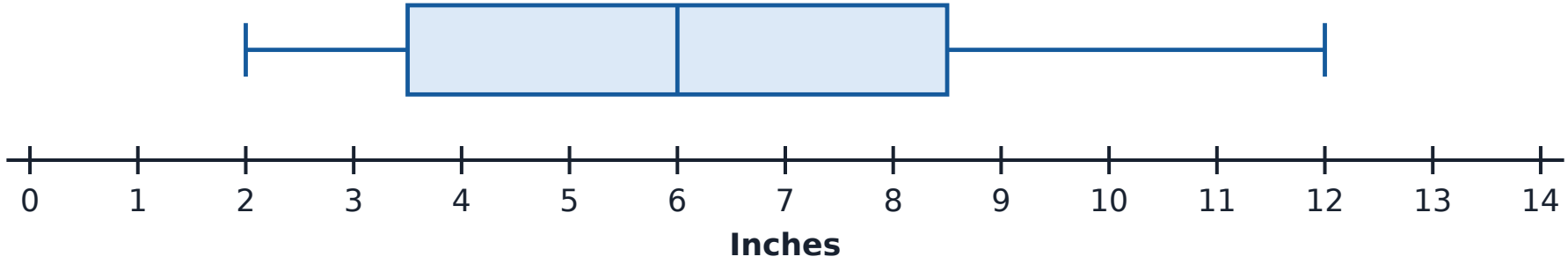

Median (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q1/5**

The box plot shows the snowfall in some storms. What is the interquartile range?




IQR (inches): _____

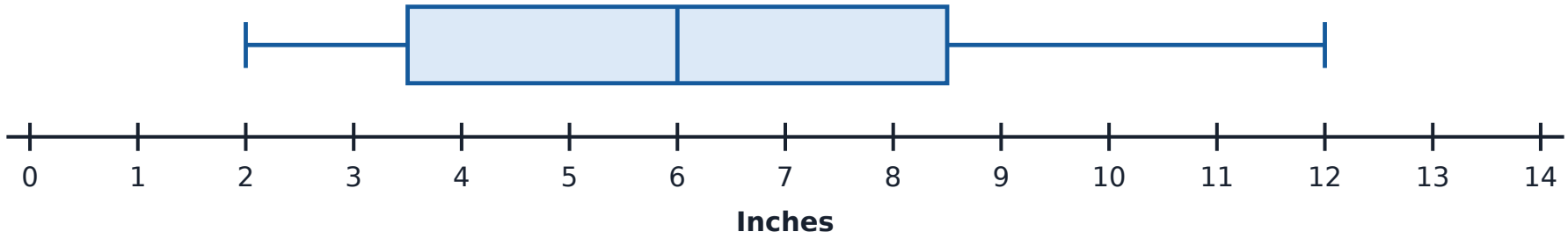
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q2/5**

**True or false?**

The box plot shows the snowfall in some storms. About half of the storms had at least 6 inches.




**A.** True

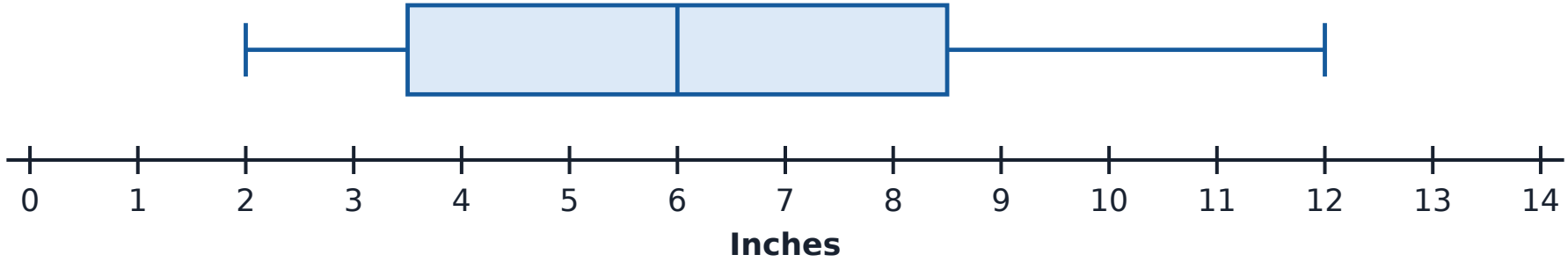
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q3/5**

The box plot shows the snowfall in some storms. What is the range?



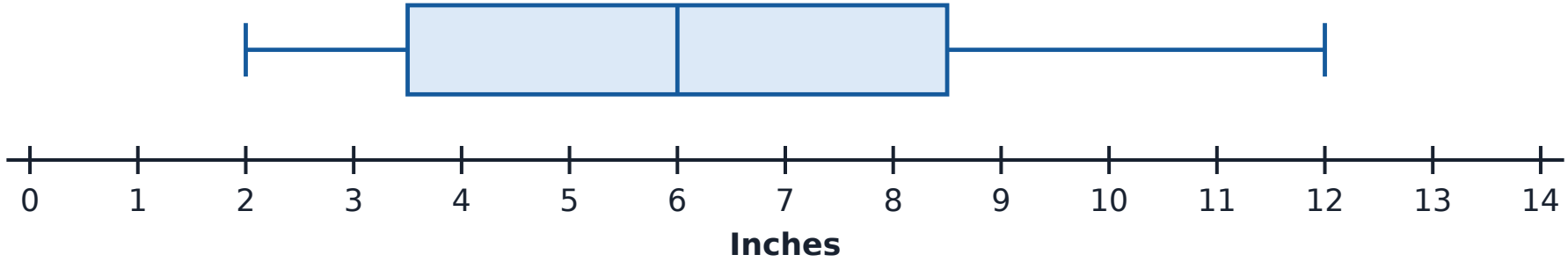

**A.** 6 inches **B.** 10 inches **C.** 12 inches **D.** 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q4/5**

The box plot shows the snowfall in some storms. What is Q1?




Q1 (inches): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 3 • BACKWARD 3 • Q5/5**

SET 3 • HSS-ID.A.1

# FORWARD 1

Mark outliers on a box plot • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

A data set has Q1 = 20 and Q3 = 30. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. What are the lower and upper fences?

Lower fence: _____

Upper fence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q1/5

Prices of 10 used bikes (dollars): 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. How many outliers are there, and where does the upper whisker end?

Number of outliers: _____

Upper whisker ends at: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F1-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q2/5**

## True or false?

In a box plot that marks outliers, a whisker ends at the fence value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q3/5

Race times (seconds) of 9 swimmers: 52, 55, 54, 31, 56, 53, 57, 55, 54. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Which value is an outlier?

**A.** 31

**B.** 57

**C.** both 31 and 57

**D.** none

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 3 • FORWARD 1 • Q4/5**

A box plot that marks outliers has Q1 = 14 and Q3 = 17, and the largest data values are 18 and 41. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Where does the upper whisker end?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

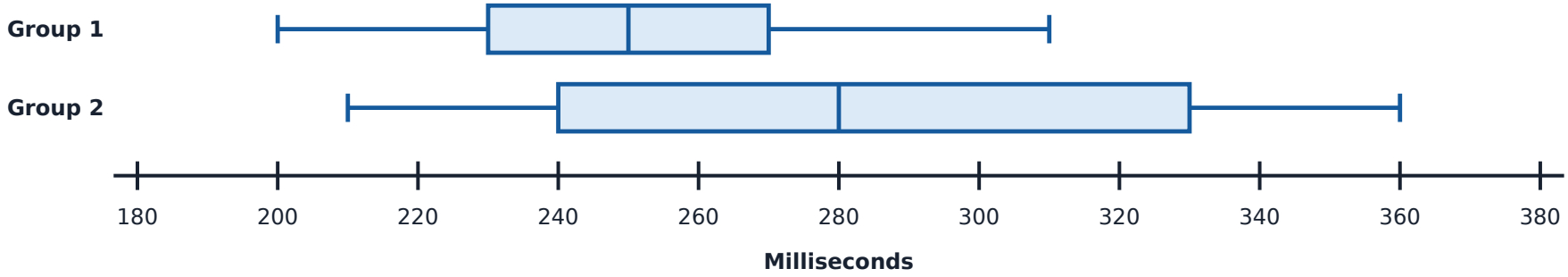
High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSS-ID.A.1

# FORWARD 2

Compare two box plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The box plots show the reaction times of two groups. Which comparison is correct?




**A.** Group 1 has the greater median and the greater IQR.

**B.** Group 1 has the greater range and the greater median.

**C.** The groups have the same median but different IQRs.

**D.** Group 2 has the greater median and the greater IQR.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 3 • FORWARD 2 • Q1/5**

Box plot 1 has median 40 and IQR 12; box plot 2 has median 46 and IQR 8. By how much do the medians differ, and which has the smaller IQR (enter 1 or 2)?

Difference of medians: _____

Smaller IQR (1 or 2): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 3 • FORWARD 2 • Q2/5

## True or false?

If two box plots have the same median, the two data sets have the same mean.

**A.** True

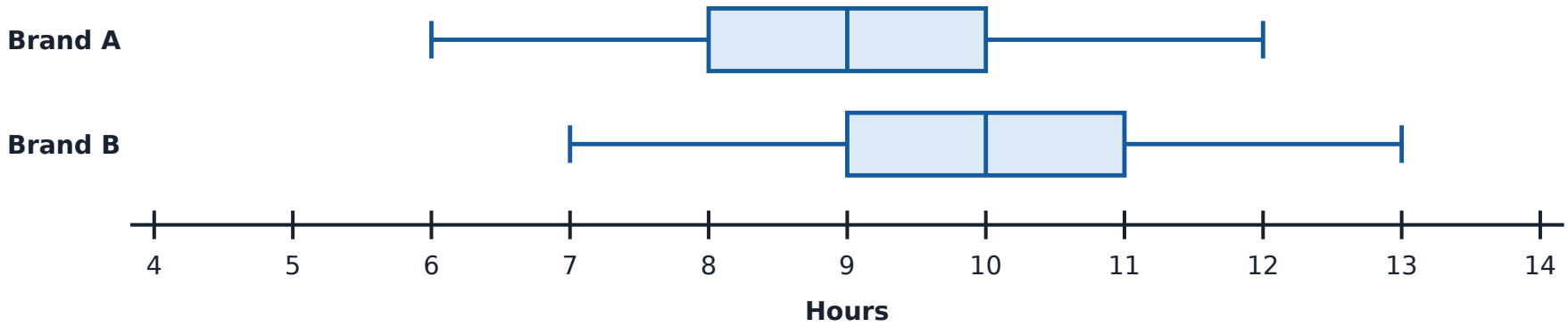
**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 3 • FORWARD 2 • Q3/5

The box plots show the battery lives of two brands. About what fraction of brand B batteries last longer than the median life of brand A?




**A.** about 1/2 **B.** about 3/4 **C.** none **D.** about 1/4

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S3-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 3 • FORWARD 2 • Q4/5**

Two box plots: class A has Q1 = 70 and Q3 = 85; class B has Q1 = 72 and Q3 = 80. Give each IQR.

IQR of A: _____

IQR of B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 3 • FORWARD 2 • Q5/5

SET 4 • HSS-ID.A.2 • INTERPRETING CATEGORICAL AND QUANTITATIVE DATA

# Compare centers with the measure suited to the shape

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.2** |
| **BACKWARD 1** | Compute a mean | **6.SP.B.5.c** |
| **BACKWARD 2** | Compute a median | **6.SP.B.5.c** |
| **BACKWARD 3** | Recognize skew and extreme values | **6.SP.A.2** |
| **FORWARD 1** | Find a missing value from a target mean or median | **HSS-ID.A.2** |
| **FORWARD 2** | Predict how changes in data move the mean and median | **HSS-ID.A.3** |

SET 4 • HSS-ID.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Sale prices (thousands of dollars) of 8 homes in each of two towns: Town A: 210, 225, 230, 240, 245, 250, 260, 900; Town B: 190, 220, 235, 255, 270, 280, 300, 310. Compare the typical prices with one measure of center suited to the shapes of both data sets. Give that measure for each town.

Town A: _____

Town B: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q1/5**

Scores of 7 students in each of two classes: class P: 70, 74, 76, 78, 80, 82, 86; class Q: 66, 70, 72, 74, 76, 78, 82. Which comparison of centers is appropriate and correct?

**A.** Compare medians: P's median 76 is 2 points above Q's median 74, so P is 2 points higher.

**B.** Compare the highest scores: P's 86 is 4 points above Q's 82, so P is 4 points higher.

**C.** Both are roughly symmetric with no outliers, so compare means: P's mean 78 is 4 points above Q's mean 74.

**D.** Both are roughly symmetric, so compare means: P's mean 76 is 2 points above Q's mean 74.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q2/5**

Delivery times (days) of 9 orders from each of two stores: store X: 2, 2, 3, 3, 3, 4, 4, 5, 14; store Y: 2, 3, 3, 4, 4, 5, 5, 6, 6. Compare the typical delivery times with one measure of center suited to the shapes of both data sets. Give that measure for each store.

Store X: _____

Store Y: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q3/5**

Heights (cm) of 7 seedlings of each of two plants: plant A: 5, 6, 6, 7, 7, 8, 30; plant B: 8, 8, 9, 9, 9, 10, 10. Which comparison of typical heights is appropriate?

**A.** Compare means: A's mean (about 9.9 cm) is greater, so A's seedlings are typically taller.

**B.** Compare means: both are about 9 cm, so the seedlings are typically the same height.

**C.** Compare maximums: A's tallest seedling (30 cm) is taller, so A's seedlings are typically taller.

**D.** A is skewed by the value 30, so compare medians: B's median 9 cm is greater than A's median 7 cm.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • MAIN • Q4/5**

Masses (grams) of 6 bags filled by each of two machines: machine 1: 498, 499, 500, 500, 501, 502; machine 2: 503, 504, 505, 505, 506, 507. Compare the typical masses with one measure of center suited to the shapes of both data sets. Give that measure for each machine.

Machine 1: _____

Machine 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSS-ID.A.2

# BACKWARD 1

Compute a mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 4, 7, 10.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q1/5**

Find the mean of 2, 3, 5, 6. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q2/5**

## True or false?

The mean of 1, 2, 3, 10 is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q3/5**

What is the mean of 5, 5, 6, 8?

**A.** 24

**B.** 5

**C.** 6

**D.** 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q4/5**

Find the mean of 1.5, 2.5, 3.5. Give a decimal.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 1 • Q5/5**

SET 4 • HSS-ID.A.2

# BACKWARD 2

Compute a median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median of 8, 2, 6.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q1/5**

Find the median of 3, 9, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q2/5**

## True or false?

The median of 1, 2, 3, 100 is 2.5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q3/5**

What is the median of 10, 12, 12, 15, 30?

**A.** 15.8

**B.** 12

**C.** 30

**D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q4/5**

Find the median of 4.5, 6, 5, 7.

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 4 • BACKWARD 2 • Q5/5**

SET 4 • HSS-ID.A.2

# BACKWARD 3

Recognize skew and extreme values • Grade 6 • Statistics & Probability • 6.SP.A.2

Which data set is skewed to the right?

**A.** 5, 5, 5, 5, 5

**B.** 1, 12, 13, 13, 14, 14, 14, 15

**C.** 1, 2, 2, 3, 3, 3, 4, 15

**D.** 1, 2, 3, 4, 5, 6, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q1/5**

## True or false?

A data set with a single value far below all the others is skewed to the left.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q2/5**

Which value is extreme in 20, 22, 21, 23, 22, 60?

**A.** 60

**B.** 23

**C.** none

**D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q3/5**

## True or false?

In 4, 5, 5, 6, 6, 6, 7, 7, 8 the values are spread evenly on both sides of 6, so the data are roughly symmetric.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q4/5**

Incomes in a town include a few very large incomes. The distribution is most likely

**A.** skewed to the right

**B.** skewed to the left

**C.** uniform

**D.** symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 4 • BACKWARD 3 • Q5/5**

SET 4 • HSS-ID.A.2

# FORWARD 1

Find a missing value from a target mean or median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Four quiz scores are 80, 85, 90, 75. What fifth score makes the mean of all five scores 84?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • FORWARD 1 • Q1/5

The values 3, 5, 8 and x have mean 7. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • FORWARD 1 • Q2/5

## True or false?

In a data set of 7 values, replacing the largest value with a larger number can change the median.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 4 • FORWARD 1 • Q3/5

The values 4, 6, 9, 10, 12 and one more value x have a median of 8. What is x?

**A.** 16

**B.** 7

**C.** 8

**D.** 6

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 4 • FORWARD 1 • Q4/5**

A team's mean score over 9 games is 70. What score in a 10th game makes the mean of all 10 games 72?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSS-ID.A.2

# FORWARD 2

Predict how changes in data move the mean and median • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The values are 10, 12, 13, 15, 40. If 40 is replaced by 20, what happens?

**A.** The mean decreases by 20 and the median stays 13.

**B.** The mean and the median both decrease by 4.

**C.** The mean decreases by 4 and the median stays 13.

**D.** The mean stays the same and the median decreases.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 4 • FORWARD 2 • Q1/5**

## True or false?

Adding 5 to every value in a data set increases both the mean and the median by 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 4 • FORWARD 2 • Q2/5

Find the mean and the median of 2, 4, 6, 8, 100.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 4 • FORWARD 2 • Q3/5

Salaries at a company include a few very high salaries. Which statement is most likely true?

**A.** The median is pulled up more than the mean.

**B.** The mean salary is greater than the median salary.

**C.** The median salary is greater than the mean salary.

**D.** The mean and the median are equal.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S4-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 4 • FORWARD 2 • Q4/5**

## True or false?

Doubling every value in a data set doubles the median but leaves the mean unchanged.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 4 • FORWARD 2 • Q5/5

# Compare spreads with the IQR or the standard deviation

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.2** |
| **BACKWARD 1** | Find the range and the interquartile range | **6.SP.B.5.c** |
| **BACKWARD 2** | Measure distances from the mean | **6.SP.B.5.c** |
| **BACKWARD 3** | Square, average, and take square roots | **8.EE.A.2** |
| **FORWARD 1** | Compare population and sample standard deviations | **HSS-ID.A.2** |
| **FORWARD 2** | Predict how adding or multiplying changes center and spread | **HSS-ID.A.2** |

SET 5 • HSS-ID.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Ages of 9 players on each of two teams: team A: 14, 15, 15, 16, 16, 17, 18, 18, 19; team B: 13, 14, 16, 16, 17, 18, 20, 22, 25. Give the interquartile range of each team. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of team A: _____

IQR of team B: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • MAIN • Q1/5**

Bus delays (minutes) on 5 days: route 1: 4, 6, 8, 10, 12; route 2: 7, 8, 8, 8, 9. Both are symmetric. Give the standard deviation of each route, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Route 1: _____

Route 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.2 • Set 5 • MAIN • Q2/5

Monthly rents (dollars) of 7 apartments in each of two buildings: A: 800, 850, 900, 950, 1000, 1050, 2400; B: 700, 800, 900, 1000, 1100, 1200, 1300. Which comparison of spread is appropriate and correct?

**A.** Compare standard deviations: A's is greater, so A's typical rents vary more than B's.

**B.** A has an outlier, so compare IQRs: B's IQR ($400) is twice A's ($200), so B's typical rents vary more.

**C.** Compare IQRs: A's IQR ($400) is twice B's ($200), so A's typical rents vary more.

**D.** Compare ranges: A's range ($1,600) is greater than B's ($600), so A's typical rents vary more.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • MAIN • Q3/5**

Lengths (cm) of 8 fish from each of two lakes: lake A: 20, 22, 23, 25, 26, 28, 30, 31; lake B: 18, 24, 24, 25, 25, 26, 26, 40. Lake B has outliers. Give the interquartile range of each lake. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

IQR of lake A: _____

IQR of lake B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • MAIN • Q4/5

Weekly savings (dollars) of 5 students in each of two groups: group 1: 10, 20, 30, 40, 50; group 2: 25, 28, 30, 32, 35. Both are symmetric. Give the standard deviation of each group, to the nearest tenth. Use the population standard deviation (divide by n). A calculator or spreadsheet may be used.

Group 1: _____

Group 2: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • MAIN • Q5/5

SET 5 • HSS-ID.A.2

# BACKWARD 1

Find the range and the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the range of 7, 3, 12, 5, 9.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q1/5**

Q1 = 18 and Q3 = 27. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q2/5**

## True or false?

The IQR is the spread of the middle half of the data.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q3/5**

For 2, 4, 6, 8, 10, 12, the IQR (with Q1 and Q3 the medians of the halves) is

**A.** 10

**B.** 6

**C.** 4

**D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q4/5**

A box plot has minimum 5, Q1 9, median 12, Q3 16, maximum 25. Give the IQR and the range.

IQR: _____

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 1 • Q5/5**

SET 5 • HSS-ID.A.2

# BACKWARD 2

Measure distances from the mean • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The mean of 2, 6, 10 is 6. What are the distances of the values from the mean?

Distance of 2: _____

Distance of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q1/5**

Find the mean absolute deviation of 1, 3, 5, 7.

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q2/5**

**True or false?**

A data set whose values are all far from the mean has a large spread.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q3/5**

Which set has the larger spread around its mean?

**A.** 0, 10, 20

**B.** cannot tell

**C.** 9, 10, 11

**D.** they are the same

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q4/5**

The values 5, 5, 5, 5 have mean 5. What is their mean absolute deviation?

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 5 • BACKWARD 2 • Q5/5**

SET 5 • HSS-ID.A.2

# BACKWARD 3

Square, average, and take square roots • Grade 8 • Expressions & Equations • 8.EE.A.2

Find $\sqrt{16} + \sqrt{9}$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q1/5**

The squared distances from a mean are 4, 0, 4. Find their average.

Average: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q2/5**

**True or false?**

$\sqrt{(9 + 16)} = \sqrt{9} + \sqrt{16}$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q3/5**

Which is $\sqrt{0.25}$?

**A.** 0.5

**B.** 0.125

**C.** 0.0625

**D.** 0.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q4/5**

Find $\sqrt{8}$ to the nearest tenth. A calculator may be used.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

**Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 5 • BACKWARD 3 • Q5/5**

SET 5 • HSS-ID.A.2

# FORWARD 1

Compare population and sample standard deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

For the data 4, 6, 8, 10, 12, give the sample standard deviation (divide by $n - 1$), to the nearest hundredth. A calculator or spreadsheet may be used.

$s =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 1 • Q1/5

## True or false?

For the same data, the sample standard deviation (divide by $n - 1$) is never smaller than the population standard deviation (divide by $n$).

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 1 • Q2/5

For 2, 4, 6, give the population and the sample standard deviations, to the nearest hundredth. A calculator or spreadsheet may be used.

Population: _____

Sample: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 1 • Q3/5

A spreadsheet gives 5.2 for STDEV.S and 4.9 for STDEV.P of the same data. Which statement is correct?

**A.** STDEV.S divides by n and treats the data as the whole population; STDEV.P divides by n − 1 for a sample.

**B.** The two functions measure different spreads: STDEV.S the range of the data and STDEV.P their IQR.

**C.** One of the two values must be an error, because a data set has only one standard deviation.

**D.** STDEV.S divides by n − 1 and treats the data as a sample; STDEV.P divides by n and treats them as the population.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • FORWARD 1 • Q4/5**

Ten values have sum of squared deviations from their mean equal to 90. Give the population and sample variances.

Population variance: _____

Sample variance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSS-ID.A.2

# FORWARD 2

Predict how adding or multiplying changes center and spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Every value in a data set is increased by 10. What happens to the standard deviation and the IQR?

**A.** Both stay the same.

**B.** The SD stays the same and the IQR increases by 10.

**C.** Both increase by 10.

**D.** Both are multiplied by 10.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • FORWARD 2 • Q1/5**

A data set has mean 20 and standard deviation 4. Every value is multiplied by 3. Give the new mean and standard deviation.

Mean: _____

Standard deviation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 2 • Q2/5

## True or false?

Converting temperatures from degrees to tenths of a degree (multiplying by 10) multiplies the IQR by 10.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 5 • FORWARD 2 • Q3/5

A data set has median 50 and IQR 8. Each value is changed to 2·value + 5. Give the new median and IQR.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.2 • Set 5 • FORWARD 2 • Q4/5

Test scores have SD 6. The teacher adds 3 bonus points to every score. The new SD is

**A.** 18

**B.** 9

**C.** 3

**D.** 6

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S5-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 5 • FORWARD 2 • Q5/5**

# Interpret differences in shape, center and spread in context

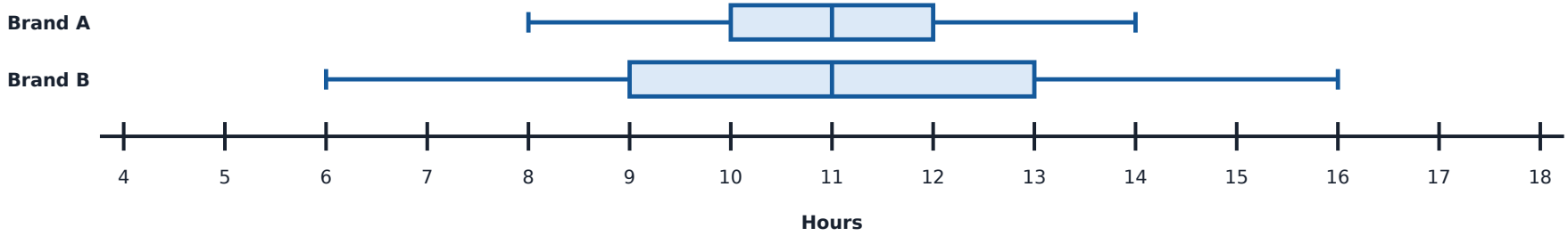
| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.3** |
| **BACKWARD 1** | Describe the shape of a distribution | **6.SP.A.2** |
| **BACKWARD 2** | What center and spread tell about data | **6.SP.A.3** |
| **BACKWARD 3** | Compare two distributions informally | **7.SP.B.4** |
| **FORWARD 1** | Measure separation in units of spread | **HSS-ID.A.3** |
| **FORWARD 2** | Judge whether a difference supports a claim | **HSS-IC.B.6** |

SET 6 • HSS-ID.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

The box plots show the battery lives (hours) of two brands. Which interpretation is correct?




**A.** Brand A's lives vary more, because its box is narrower than brand B's box is.

**B.** Typical lives are the same (both medians 11 hours), but brand B's lives vary more (IQR 4 versus 2 hours).

**C.** Brand B lasts longer, because its longest-lasting battery runs longer than brand A's does.

**D.** Brand B is better in every way, because its whiskers and its box are both longer.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q1/5**

Waiting times at two stores: store 1 is roughly symmetric with mean 6 minutes and standard deviation 1.5 minutes; store 2 is skewed right with median 5 minutes, IQR 6 minutes, and a few waits over 20 minutes. Which interpretation is correct?

**A.** Store 2 often has a shorter wait, but its waits are much less predictable, with occasional very long waits.

**B.** Store 2 always has shorter waits than store 1, because its median is 1 minute lower than the mean of store 1.

**C.** The stores cannot be compared at all, because they are summarized with different measures.

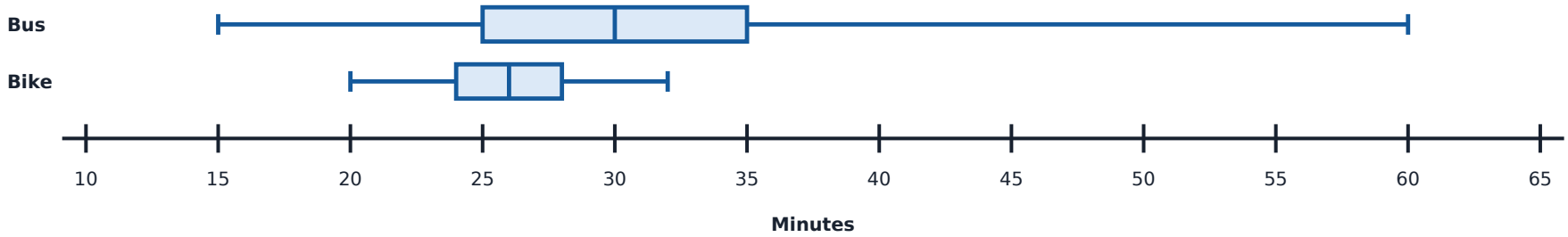
**D.** Store 1's waits vary more, because its waits are described with a mean and a standard deviation.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q2/5**

The box plots show commute times (minutes) by bus and by bike for the same group of workers. Which interpretation is correct?




**A.** Bus trips are shorter in general, because the bus box plot starts at a lower time than the bike box plot.

**B.** Bike trips vary more, because the bike box plot is shorter than the bus box plot on the scale.

**C.** Both ways take the same time, because both box plots contain the times from 25 to 28 minutes.

**D.** Bike trips are more consistent; bus trips take a little longer in the middle and vary much more, including long trips.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q3/5**

Two classes both have mean score 70 on a test. Class X's scores form two clusters, around 55 and around 85; class Y's form one mound around 70. Which interpretation is correct?

**A.** Class X did better, because some of its students scored around 85, which is above every score in class Y.

**B.** Class Y has more spread, because its scores are all close to the middle of the range of scores.

**C.** The equal means hide a difference: class X seems to contain two groups doing very differently; class Y is more alike.

**D.** The classes performed the same way, because their means are equal and the mean summarizes the whole class.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q4/5**

Apples from two farms both have mean weight 150 g; farm A's weights have standard deviation 5 g and farm B's have standard deviation 20 g. Both distributions are roughly symmetric. Which interpretation is correct?

**A.** Farm B's apples are heavier, because a standard deviation of 20 g is larger than one of 5 g.

**B.** Farm A's apples are more uniform in size; on average, apples from both farms weigh about the same.

**C.** Farm A's apples are lighter, because a smaller standard deviation means a smaller mean weight.

**D.** The farms' apples are the same in every way, because their mean weights are both 150 g.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • MAIN • Q5/5**

SET 6 • HSS-ID.A.3

# BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

A histogram has one peak in the middle and falls off evenly on both sides. Its shape is

**A.** symmetric

**B.** bimodal

**C.** skewed left

**D.** skewed right

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q1/5**

## True or false?

A histogram with two separate peaks is called bimodal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q2/5**

A histogram has a long tail toward small values. Its shape is

**A.** uniform

**B.** symmetric

**C.** skewed right

**D.** skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q3/5**

## True or false?

A distribution whose bars are all about the same height is skewed right.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q4/5**

Waiting times at a busy clinic are mostly short with a few very long waits. The shape is most likely

**A.** skewed left

**B.** symmetric

**C.** bimodal

**D.** skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 6 • BACKWARD 1 • Q5/5**

SET 6 • HSS-ID.A.3

# BACKWARD 2

What center and spread tell about data • Grade 6 • Statistics & Probability • 6.SP.A.3

## True or false?

A measure of center summarizes all the values of a data set with a single number.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q1/5**

## True or false?

A measure of spread tells how much the values vary.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q2/5**

Two data sets have the same median but different IQRs. Which is true?

**A.** Their typical values are similar, but one varies more.

**B.** One of them has a larger typical value than the other.

**C.** They are identical data sets with the same values.

**D.** They have the same range and the same maximum.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q3/5**

Which describes spread?

**A.** the median

**B.** the mean

**C.** the IQR

**D.** the mode

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q4/5**

## True or false?

A data set with a larger mean must have a larger standard deviation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 6 • BACKWARD 2 • Q5/5**

SET 6 • HSS-ID.A.3

# BACKWARD 3

Compare two distributions informally • Grade 7 • Statistics & Probability • 7.SP.B.4

Group 1 has median 12 and IQR 3; group 2 has median 15 and IQR 3. Which is true?

**A.** Group 2 varies more, since its median is higher.

**B.** The groups are the same, since the IQRs are equal.

**C.** Group 2's typical value is 3 higher, with similar spread.

**D.** Group 1's values are higher, since its IQR is the same.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q1/5**

## True or false?

If two groups have very different medians and small IQRs, their distributions overlap very little.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q2/5**

Group A has mean 40 and group B has mean 46. By how much are the means different?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q3/5**

Which pair of groups overlaps the most?

**A.** medians 20 and 30, both IQRs 1

**B.** medians 20 and 21, both IQRs 10

**C.** medians 20 and 40, both IQRs 2

**D.** medians 10 and 50, both IQRs 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q4/5**

## True or false?

Two groups with the same center must have the same spread.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

**Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 6 • BACKWARD 3 • Q5/5**

SET 6 • HSS-ID.A.3

# FORWARD 1

Measure separation in units of spread • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Two roughly symmetric groups have means 70 and 76; both have standard deviation 4. How many standard deviations apart are the means?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 6 • FORWARD 1 • Q1/5

Group medians are 30 and 45; both IQRs are 10. How many IQRs apart are the medians?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 6 • FORWARD 1 • Q2/5

Pair 1: means 50 and 55, both SDs 10. Pair 2: means 50 and 53, both SDs 1. Which pair is more clearly separated?

**A.** pair 1, 5 points apart

**B.** pair 1, 0.5 SDs apart

**C.** pair 2, 3 SDs apart

**D.** they are equally separated

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 6 • FORWARD 1 • Q3/5**

## True or false?

Two groups whose means are 0.2 standard deviations apart overlap a great deal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 6 • FORWARD 1 • Q4/5

Why measure a difference in units of spread?

**A.** Units of spread show which group has more values in it.

**B.** Units of spread remove the need to compare the centers of the groups.

**C.** A gap matters only relative to how much values vary within each group.

**D.** Units of spread make every difference look larger than it really is.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 6 • FORWARD 1 • Q5/5**

SET 6 • HSS-ID.A.3

# FORWARD 2

Judge whether a difference supports a claim • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6

A report says that brand X bulbs "last much longer" than brand Y: means 1,010 and 1,000 hours, both with standard deviation 80 hours. Evaluate.

**A.** Weak: the claim would be supported only if the standard deviations were different.

**B.** Strong: both brands have the same standard deviation, so the difference in means must be large.

**C.** Weak: the means differ by about 0.13 standard deviations, so the lives overlap almost completely.

**D.** Strong: 1,010 is greater than 1,000, so every brand X bulb lasts longer than any brand Y bulb.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F2-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q1/5**

## True or false?

A difference of 5 points between group means is more convincing when each group's standard deviation is 2 than when it is 20.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q2/5

A coach claims the new training "improved race times a lot": median time dropped from 52 to 50 seconds; IQRs 8 seconds. Which evaluation is best?

**A.** The drop is small relative to the spread (a quarter of an IQR), so "a lot" overstates it.

**B.** The claim cannot be judged, because the IQRs of the two groups are equal.

**C.** The claim is wrong, because a drop in the median time means the athletes slowed down.

**D.** The drop is large, because two seconds is a long time in a race of 52 seconds.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S6-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q3/5**

Two groups have means 120 and 135 and the same standard deviation 10. How many standard deviations apart are they?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q4/5

## True or false?

If two groups' means differ by less than one standard deviation, the groups cannot differ in any important way.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.6 • Set 6 • FORWARD 2 • Q5/5

# Identify outliers and account for their effects

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.3** |
| **BACKWARD 1** | Find the interquartile range | **6.SP.B.5.c** |
| **BACKWARD 2** | Compute the fences | **6.NS.B.3** |
| **BACKWARD 3** | How an extreme value affects the mean and the median | **6.SP.B.5.d** |
| **FORWARD 1** | Decide whether an outlier is an error or a real value | **HSS-ID.A.3** |
| **FORWARD 2** | Report resistant summaries after checking outliers | **HSS-ID.A.3** |

SET 7 • HSS-ID.A.3

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q1/5

Points a player scored in 11 games: 12, 15, 14, 18, 16, 13, 17, 15, 41, 14, 16. The value 41 is an outlier. Find the mean with and without it, each to the nearest tenth.

Mean with 41: _____

Mean without 41: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • MAIN • Q2/5

Hours of screen time 11 students reported for one weekend: 6, 8, 7, 9, 7, 8, 30, 6, 7, 8, 1. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the lower fence, the upper fence, and the number of outliers.

Lower fence: _____

Upper fence: _____

Number of outliers: _____

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • MAIN • Q3/5**

Prices (dollars) of 10 used bikes: 180, 210, 195, 205, 190, 600, 200, 185, 215, 220. The $600 bike is an outlier. Which describes the effect of removing it?

**A.** The mean and the median both drop by $40.

**B.** The median drops by $40, while the mean drops by only $2.50.

**C.** The mean drops by $40, while the median drops by only $2.50.

**D.** Neither the mean nor the median changes.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • MAIN • Q4/5**

Support calls received in each of 10 hours: 3, 5, 4, 6, 5, 4, 22, 5, 3, 4. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves. An outlier is a value below Q1 − 1.5·IQR or above Q3 + 1.5·IQR. Give the upper fence and the number of outliers.

Upper fence: _____

Number of outliers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • MAIN • Q5/5

SET 7 • HSS-ID.A.3

# BACKWARD 1

Find the interquartile range • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Q1 = 12 and Q3 = 20. Find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q1/5**

For 1, 3, 4, 6, 8, 9, find the IQR (Q1 and Q3 are the medians of the halves).

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q2/5**

## True or false?

The IQR is the difference between the maximum and the minimum.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q3/5**

A box plot has Q1 = 30 and Q3 = 42. The IQR is

**A.** 36

**B.** 42

**C.** 72

**D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q4/5**

For 2, 4, 5, 7, 9, 10, 12 (odd count; leave out the median), find the IQR.

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 7 • BACKWARD 1 • Q5/5**

SET 7 • HSS-ID.A.3

# BACKWARD 2

Compute the fences • Grade 6 • The Number System • 6.NS.B.3

Find 1.5 × 6.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q1/5**

Find 1.5 × 2.5.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q2/5**

**True or false?**

If Q1 = 10 and the IQR is 4, the lower fence is $10 - 6 = 4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q3/5**

If Q3 = 25 and the IQR is 10, the upper fence is

**A.** 15

**B.** 35

**C.** 40

**D.** 37.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q4/5**

Q1 = 7.5 and the IQR is 3. Find the lower fence.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

**Grade 6 • The Number System • 6.NS.B.3 • Set 7 • BACKWARD 2 • Q5/5**

SET 7 • HSS-ID.A.3

# BACKWARD 3

How an extreme value affects the mean and the median • Grade 6 • Statistics & Probability • 6.SP.B.5.d

In 3, 4, 5, 6, 50, which is pulled more by the value 50?

**A.** the mean

**B.** neither

**C.** the median

**D.** both equally

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q1/5**

## True or false?

The median of 2, 3, 4, 5, 100 is 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q2/5**

Find the mean of 2, 3, 4, 5, 100.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q3/5**

## True or false?

Removing an extreme value always changes the median more than the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q4/5**

For skewed data with outliers, the better summary of a typical value is usually

**A.** the maximum

**B.** the range

**C.** the median

**D.** the mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 7 • BACKWARD 3 • Q5/5**

SET 7 • HSS-ID.A.3

# FORWARD 1

Decide whether an outlier is an error or a real value • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

Heights of 9 adults in a study are listed in centimeters; one value is 17.5. Which is the most reasonable judgment?

**A.** Outliers are always errors, so every outlier in the data should be removed.

**B.** It is a real height, so it must be kept exactly as it is in the data.

**C.** It is probably a recording error (likely 175 cm), so check the record before using it.

**D.** It is an outlier, so it should be deleted without any further checking.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F1-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 1 • Q1/5**

## True or false?

A value that the 1.5·IQR rule flags as an outlier must be removed from the data.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q2/5

Among 30 home prices, one mansion sells for 10 times the median. Which is correct?

**A.** It must be removed, because the outlier rule requires it to be deleted.

**B.** It is a legitimate unusual value; report resistant summaries such as the median and the IQR.

**C.** It is a recording error, because no real home could cost ten times the median price.

**D.** It should replace the median, because it is the most informative value in the data.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • FORWARD 1 • Q3/5**

## True or false?

An outlier can be a correct measurement of an unusual case.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 1 • Q4/5

A thermometer log shows 18, 19, 21, 20, 210, 19 °C in a classroom. What is the best next step?

**A.** Check the 210 reading, since a decimal point was likely lost (21.0), before computing summaries.

**B.** Report the mean including 210, because every reading is data and must be used as recorded.

**C.** Delete the 19s, because they are the most common values and do not add information.

**D.** Report the 210 reading as the median, because it is the most unusual value in the list.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.3 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSS-ID.A.3

# FORWARD 2

Report resistant summaries after checking outliers • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.3

After checking, the $600 bike price is real. Which summaries best describe a typical used bike and the spread?

**A.** the median and the IQR

**B.** the mean and the range

**C.** the maximum and the minimum

**D.** the mean and the standard deviation

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 2 • Q1/5**

After a recording error is corrected, the data are 12, 15, 14, 18, 16, 13, 17, 15, 14, 14, 16 (the 41 was really 14). Find the new mean to the nearest tenth and the new median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • FORWARD 2 • Q2/5

## True or false?

The IQR barely changes when a single extreme value is removed from a large data set.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.3 • Set 7 • FORWARD 2 • Q3/5

Which summary is most changed by one extreme value?

**A.** Q1

**B.** the standard deviation

**C.** the median

**D.** the IQR

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S7-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.3 • Set 7 • FORWARD 2 • Q4/5**

Data 5, 6, 6, 7, 8, 30: find the median and the IQR (Q1 and Q3 are the medians of the halves).

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.A.3 • Set 7 • FORWARD 2 • Q5/5

# Estimate percentages with the 68-95-99.7 rule

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.4** |
| **BACKWARD 1** | Count standard deviations from the mean | **HSS-ID.A.2** |
| **BACKWARD 2** | Find a percent of a population | **6.RP.A.3.c** |
| **BACKWARD 3** | Use the symmetry of the normal curve | **6.SP.A.2** |
| **FORWARD 1** | Find a cutoff from a percent | **HSS-ID.A.4** |
| **FORWARD 2** | Compare positions in two populations with standardized values | **HSS-ID.A.4** |

SET 8 • HSS-ID.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights of adult women in a population are approximately normal with mean 64 inches and standard deviation 2.5 inches. Use the 68-95-99.7 rule. What percent are between 61.5 and 66.5 inches, and what percent are between 59 and 69 inches?

61.5 to 66.5: _____

59 to 69: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q1/5

Battery lives are approximately normal with mean 40 hours and standard deviation 4 hours. Use the 68-95-99.7 rule. What percent of batteries last more than 48 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q2/5

Commute times are approximately normal with mean 30 minutes and standard deviation 5 minutes. Use the 68-95-99.7 rule. What percent are between 25 and 40 minutes?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q3/5

Egg masses are approximately normal with mean 58 g and standard deviation 3 g. Use the 68-95-99.7 rule. What percent are less than 52 g, and what percent are more than 61 g?

Less than 52 g: _____

More than 61 g: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q4/5

Scores on a test are approximately normal with mean 500 and standard deviation 100. Use the 68-95-99.7 rule. What percent are between 300 and 600, and what percent are below 400?

300 to 600: _____

Below 400: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSS-ID.A.4

# BACKWARD 1

Count standard deviations from the mean • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

The mean is 50 and the standard deviation is 5. How many standard deviations above the mean is 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q1/5

The mean is 12 and the standard deviation is 3. How many standard deviations below the mean is 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q2/5

## True or false?

With mean 100 and standard deviation 15, the value 70 is 2 standard deviations below the mean.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q3/5

With mean 64 and standard deviation 2.5, which value is 1 standard deviation above the mean?

**A.** 62.5

**B.** 66.5

**C.** 65

**D.** 67

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-B1-Q4**

**High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q4/5**

The mean is 30 and the standard deviation is 5. What value is 2 standard deviations below the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

# BACKWARD 2

Find a percent of a population • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

What is 68% of 500?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q1/5**

Half of 5% is what percent?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q2/5**

## True or false?

34% + 13.5% = 47.5%.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q3/5**

If 95% of values are inside an interval, what percent are outside it?

**A.** 5%

**B.** 95%

**C.** 0%

**D.** 2.5%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q4/5**

What is 100% minus 68%, divided by 2?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 8 • BACKWARD 2 • Q5/5**

SET 8 • HSS-ID.A.4

# BACKWARD 3

Use the symmetry of the normal curve • Grade 6 • Statistics & Probability • 6.SP.A.2

## True or false?

In a normal distribution, half of the values are above the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q1/5**

**True or false?**

In a normal distribution, more values are between the mean and 1 SD above it than between the mean and 1 SD below it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q2/5**

What percent of a normal distribution lies between the mean and 2 standard deviations above it?

**A.** 95%

**B.** 50%

**C.** 47.5%

**D.** 34%

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q3/5**

What percent of a normal distribution lies between 1 SD below the mean and the mean?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q4/5**

What percent of a normal distribution is more than 1 standard deviation below the mean?

**A.** 16%

**B.** 34%

**C.** 2.5%

**D.** 32%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 8 • BACKWARD 3 • Q5/5**

SET 8 • HSS-ID.A.4

# FORWARD 1

Find a cutoff from a percent • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Heights are approximately normal with mean 64 in and standard deviation 2.5 in. Use the 68-95-99.7 rule. Above what height are the tallest 2.5%?

Height (in): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q1/5

Times are approximately normal with mean 30 min and standard deviation 5 min. Use the 68-95-99.7 rule. Below what time are the fastest 16%?

Time (min): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q2/5

## True or false?

In a normal distribution with mean 200 and standard deviation 20, the middle 95% is from 160 to 240.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q3/5

Test scores are approximately normal with mean 70 and SD 8. Which score separates the top 16%?

**A.** 78

**B.** 74

**C.** 62

**D.** 86

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q4/5**

Lengths are approximately normal with mean 12 cm and standard deviation 0.5 cm. Use the 68-95-99.7 rule. Give the interval that holds the middle 68%.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSS-ID.A.4

# FORWARD 2

Compare positions in two populations with standardized values • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Ana scored 85 on a test with mean 75 and SD 5; Ben scored 90 on a test with mean 80 and SD 10. How many standard deviations above the mean is each score?

Ana: _____

Ben: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q1/5

Using the scores above (Ana 2 SDs above, Ben 1 SD above, both tests normal), which is correct?

**A.** Ana's score is relatively higher: about 2.5% scored above her, but about 16% scored above Ben.

**B.** Ben's score is relatively higher, because 90 is a larger number than 85.

**C.** The scores are equally good, because both are 10 points above their means.

**D.** They cannot be compared, because the two tests had different means.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q2/5**

## True or false?

A value 1 SD below the mean in one normal population and a value 1 SD below the mean in another are at the same percentile.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q3/5

A 100-m time of 12.0 s comes from a population with mean 13.0 s and SD 0.5 s. How many standard deviations below the mean is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q4/5

Two normal populations: in A, mean 50, SD 5; in B, mean 50, SD 10. Which value is more unusual?

**A.** they are equally unusual

**B.** 60 in population A

**C.** 55 in population B

**D.** 60 in population B

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S8-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 8 • FORWARD 2 • Q5/5**

# Estimate normal areas with a table or calculator

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.4** |
| **BACKWARD 1** | Compute a standardized value | **7.NS.A.3** |
| **BACKWARD 2** | Read areas as proportions | **7.SP.C.5** |
| **BACKWARD 3** | Find right tails and areas between from left areas | **HSS-ID.A.4** |
| **FORWARD 1** | Find the cutoff for a percentile | **HSS-ID.A.4** |
| **FORWARD 2** | Make decisions with normal areas | **HSS-ID.A.4** |

SET 9 • HSS-ID.A.4

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are approximately normal with mean 100 and standard deviation 15. What proportion of people score below 118? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | 1.0 | 1.2 | 1.5 | 2.0 |
|---|---|---|---|---|
| Area to the left | 0.8413 | 0.8849 | 0.9332 | 0.9772 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q1/5

Lengths of calls are approximately normal with mean 50 seconds and standard deviation 8 seconds. What proportion of calls last more than 60 seconds? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | 1.0 | 1.25 | 1.5 | 2.0 |
|---|---|---|---|---|
| Area to the left | 0.8413 | 0.8944 | 0.9332 | 0.9772 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q2/5

Daily water use is approximately normal with mean 20 gallons and standard deviation 2.5 gallons. What proportion of days use between 18 and 23 gallons? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | −1.0 | −0.8 | 0.8 | 1.2 |
|---|---|---|---|---|
| Area to the left | 0.1587 | 0.2119 | 0.7881 | 0.8849 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q3/5

Grade-point averages at a school are approximately normal with mean 3.0 and standard deviation 0.4. What proportion are below 2.5? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | −1.5 | −1.25 | −1.0 | −0.5 |
|---|---|---|---|---|
| Area to the left | 0.0668 | 0.1056 | 0.1587 | 0.3085 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q4/5

Heights of adult men are approximately normal with mean 175 cm and standard deviation 7 cm. What proportion are between 168 and 189 cm? Use the table values printed below (areas to the left of $z$). Give four decimal places.

| $z$ | $-2.0$ | $-1.0$ | $1.0$ | $2.0$ |
|---|---|---|---|---|
| Area to the left | 0.0228 | 0.1587 | 0.8413 | 0.9772 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • MAIN • Q5/5

SET 9 • HSS-ID.A.4

# BACKWARD 1

Compute a standardized value • Grade 7 • The Number System • 7.NS.A.3

Compute (130 − 100)/15.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q1/5**

Compute $(42 - 50)/4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q2/5**

**True or false?**

$(9.5 - 10)/0.5 = -1$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q3/5**

Compute $(61 - 58)/2$.

**A.** 0.67

**B.** 3

**C.** 1.5

**D.** $-1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q4/5**

Compute $(2.2 - 3.0)/0.4$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

**Grade 7 • The Number System • 7.NS.A.3 • Set 9 • BACKWARD 1 • Q5/5**

SET 9 • HSS-ID.A.4

# BACKWARD 2

Read areas as proportions • Grade 7 • Statistics & Probability • 7.SP.C.5

**True or false?**

An area of 0.8413 to the left of $z = 1$ means about 84% of the population is below 1 SD above the mean.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q1/5**

The table gives area 0.0228 to the left of $z = -2$. What does this mean?

**A.** Exactly 2 values are below the mean.

**B.** About 22.8% of values are below the mean.

**C.** About 2.28% of values are above the mean.

**D.** About 2.28% of values are below 2 SDs below the mean.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q2/5**

Write 0.1587 as a percent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q3/5**

## True or false?

A proportion of a population can be greater than 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q4/5**

Which is the largest proportion?

**A.** 0.5

**B.** 0.8413

**C.** 0.9772

**D.** 0.0228

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.5 • Set 9 • BACKWARD 2 • Q5/5**

SET 9 • HSS-ID.A.4

# BACKWARD 3

Find right tails and areas between from left areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

The area to the left of $z = 1.5$ is 0.9332. What is the area to the right of $z = 1.5$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q1/5

Areas to the left: $z = -1$ gives 0.1587; $z = 1$ gives 0.8413. Find the area between $z = -1$ and $z = 1$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q2/5

## True or false?

The area to the right of $z = 0$ is 0.5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q3/5

The area to the left of $z = 0.5$ is 0.6915. The area between $z = 0$ and $z = 0.5$ is

**A.** 0.3085

**B.** 0.6915

**C.** 0.1915

**D.** 0.5

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S9-B3-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q4/5**

The area to the left of $z = 2$ is 0.9772. What is the area to the left of $z = -2$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSS-ID.A.4

# FORWARD 1

Find the cutoff for a percentile • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

IQ scores are normal with mean 100 and SD 15. The table gives area 0.9332 to the left of $z = 1.5$. What score is at the 93.32nd percentile?

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q1/5

Call lengths are normal with mean 50 s and SD 8 s. The table gives area 0.1056 to the left of $z = -1.25$. What length is at the 10.56th percentile?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q2/5

## True or false?

A value with $z = 0$ is at the 50th percentile of a normal distribution.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q3/5

A factory wants only the longest 2.28% of rods to be rejected. Rod lengths are normal with mean 10 cm and SD 0.2 cm; the table gives area 0.9772 to the left of $z = 2$. What cutoff should it use?

**A.** 10.2 cm

**B.** 12 cm

**C.** 10.4 cm

**D.** 9.6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q4/5

Heights are normal with mean 175 cm and SD 7 cm; area 0.8413 is to the left of $z = 1$. What height separates the tallest 15.87%?

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSS-ID.A.4

# FORWARD 2

Make decisions with normal areas • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

A bolt fits if its diameter is between 9.8 and 10.2 mm. Diameters are normal with mean 10.0 mm and SD 0.1 mm. Table: area 0.9772 left of $z = 2$ and 0.0228 left of $z = -2$. What proportion of bolts fit?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q1/5

Machine A fills bottles with mean 500 mL, SD 4 mL; machine B with mean 502 mL, SD 1 mL. A bottle is underfilled below 498 mL. Table: area 0.3085 to the left of $z = -0.5$; the area to the left of $z = -4$ is less than 0.0001. Which machine underfills fewer bottles?

**A.** A, because its standard deviation is larger

**B.** B: under 0.01% versus about 31% for A

**C.** A, because its mean is closer to 498 mL

**D.** They underfill the same proportion

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q2/5

## True or false?

Reducing a process's standard deviation, with the mean unchanged, reduces the proportion of items far from the mean.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q3/5

Scores are normal with mean 70 and SD 10. A prize goes to scores of 85 or more. Table: area 0.9332 left of z = 1.5. What proportion of students win?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q4/5

Two normal processes have the same mean. Process 1 has SD 2; process 2 has SD 5. Which has more items within 4 units of the mean?

**A.** process 2

**B.** the same

**C.** cannot tell

**D.** process 1

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S9-F2-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 9 • FORWARD 2 • Q5/5**

# Recognize when a normal model is not appropriate

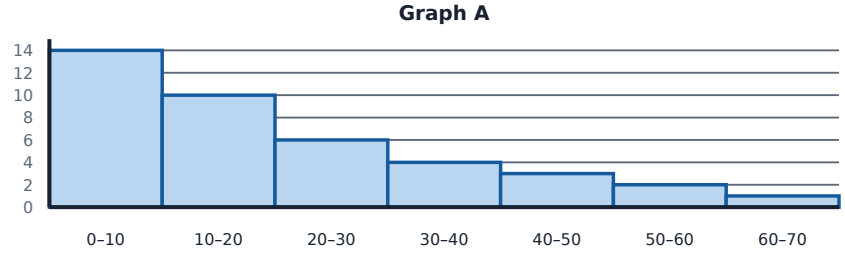
| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.A.4** |
| **BACKWARD 1** | Describe the shape of a distribution | **6.SP.A.2** |
| **BACKWARD 2** | Relate the mean and median to the shape | **6.SP.B.5.d** |
| **BACKWARD 3** | Test a normal model against the data | **HSS-ID.A.4** |
| **FORWARD 1** | Compare actual percents with the 68-95-99.7 rule | **HSS-ID.A.4** |
| **FORWARD 2** | Choose summaries for data that are not normal | **HSS-ID.A.2** |

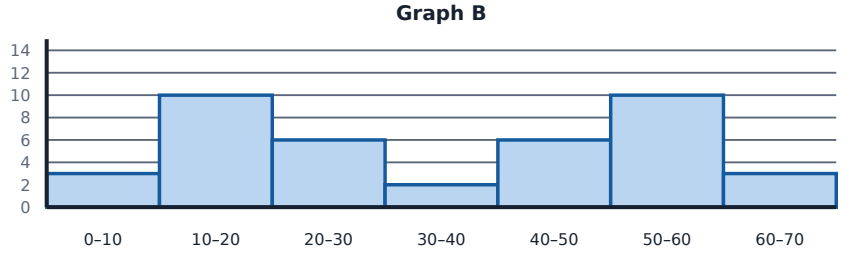
SET 10 • HSS-ID.A.4

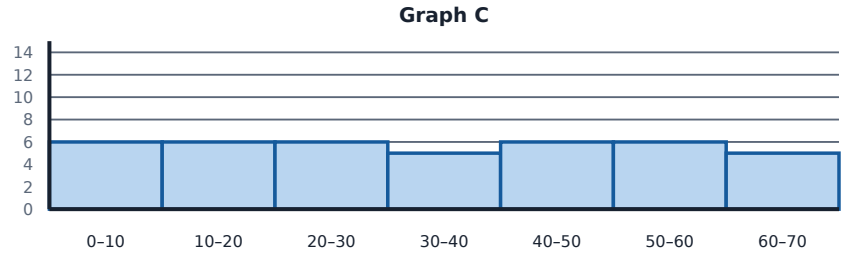
# MAIN

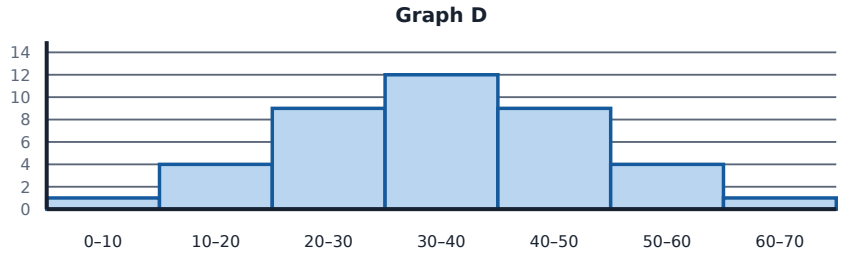
Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Each histogram shows 40 measurements on the same scale. For which data set is a normal model most reasonable? In each histogram, the horizontal axis shows value and the vertical axis shows frequency.




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q1/5**

Waiting times at a clinic can never be negative. Their mean is 3 minutes and their standard deviation is 4 minutes. If a normal model with this mean and standard deviation were used, what proportion of waiting times would it place below 0 minutes? Use the table values printed below (areas to the left of z). Give four decimal places.

| z | −1.0 | −0.75 | −0.5 | 0.5 |
|---|---|---|---|---|
| Area to the left | 0.1587 | 0.2266 | 0.3085 | 0.6915 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q2/5

Waiting times can never be negative, yet a normal model with their mean (3 minutes) and standard deviation (4 minutes) places about 23% of times below 0. What does this show?

**A.** The times are normal, but 23% of them were recorded wrong.

**B.** The standard deviation must have been computed incorrectly.

**C.** A normal model fits, because the mean is a positive number.

**D.** A normal model is not appropriate for these waiting times.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q3/5**

Which data set is least likely to be approximately normal?

**A.** the masses of eggs from one large farm

**B.** repeated careful measurements of one object's length

**C.** the yearly incomes of all households in a large city

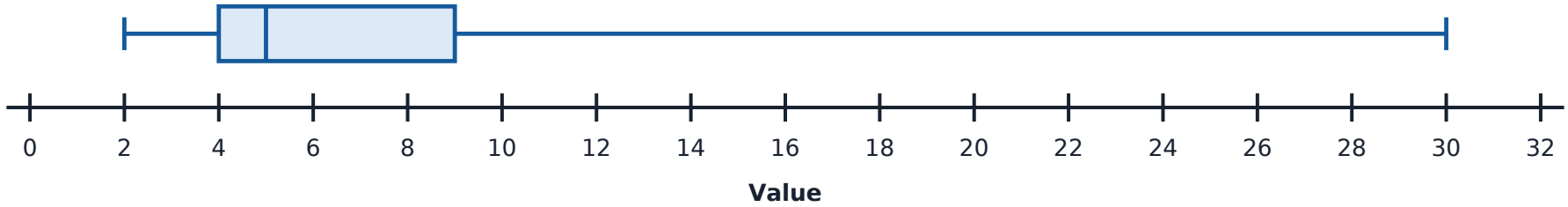
**D.** the heights of adult women in one country

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q4/5**

A box plot of 60 values shows minimum 2, Q1 4, median 5, Q3 9 and maximum 30. Which conclusion is correct?




**A.** The data are roughly symmetric, so a normal model fits well.

**B.** The data are skewed left, so a normal model is not a good fit.

**C.** The data are skewed right, so a normal model is not a good fit.

**D.** A median of 5 shows that the data must be normal.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • MAIN • Q5/5**

SET 10 • HSS-ID.A.4

# BACKWARD 1

Describe the shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

## True or false?

A histogram with a long tail to the right is called skewed right.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q1/5**

What is the shape of the data 1, 1, 2, 2, 2, 3, 3, 4, 6, 9?

**A.** skewed right

**B.** skewed left

**C.** symmetric

**D.** uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q2/5**

## True or false?

A distribution with two separate peaks is called unimodal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q3/5**

Which list of values is roughly symmetric?

**A.** 1, 10, 18, 19, 19, 20

**B.** 3, 3, 3, 3, 3, 30

**C.** 1, 1, 1, 2, 9, 20

**D.** 2, 4, 5, 5, 6, 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q4/5**

For the data 4, 5, 5, 6, 6, 6, 7, 7, 8, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 10 • BACKWARD 1 • Q5/5**

SET 10 • HSS-ID.A.4

# BACKWARD 2

Relate the mean and median to the shape • Grade 6 • Statistics & Probability • 6.SP.B.5.d

**True or false?**

In a distribution skewed right, the mean is usually greater than the median.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q1/5**

A data set has mean 52 and median 40. Which shape is most likely?

**A.** skewed right

**B.** skewed left

**C.** uniform

**D.** symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q2/5**

For the data 1, 2, 2, 3, 22, find the mean and the median.

Mean: _____

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q3/5**

**True or false?**

For strongly skewed data, the mean and standard deviation are the best summaries of center and spread.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q4/5**

For data skewed right, which measure best describes a typical value?

**A.** median

**B.** mean

**C.** midrange

**D.** maximum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 10 • BACKWARD 2 • Q5/5**

SET 10 • HSS-ID.A.4

# BACKWARD 3

Test a normal model against the data • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Counts of errors per page (never negative) have mean 8 and standard deviation 4. How many standard deviations below the mean is 0, and what percent would a normal model place below 0 by the 68-95-99.7 rule?

Standard deviations: _____

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q1/5

## True or false?

A data set with any mean and any standard deviation is approximately normal.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q2/5

Of 20 values with mean 50 and standard deviation 10, 19 lie between 40 and 60. Is this consistent with a normal model?

**A.** Yes: a normal model predicts about 95% within 1 SD.

**B.** Yes: every value is within 3 SDs of the mean.

**C.** No: a normal model predicts about 68% within 1 SD, but 95% are.

**D.** No: a normal model predicts all values within 1 SD.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-B3-Q3**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q3/5**

A normal model is used for 200 values. About how many values does it predict within 2 standard deviations of the mean?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q4/5

## True or false?

A normal model with mean 3 and SD 4 places about 16% of its values below −1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSS-ID.A.4

# FORWARD 1

Compare actual percents with the 68-95-99.7 rule • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4

Of 40 values, 27 lie within 1 standard deviation of the mean and 38 lie within 2. What percent is each?

Within 1 SD: _____

Within 2 SD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q1/5

Of 40 values, 67.5% are within 1 SD of the mean and 95% within 2 SDs, and a histogram is one symmetric mound. Which conclusion is reasonable?

**A.** A normal model is not reasonable, because 67.5% is not 68%.

**B.** The data must be skewed right.

**C.** A normal model is reasonable for these data.

**D.** No conclusion is possible without the raw values.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F1-Q2**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q2/5**

## True or false?

If 92% of a data set lies within 1 SD of its mean, the data fit a normal model well.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q3/5

Of 50 values, 46 lie within 1 standard deviation of the mean. What percent is that?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q4/5

Which result most strongly suggests a data set is not normal?

**A.** About half of the values lie above the mean.

**B.** 96% of the values lie within 2 SDs of the mean.

**C.** 92% of the values lie within 1 SD of the mean.

**D.** 70% of the values lie within 1 SD of the mean.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.4 • Set 10 • FORWARD 1 • Q5/5**

SET 10 • HSS-ID.A.4

# FORWARD 2

Choose summaries for data that are not normal • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.A.2

Home prices in a town are strongly skewed right. Which pair of summaries is most appropriate?

**A.** mode and maximum

**B.** mean and range

**C.** mean and standard deviation

**D.** median and IQR

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 10 • FORWARD 2 • Q1/5**

Home prices (thousands of dollars) are 150, 160, 170, 180, 190, 200, 900. Find the median and the IQR. Quartiles: Q1 is the median of the lower half and Q3 the median of the upper half of the ordered data; when the number of values is odd, the median is left out of both halves.

Median: _____

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.A.2 • Set 10 • FORWARD 2 • Q2/5

## True or false?

Using a normal model for strongly skewed data can give misleading percentages.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q3/5

For the prices 150, 160, 170, 180, 190, 200, 900 (thousands), which statement is correct?

**A.** The value 900 pulls the median above the mean.

**B.** The mean and the median are equal.

**C.** The value 900 pulls the mean above the median.

**D.** The value 900 changes the IQR more than the mean.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S10-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.A.2 • Set 10 • FORWARD 2 • Q4/5**

## True or false?

The standard deviation is resistant to one extreme value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.A.2 • Set 10 • FORWARD 2 • Q5/5

# Joint and marginal frequencies in two-way tables

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.5** |
| **BACKWARD 1** | Complete and read a two-way frequency table | **8.SP.A.4** |
| **BACKWARD 2** | Write a part of a whole as a fraction, decimal and percent | **7.NS.A.2.d** |
| **FORWARD 1** | Build a joint relative-frequency table | **HSS-ID.B.5** |
| **FORWARD 2** | Look for an association by comparing rows | **HSS-ID.B.5** |

SET 11 • HSS-ID.B.5

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were asked how they prefer to take a class. What fraction of all students are 10th graders who chose online? Give a fraction in lowest terms.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q1/5

160 students were asked how they prefer to take a class. What is the marginal relative frequency of choosing online? Give a fraction in lowest terms.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 11 • MAIN • Q2/5

A survey of 120 students: 50 are 7th graders and 70 are 8th graders. 18 of the 7th graders play an instrument, and 20 of the 8th graders do not. How many 7th graders do not play, and how many 8th graders do play?

7th graders who do not play: _____

8th graders who play: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q3/5

160 students were surveyed (table). Which statement gives a marginal relative frequency?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

**A.** 28.125% of all students are 10th graders who chose online.

**B.** 53.125% of all students chose in person.

**C.** 56.25% of the 10th graders chose online.

**D.** 37.5% of the 9th graders chose online.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • MAIN • Q4/5**

A dealer has 100 cars in two lots (table). What percent of all the cars are gray, and what percent of all the cars are red cars in lot B?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Gray: _____

Red in lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • MAIN • Q5/5

SET 11 • HSS-ID.B.5

# BACKWARD 1

Complete and read a two-way frequency table • Grade 8 • Statistics & Probability • 8.SP.A.4

120 adults were asked whether they own a dog and whether they walk daily (table). How many walk daily?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q1/5**

From the dog-and-walking table, how many adults have no dog and do not walk daily?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q2/5**

Of 50 students, 20 are in band and 12 of the band members are 9th graders. 30 of the 50 students are 9th graders. How many 9th graders are not in band?

**A.** 30

**B.** 8

**C.** 18

**D.** 12

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q3/5**

## True or false?

In a two-way table, the row totals add up to the grand total.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q4/5**

## True or false?

In a two-way table, a cell count can be greater than its row total.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 11 • BACKWARD 1 • Q5/5**

SET 11 • HSS-ID.B.5

# BACKWARD 2

Write a part of a whole as a fraction, decimal and percent • Grade 7 • The Number System • 7.NS.A.2.d

Write 45/160 as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q1/5**

Write 24/120 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q2/5**

## True or false?

3/8 = 0.375.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q3/5**

Write 15/32 as a percent.

**A.** 15.32%

**B.** 47%

**C.** 2.13%

**D.** 46.875%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q4/5**

Write 30/160 in lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

**Grade 7 • The Number System • 7.NS.A.2.d • Set 11 • BACKWARD 2 • Q5/5**

SET 11 • HSS-ID.B.5

# FORWARD 1

Build a joint relative-frequency table • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

Convert the online-class table to relative frequencies of the grand total (160). What are the joint relative frequencies for 9th grade online and 9th grade in person? Give decimals.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

9th online: _____

9th in person: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q1/5

## True or false?

In a table of joint relative frequencies of the grand total, all the cells add to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q2/5

In a joint relative-frequency table, the 9th-grade cells are 0.1875 and 0.3125. What is the 9th-grade marginal relative frequency?

**A.** 0.5

**B.** 0.1875

**C.** 0.6

**D.** 0.125

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 1 • Q3/5**

In the car table (100 cars), what are the joint relative frequencies for white cars in lot A and gray cars in lot B? Give decimals.

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

White, lot A: _____

Gray, lot B: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 1 • Q4/5**

## True or false?

A marginal relative frequency is always smaller than every joint relative frequency in its row.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSS-ID.B.5

# FORWARD 2

Look for an association by comparing rows • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the online-class table, what percent of 9th graders and what percent of 10th graders chose online?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

9th: _____

10th: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q1/5

If grade and choice of class were not associated, what would be true?

**A.** Every cell of the table would contain the same count.

**B.** The percent choosing online would be about the same in each grade.

**C.** The number choosing online would be 0 in one grade.

**D.** The grade totals would have to be unequal.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 2 • Q2/5**

## True or false?

Because more 10th graders (45) than 9th graders (30) chose online, the counts alone prove an association, even if the grades had very different sizes.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 11 • FORWARD 2 • Q3/5

In the car table, what percent of lot A cars and what percent of lot B cars are gray?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Lot A: _____

Lot B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 11 • FORWARD 2 • Q4/5

Gray makes up 25% of lot A and 50% of lot B. Which is correct?

**A.** Lot B has exactly twice as many cars as lot A.

**B.** Gray is the most common color in lot A.

**C.** Lot and color are associated: gray is more common in lot B.

**D.** Lot and color are not associated.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S11-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 11 • FORWARD 2 • Q5/5**

# Conditional relative frequencies and association

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.5** |
| **BACKWARD 1** | Choose the denominator for a conditional relative frequency | **8.SP.A.4** |
| **BACKWARD 2** | Find a percent of a group | **6.RP.A.3.c** |
| **BACKWARD 3** | Compare two fractions | **4.NF.A.2** |
| **FORWARD 1** | Distinguish the two directions of a conditional rate | **HSS-ID.B.5** |
| **FORWARD 2** | Recognize when combining groups reverses a comparison | **HSS-ID.B.5** |

SET 12 • HSS-ID.B.5

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

160 students were surveyed (table). What fraction of the 9th graders chose online, and what fraction of the 10th graders chose online? Give fractions in lowest terms.

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

9th graders: _____

10th graders: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q1/5

In the survey, 37.5% of 9th graders and 56.25% of 10th graders chose online. Which statement is correct?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

**A.** 9th graders chose online at a higher rate, so grade and choice are associated.

**B.** The rates are equal, so grade and choice are not associated.

**C.** 10th graders chose online at a higher rate, so grade and choice are associated.

**D.** Each grade has 80 students, so grade and choice cannot be associated.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • MAIN • Q2/5**

160 students were surveyed (table). Of the students who chose online, what percent were 10th graders?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • MAIN • Q3/5

120 adults were surveyed (table). What percent of dog owners walk daily, and what percent of adults without a dog walk daily?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

Dog owners: _____

No dog: _____

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • MAIN • Q4/5**

In the survey, 60% of dog owners and 25% of adults without a dog walk daily. Which conclusion is supported?

| | Walks daily | Does not | Total |
|---|---|---|---|
| Has a dog | 36 | 24 | 60 |
| No dog | 15 | 45 | 60 |
| Total | 51 | 69 | 120 |

**A.** Dog ownership and walking daily are not associated.

**B.** Owning a dog causes people to walk daily.

**C.** Owning a dog and walking daily are associated in this sample.

**D.** Adults without a dog walk daily more often.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • MAIN • Q5/5**

SET 12 • HSS-ID.B.5

# BACKWARD 1

Choose the denominator for a conditional relative frequency • Grade 8 • Statistics & Probability • 8.SP.A.4

Of the 60 dog owners, 36 walk daily. What fraction of dog owners walk daily? Give lowest terms.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q1/5**

**True or false?**

To find the percent of 9th graders who chose online, divide the 9th-grade online count by the grand total of all students.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q2/5**

In the online-class table, which number is the denominator for the percent of online-choosers who are 9th graders?

| | Online | In person | Total |
|---|---|---|---|
| 9th grade | 30 | 50 | 80 |
| 10th grade | 45 | 35 | 80 |
| Total | 75 | 85 | 160 |

**A.** 80, the 9th-grade total

**B.** 75, the online total

**C.** 30, the cell count

**D.** 160, the grand total

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q3/5**

Write 30 out of 80 as a percent.

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q4/5**

## True or false?

A row relative frequency uses that row's total as the denominator.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 12 • BACKWARD 1 • Q5/5**

SET 12 • HSS-ID.B.5

# BACKWARD 2

Find a percent of a group • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c

45 is what percent of 80?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q1/5**

What is 25% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q2/5**

18 out of 24 is what percent?

**A.** 18%

**B.** 75%

**C.** 133%

**D.** 24%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q3/5**

## True or false?

9/16 = 56.25%.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q4/5**

## True or false?

45 is 60% of 80.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

**Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 12 • BACKWARD 2 • Q5/5**

SET 12 • HSS-ID.B.5

# BACKWARD 3

Compare two fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which is larger, 3/8 or 9/16?

**A.** 3/8

**B.** they cannot be compared

**C.** they are equal

**D.** 9/16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q1/5**

## True or false?

3/5 > 1/4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q2/5**

Write 3/8 as an equivalent fraction with denominator 16. What is the numerator?

Numerator: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q3/5**

## True or false?

2/5 > 3/4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q4/5**

Order 1/4, 3/5 and 3/8 from least to greatest.

**A.** 1/4, 3/5, 3/8

**B.** 3/8, 1/4, 3/5

**C.** 1/4, 3/8, 3/5

**D.** 3/5, 3/8, 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

**Grade 4 • Number & Operations–Fractions • 4.NF.A.2 • Set 12 • BACKWARD 3 • Q5/5**

SET 12 • HSS-ID.B.5

# FORWARD 1

Distinguish the two directions of a conditional rate • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

In the car table, what percent of the gray cars are in lot B, and what percent of the lot B cars are gray?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Gray cars in lot B: _____

Lot B cars that are gray: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q1/5

## True or false?

The percent of lot B cars that are gray equals the percent of gray cars that are in lot B.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 1 • Q2/5

In the car table, what percent of the white cars are in lot A?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

**A.** 30%

**B.** 45%

**C.** 18%

**D.** 40%

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • FORWARD 1 • Q3/5**

In the car table, what percent of the lot A cars are red?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 1 • Q4/5

Which statement uses a column total of the car table as its denominator?

| | White | Red | Gray | Total |
|---|---|---|---|---|
| Lot A | 18 | 12 | 10 | 40 |
| Lot B | 22 | 8 | 30 | 60 |
| Total | 40 | 20 | 40 | 100 |

**A.** 12% of all cars are red cars in lot A.

**B.** Of the lot A cars, 30% are red.

**C.** Of the red cars, 60% are in lot A.

**D.** 40% of all cars are in lot A.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-F1-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • FORWARD 1 • Q5/5**

SET 12 • HSS-ID.B.5

# FORWARD 2

Recognize when combining groups reverses a comparison • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.5

Use the hospital table. What is the survival rate for mild cases at hospital A and at hospital B?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 2 • Q1/5

Use the hospital table. What is the survival rate for severe cases at hospital A and at hospital B?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 2 • Q2/5

Use the hospital table. What is the overall survival rate (all 500 patients) at hospital A and at hospital B?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

A: _____

B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.5 • Set 12 • FORWARD 2 • Q3/5

Hospital A has the higher survival rate for mild cases and for severe cases, but the lower rate overall. Why?

| Hospital | Mild: survived / patients | Severe: survived / patients |
|---|---|---|
| A | 95 / 100 | 120 / 400 |
| B | 360 / 400 | 25 / 100 |

**A.** The two hospitals treated different total numbers of patients.

**B.** The overall rates were computed with the wrong denominators.

**C.** A treats many more mild cases, and mild cases survive less often.

**D.** A treats many more severe cases, and severe cases survive less often.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S12-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.5 • Set 12 • FORWARD 2 • Q4/5**

## True or false?

If one group has the higher rate within every category, it must have the higher rate overall.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.5 • Set 12 • FORWARD 2 • Q5/5

# Represent two quantitative variables on a scatter plot

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6** |
| **BACKWARD 1** | Plot and read points in the first quadrant | **5.G.A.2** |
| **BACKWARD 2** | Describe patterns of association | **8.SP.A.1** |
| **FORWARD 1** | Compare the direction, form and strength of associations | **HSS-ID.B.6** |
| **FORWARD 2** | Distinguish association from causation | **HSS-ID.C.9** |

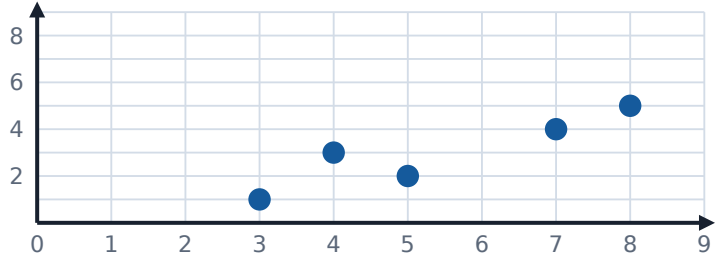
SET 13 • HSS-ID.B.6

# MAIN

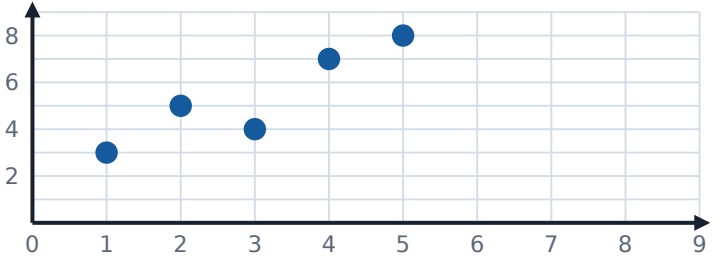
Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

A seedling was measured weekly: (week, height in cm) = (1, 3), (2, 5), (3, 4), (4, 7), (5, 8). Which scatter plot shows the data?

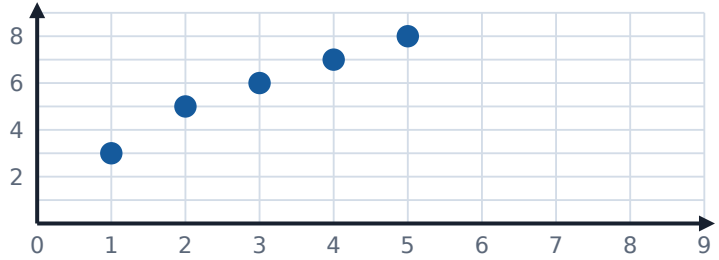
**A**



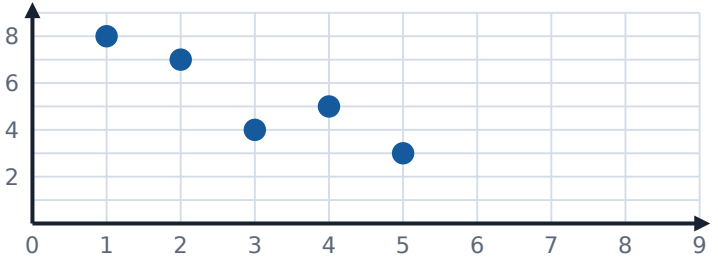

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q1/5**

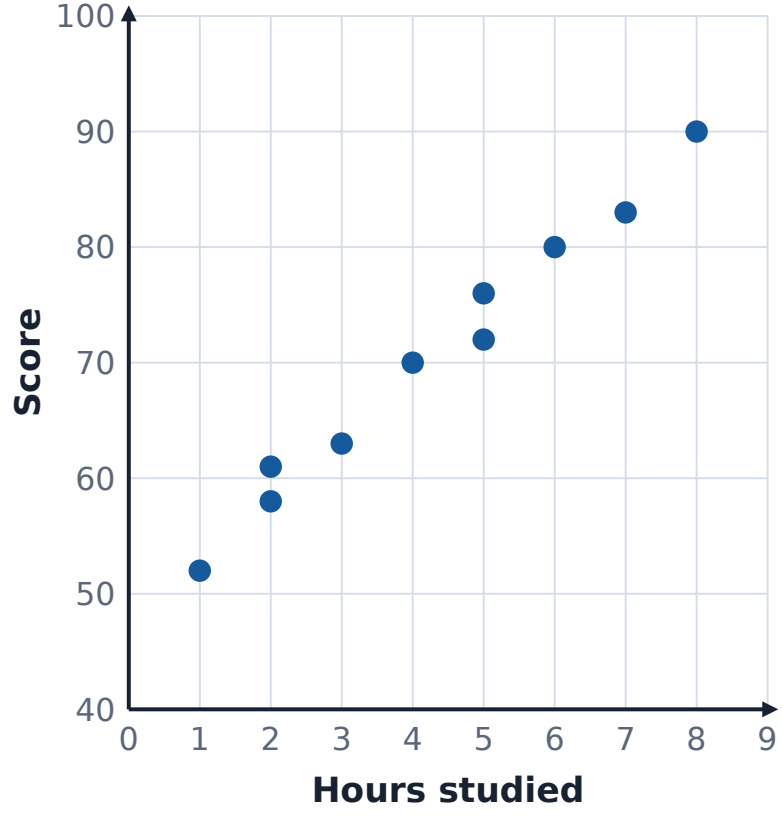
The scatter plot shows hours studied and test scores. Which describes the association?

**A.** curved, not linear

**B.** no clear association

**C.** strong, positive, linear

**D.** strong, negative, linear




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q2/5

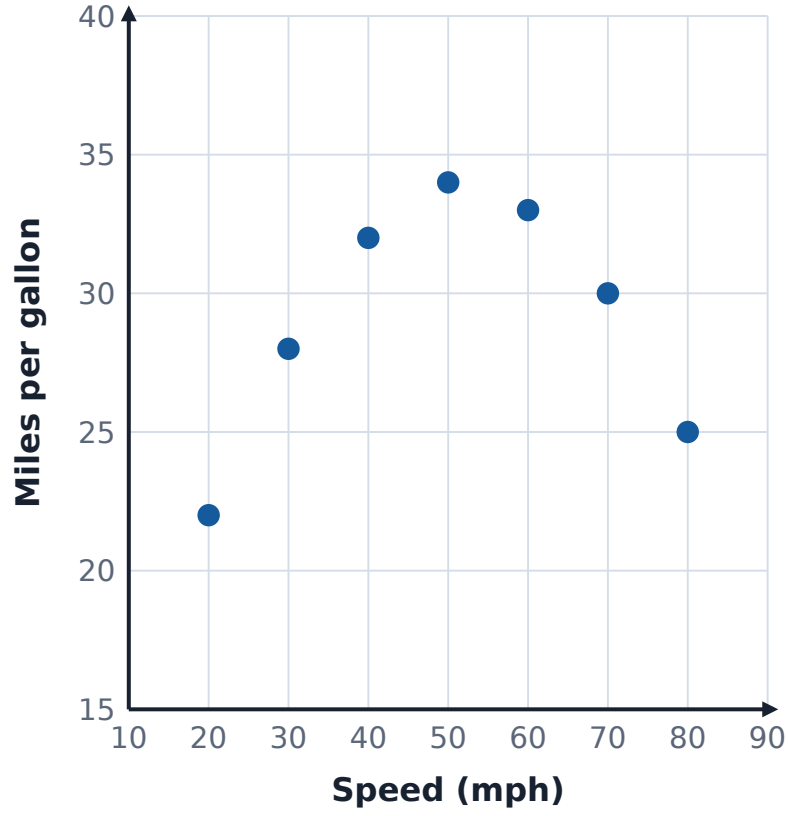
The scatter plot shows a car's speed and fuel efficiency. Which describes the relationship?

**A.** curved: rises, then falls

**B.** no association at all

**C.** strong, negative, linear

**D.** strong, positive, linear




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

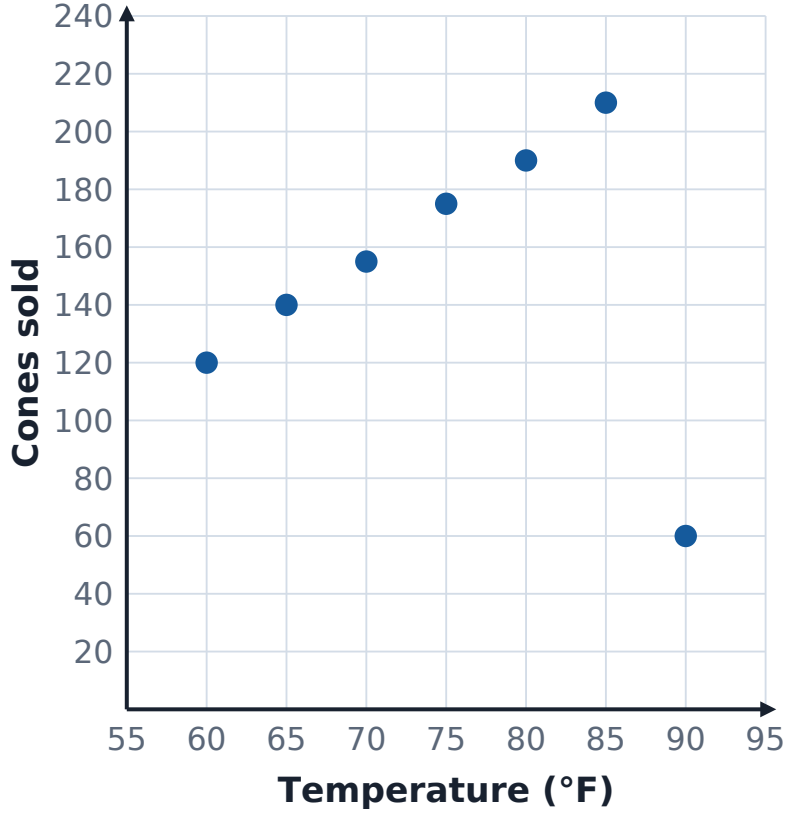
ID S13-M-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q3/5

The scatter plot shows daily high temperature and ice-cream cones sold. One day does not follow the pattern. Give its temperature and its cones sold.

Temperature (°F): _____

Cones sold: _____




**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q4/5**

Which pair of variables would most likely show a negative association?

**A.** the height and arm span of adults

**B.** hours studied and test score

**C.** the age of a car and its resale value

**D.** shoe size and quiz score of adults

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • MAIN • Q5/5**

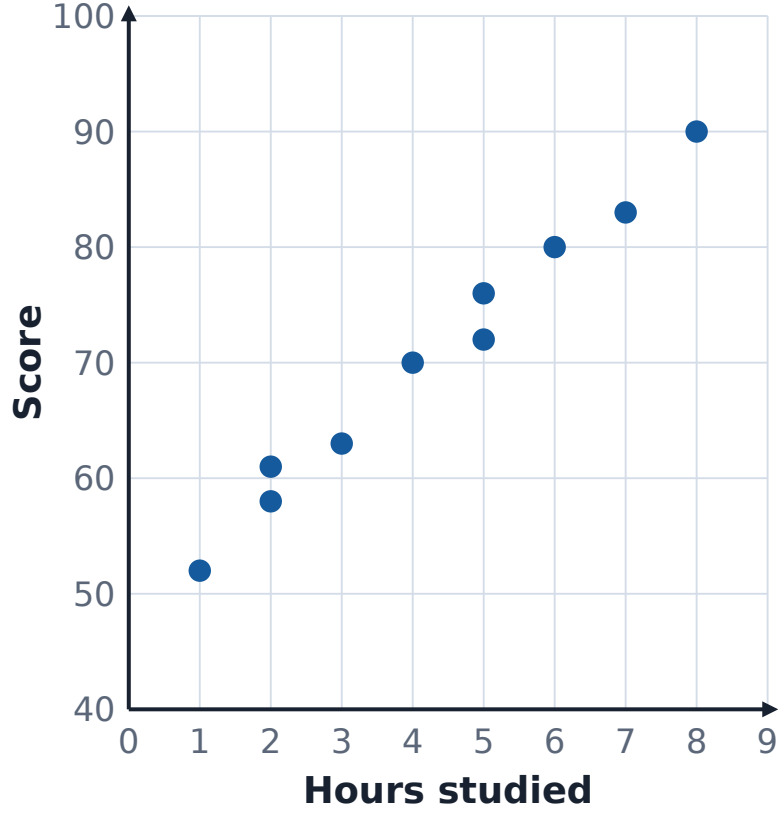
SET 13 • HSS-ID.B.6

# BACKWARD 1

Plot and read points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

In the study scatter plot, what score did the student who studied 7 hours get?

Score: _____



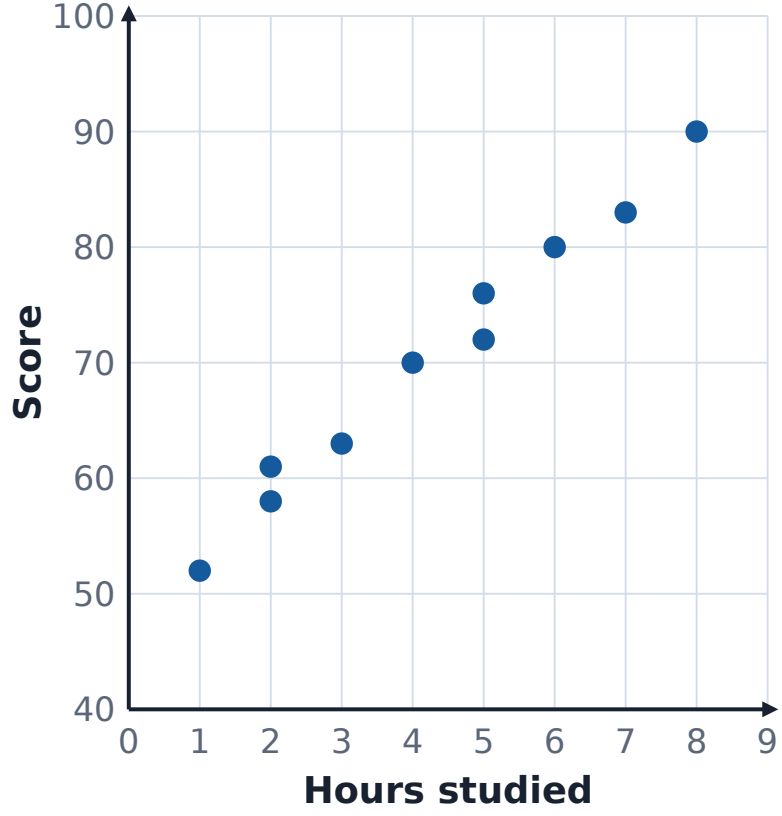

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q1/5**

In the study scatter plot, how many students studied 2 hours?

Number: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q2/5**

In the study data, what does the point (4, 70) mean?

**A.** a student studied 4 hours and scored 70

**B.** a student studied 70 hours and scored 4

**C.** 4 students scored 70

**D.** the average score was 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q3/5**

**True or false?**

In the ordered pair (3, 63), 63 is the x-coordinate.

**A.** True

**B.** False

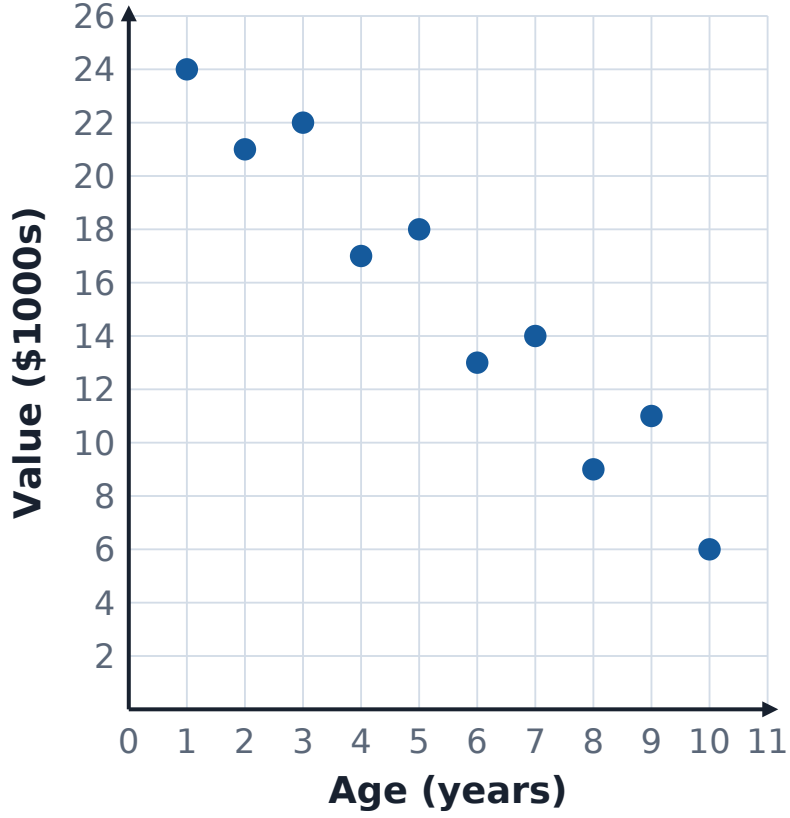
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q4/5**

In the car scatter plot, what is the resale value of the 8-year-old car, in thousands of dollars?

Value: _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

**Grade 5 • Geometry • 5.G.A.2 • Set 13 • BACKWARD 1 • Q5/5**

SET 13 • HSS-ID.B.6

# BACKWARD 2

Describe patterns of association • Grade 8 • Statistics & Probability • 8.SP.A.1

## True or false?

When y tends to increase as x increases, the association is positive.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q1/5**

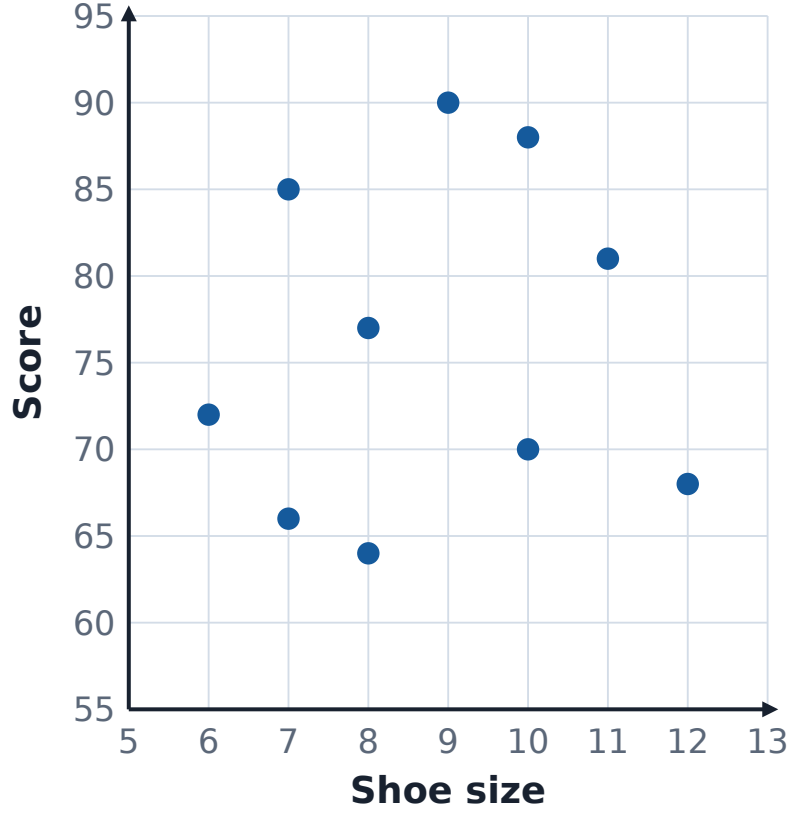
The scatter plot shows shoe size and quiz score. Which describes it?

**A.** curved

**B.** strong, positive, linear

**C.** strong, negative, linear

**D.** no clear association




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q2/5**

## True or false?

A scatter plot whose points lie close to a U-shaped curve shows a linear association.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q3/5**

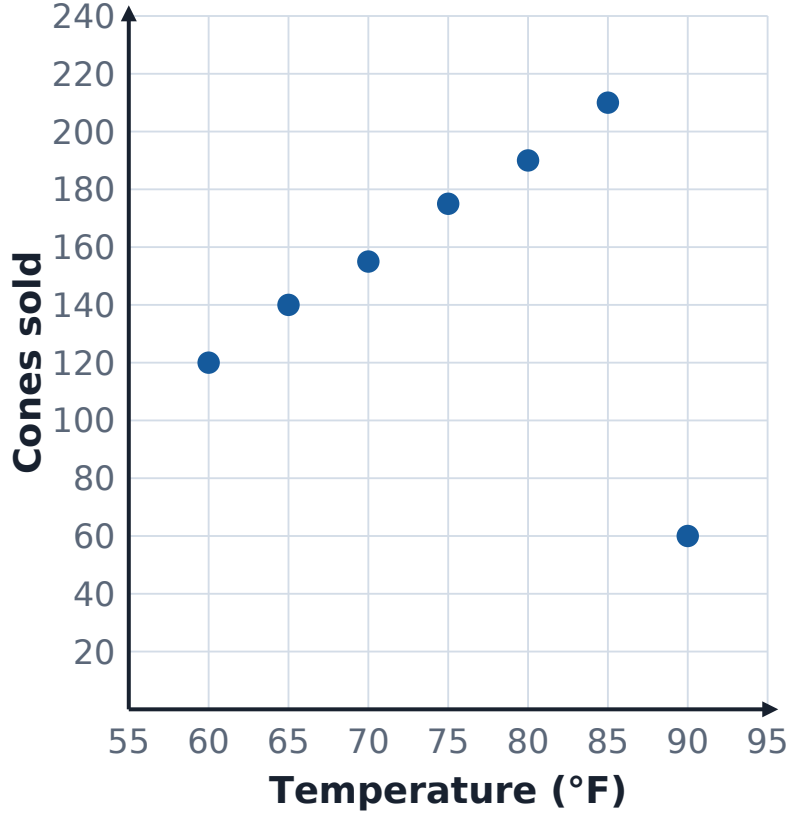
In the ice-cream scatter plot, the point (90, 60) is best described as

**A.** the trend line

**B.** an outlier

**C.** a cluster

**D.** the y-intercept



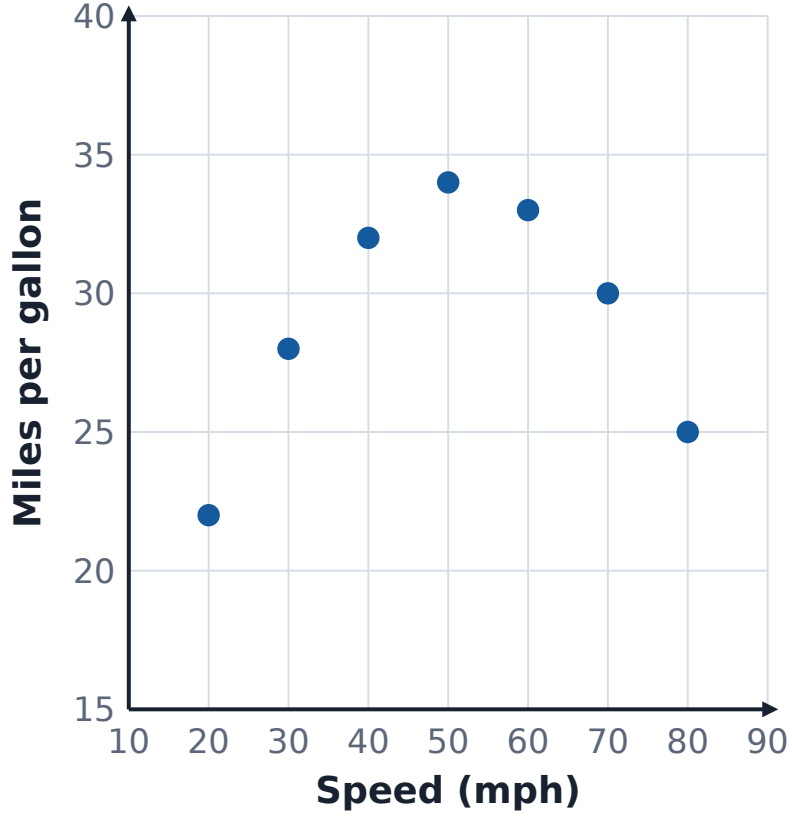

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q4/5**

In the fuel-efficiency scatter plot, at what speed is the fuel efficiency greatest?

Speed (mph): _____




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 13 • BACKWARD 2 • Q5/5**

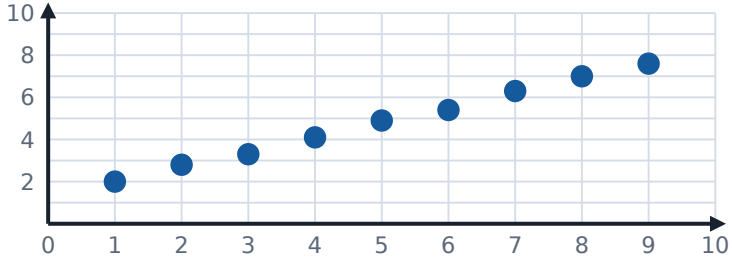
SET 13 • HSS-ID.B.6

# FORWARD 1

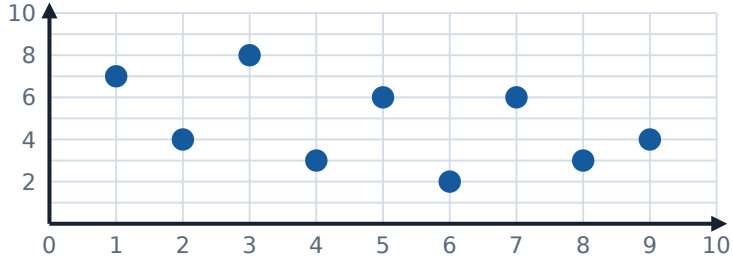
Compare the direction, form and strength of associations • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

Which scatter plot shows a strong, negative, linear association?

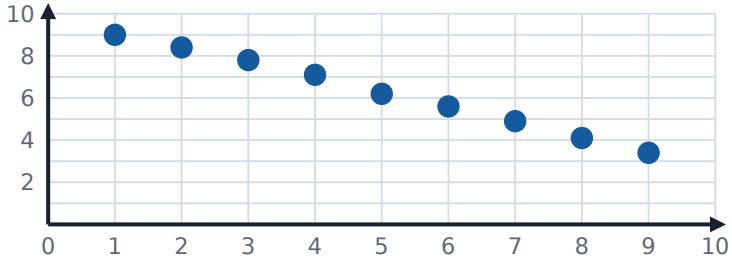
**A**



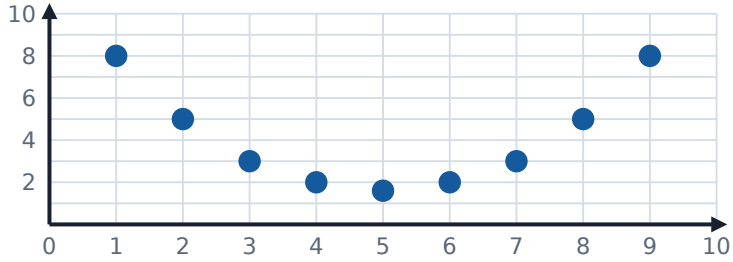

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q1/5**

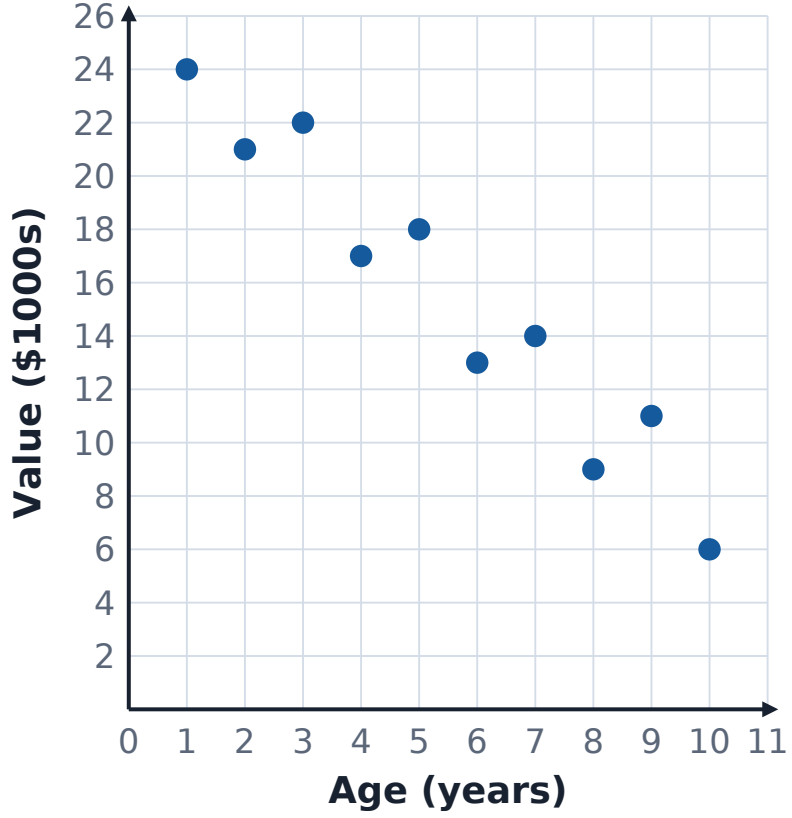
The scatter plot shows the age and resale value of cars. Which describes it?

**A.** curved, then level

**B.** strong, negative, linear

**C.** weak, positive, linear

**D.** strong, positive, linear




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q2/5

## True or false?

A pattern can show a strong association even when it is not linear.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q3/5

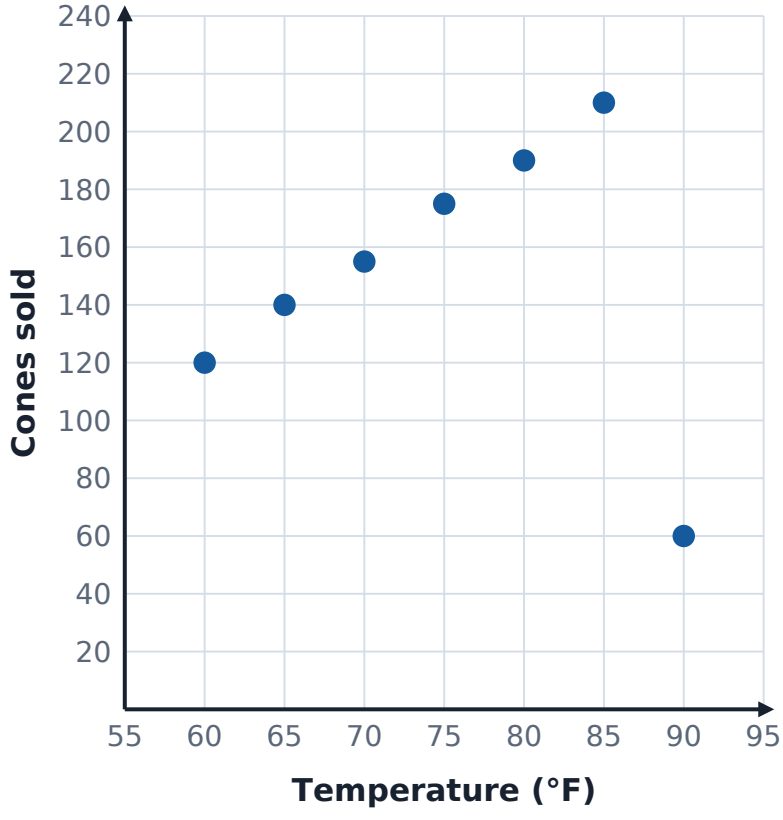
In the ice-cream scatter plot, if the outlier (90, 60) is ignored, the other six days show

**A.** no association

**B.** a strong, negative, linear association

**C.** a U-shaped association

**D.** a strong, positive, linear association




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q4/5

## True or false?

Removing an outlier can never change how strong an association looks.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSS-ID.B.6

# FORWARD 2

Distinguish association from causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Cities with more fire stations tend to have more fires. Which is the best explanation?

**A.** Fires cause cities to build fewer stations.

**B.** Fire stations cause fires to start.

**C.** Larger cities have both more stations and more fires.

**D.** The association must be a mistake in the data.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 13 • FORWARD 2 • Q1/5**

## True or false?

A strong association between two variables proves that one causes the other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 13 • FORWARD 2 • Q2/5

To show that a fertilizer causes taller plants, which study is needed?

**A.** a survey of gardeners about their plants

**B.** a scatter plot of plant heights found in gardens

**C.** a randomized experiment with fertilized and unfertilized plants

**D.** measuring only the plants that received fertilizer

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F2-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.9 • Set 13 • FORWARD 2 • Q3/5**

Ice-cream sales and swimming accidents both rise during the year. What best explains the association?

**A.** The two variables are unrelated.

**B.** Swimming accidents cause ice-cream sales.

**C.** Eating ice cream causes swimming accidents.

**D.** Hot weather increases both.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S13-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 13 • FORWARD 2 • Q4/5**

## True or false?

In an observational study, a lurking variable can create an association between two variables.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.9 • Set 13 • FORWARD 2 • Q5/5

# Use a function fitted to data to solve problems

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.a** |
| **BACKWARD 1** | Evaluate a function at an input | **HSF-IF.A.2** |
| **BACKWARD 2** | Evaluate expressions with exponents | **6.EE.A.2.c** |
| **BACKWARD 3** | Solve a linear equation for the input | **HSA-REI.B.3** |
| **FORWARD 1** | Judge whether a prediction is reasonable | **HSS-ID.B.6.a** |
| **FORWARD 2** | Compare predictions from two fitted models | **HSS-ID.B.6.a** |

SET 14 • HSS-ID.B.6.a

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

For a group of students, a line fitted to the data is $y = 2.5x + 12$, where $x$ is hours of practice and $y$ is the predicted number of free throws made out of 50. Predict $y$ for 8 hours of practice.

Predicted $y$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q1/5

A bacteria culture is modeled by $P = 200(1.15)^t$, where t is hours. Predict the population after 6 hours, to the nearest whole number. A calculator or spreadsheet may be used.

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 14 • MAIN • Q2/5

A ball's height is modeled by $h = -4.9t^2 + 20t + 1.5$ (meters, t in seconds). Predict the height at t = 2 seconds.

Height (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 14 • MAIN • Q3/5

A line fitted to data is $y = 0.8x + 5$, where $x$ is the temperature (°C) and $y$ is the predicted number of visitors (hundreds). For what temperature does the model predict 29 hundred visitors?

Temperature (°C): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q4/5

A car's value is modeled by $V = 18000(0.85)^t$, where t is years. Predict the value after 4 years, to the nearest dollar. A calculator or spreadsheet may be used.

Value ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • MAIN • Q5/5

SET 14 • HSS-ID.B.6.a

# BACKWARD 1

Evaluate a function at an input • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

$f(x) = 3x - 4$. Find $f(6)$.

$f(6)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q1/5**

$g(t) = 100(2)^t$. Find $g(3)$.

$g(3)$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q2/5**

## True or false?

If $h(t) = -5t^2 + 10t$, then $h(1) = 5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q3/5**

f(x) = 0.5x + 7. What is f(10)?

**A.** 7.5

**B.** 17

**C.** 57

**D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q4/5**

## True or false?

If $p(t) = 50(1.1)^t$, then $p(0) = 55$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

# BACKWARD 2

Evaluate expressions with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $1.5^2$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q1/5**

Evaluate $4 \cdot 3^2 - 6$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q2/5**

## True or false?

$2 \cdot 5^2 = 100$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q3/5**

Evaluate $1000(0.9)^2$.

**A.** 1800

**B.** 900

**C.** 81

**D.** 810

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q4/5**

## True or false?

$(-3)^2 = 9$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

**Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 14 • BACKWARD 2 • Q5/5**

SET 14 • HSS-ID.B.6.a

# BACKWARD 3

Solve a linear equation for the input • High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3

Solve 0.8x + 5 = 21.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q1/5

Solve 2.5x + 12 = 47.

x: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q2/5

**True or false?**

The solution of $3x - 4 = 11$ is $x = 5$.

**A.** True

**B.** False

Solve 1.2x + 6 = 30.

**A.** 28.8

**B.** 25

**C.** 20

**D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q4/5

## True or false?

The solution of $0.5x + 3 = 8$ is $x = 5.5$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

High School — Algebra I • Algebra: Reasoning with Equations & Inequalities • HSA-REI.B.3 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSS-ID.B.6.a

# FORWARD 1

Judge whether a prediction is reasonable • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The line $y = 2.5x + 12$ was fitted to students who practiced 0 to 12 hours ($y$ = free throws made out of 50). It predicts $y = 87$ for 30 hours. Which is correct?

**A.** Reasonable, because 30 hours is a whole number of hours.

**B.** Unreasonable, because a fitted line must have a negative slope.

**C.** Unreasonable: 30 hours is far outside the data, and 87 exceeds 50.

**D.** Reasonable, because the line fits the data from 0 to 12 hours.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F1-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q1/5**

A line fitted to data for x from 10 to 40 is $y = 3x - 5$. Predict y at $x = 25$, and state whether $x = 25$ is inside the data range by entering yes or no.

Predicted y: _____

Inside the range (yes or no): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q2/5

## True or false?

Predicting far outside the range of the data (extrapolation) is usually less reliable than predicting inside it.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q3/5

The model $V = 18000(0.85)^t$ fits a car's value for t = 0 to 8 years. What value does it predict at t = 0?

V ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q4/5

## True or false?

The model $h = -4.9t^2 + 20t + 1.5$ gives sensible heights for every value of t, including t = 10.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSS-ID.B.6.a

# FORWARD 2

Compare predictions from two fitted models • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Two models fit a town's population (thousands), t years after 2020: linear $L = 2t + 40$ and exponential $E = 40(1.05)^t$. Predict each for $t = 2$. Round E to two decimal places.

L: _____

E: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q1/5

For the models $L = 2t + 40$ and $E = 40(1.05)^t$, which statement about large t is correct?

**A.** E eventually predicts less, because 1.05 is close to 1.

**B.** The two models always predict the same population.

**C.** L always predicts more, because its slope is 2.

**D.** E eventually predicts more, because it grows by a constant percent.

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q2/5**

Using L = 2t + 40 (thousands), in what year t after 2020 does L predict 60 thousand?

t: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q3/5

## True or false?

If two models fit the data equally well inside the data range, they must give the same prediction far outside it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q4/5

A residual check shows the exponential model leaves random scatter and the linear model leaves a curved pattern. Which model should be used for predictions?

**A.** the exponential model

**B.** neither, because residuals exist

**C.** the linear model

**D.** either, equally

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S14-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 14 • FORWARD 2 • Q5/5**

# Choose and fit a linear, quadratic or exponential function

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.a** |
| **BACKWARD 1** | Tell equal differences from equal factors | **HSF-LE.A.1.a** |
| **BACKWARD 2** | Write a linear function from two points | **8.F.B.4** |
| **BACKWARD 3** | Write an exponential function from two points | **HSF-LE.A.2** |
| **FORWARD 1** | Fit a model with regression technology | **HSS-ID.B.6.a** |
| **FORWARD 2** | Check a fitted model with residuals | **HSS-ID.B.6.b** |

SET 15 • HSS-ID.B.6.a

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

The data fit an exponential function $y = a \cdot b^x$ exactly. Find a and b.

| x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| y | 3 | 6 | 12 | 24 | 48 |

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q1/5

The data fit a linear function $y = mx + b$ exactly. Find $m$ and $b$.

| x | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| y | 5 | 8 | 11 | 14 | 17 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q2/5

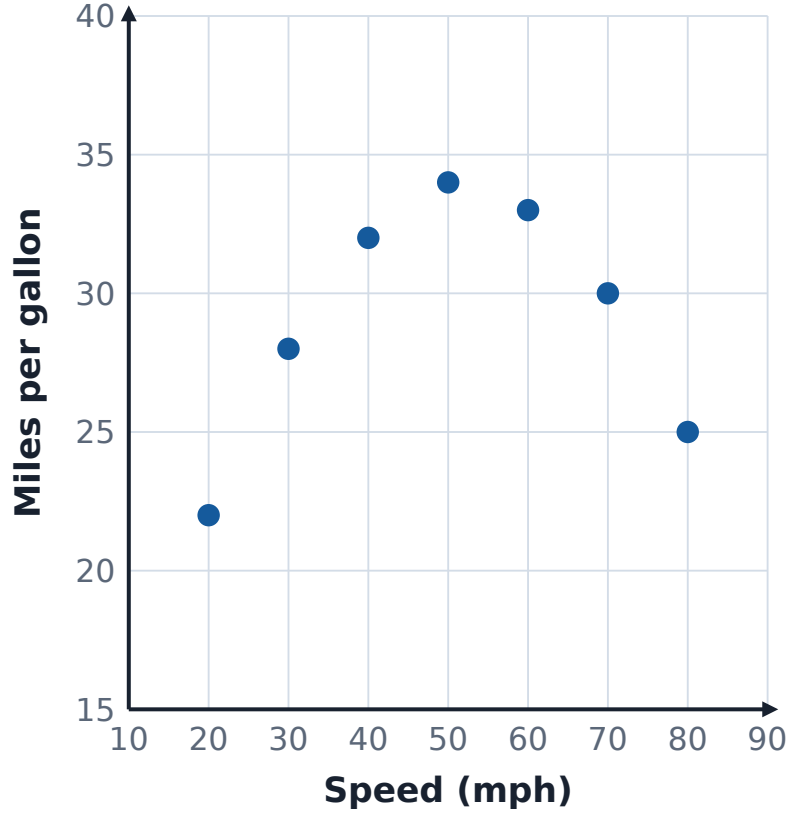
The scatter plot shows a car's speed and fuel efficiency. Which family of functions best fits the data?

**A.** quadratic

**B.** exponential

**C.** linear

**D.** constant




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 15 • MAIN • Q3/5

A town has 12,000 people and its population grows by about 8% each year. Which model fits, with t in years?

**A.** $P = 12000 + 0.08t$

**B.** $P = 12000(1.08)^t$

**C.** $P = 12000 + 1.08t$

**D.** $P = 12000(0.08)^t$

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S15-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 15 • MAIN • Q4/5**

An exponential function $y = a \cdot b^x$ passes through (0, 500) and (1, 400). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • MAIN • Q5/5

SET 15 • HSS-ID.B.6.a

# BACKWARD 1

Tell equal differences from equal factors • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a

## True or false?

In $y = 3x + 5$, $y$ changes by equal differences over equal intervals of $x$.

**A.** True

**B.** False

For x = 0, 1, 2, 3 the values of y are 2, 6, 18, 54. What is the pattern?

**A.** squared each step

**B.** add 3 each step

**C.** add 4 each step

**D.** multiplied by 3 each step

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q2/5

**True or false?**

The values 4, 7, 12, 19 (for x = 0 to 3) have constant first differences.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q3/5

For x = 0 to 4, y = 4, 7, 12, 19, 28. What is the constant second difference?

Second difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q4/5

Which list of y-values (for x = 0, 1, 2, 3) is linear?

**A.** 1, 2, 4, 8

**B.** 0, 1, 4, 9

**C.** 5, 5.5, 6.5, 8

**D.** 10, 7, 4, 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

# BACKWARD 2

Write a linear function from two points • Grade 8 • Functions • 8.F.B.4

A line passes through (0, 5) and (2, 11). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q1/5**

Find the slope of the line through (1, 4) and (5, 12).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q2/5**

**True or false?**

The table $x = 0, 1, 2$ and $y = 7, 4, 1$ fits $y = -3x + 7$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q3/5**

Which function fits x = 0, 1, 2 and y = 2, 4.5, 7?

**A.** y = 2x + 2.5

**B.** y = 2.5x + 2

**C.** y = 4.5x

**D.** y = 2.5x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q4/5**

**True or false?**

The line through (0, 4) and (1, 10) has slope 4.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 15 • BACKWARD 2 • Q5/5**

SET 15 • HSS-ID.B.6.a

# BACKWARD 3

Write an exponential function from two points • High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

An exponential function $y = a \cdot b^x$ passes through (0, 200) and (1, 300). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q1/5

An exponential function $y = a \cdot b^x$ with $b > 0$ passes through (0, 80) and (2, 20). Find a and b.

a: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q2/5

## True or false?

$y = 50(1.2)^x$ passes through (1, 60).

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q3/5

Which exponential function passes through (0, 4) and (1, 12)?

**A.** $y = 3(4)^x$

**B.** $y = 4(3)^x$

**C.** $y = 4 + 8x$

**D.** $y = 12(4)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

High School — Algebra I • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 15 • BACKWARD 3 • Q4/5

## True or false?

$y = 10(0.5)^x$ passes through (2, 5).

**A.** True

**B.** False

SET 15 • HSS-ID.B.6.a

# FORWARD 1

Fit a model with regression technology • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

Find the least-squares line $y = mx + b$ for the spring data. Round $m$ and $b$ to two decimal places. A calculator or spreadsheet may be used.

| Mass (g) | 2 | 4 | 5 | 7 | 8 | 10 |
|---|---|---|---|---|---|---|
| Length (cm) | 15 | 19 | 24 | 26 | 31 | 35 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q1/5

The points (0, 2), (1, 5), (2, 10), (3, 17) lie exactly on a quadratic $y = ax^2 + bx + c$. Find a, b and c. A calculator or spreadsheet may be used.

a: _____

b: _____

c: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q2/5

Data on y rise faster and faster as x increases, y is always positive, and each step multiplies y by about 1.3. Which regression should be used?

**A.** linear

**B.** exponential

**C.** none of these

**D.** quadratic

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S15-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q3/5**

Using the fitted line y = 2.52x + 9.86, predict y when x = 6.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q4/5

## True or false?

Height-versus-time data for a thrown ball are better fitted by a quadratic than by a line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.a • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSS-ID.B.6.a

# FORWARD 2

Check a fitted model with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

For the model $y = 2.52x + 9.86$, find the residual (observed minus predicted) for the data point $(5, 24)$.

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q1/5

For the model $y = 2.52x + 9.86$, find the residual (observed minus predicted) for the data point $(4, 19)$.

Residual: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q2/5

## True or false?

A positive residual means the data point lies above the fitted line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q3/5

A line is fitted to data that actually grow exponentially. What pattern do the residuals usually show?

**A.** every residual positive

**B.** a curved (U-shaped) pattern

**C.** random scatter around 0

**D.** all residuals equal to 0

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S15-F2-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q4/5**

## True or false?

If the residuals of a linear model are small but form a clear curve, a linear model is still the best family.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 15 • FORWARD 2 • Q5/5

# Assess the fit of a function with residuals

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.b** |
| **BACKWARD 1** | Subtract with signed decimals | **7.NS.A.1.c** |
| **BACKWARD 2** | Find a predicted value from a model | **HSF-IF.A.2** |
| **BACKWARD 3** | Compare an observed value with a prediction | **HSS-ID.B.6.a** |
| **FORWARD 1** | Compute least-squares residuals | **HSS-ID.B.6.b** |
| **FORWARD 2** | Compare two models with residuals | **HSS-ID.B.6.b** |

SET 16 • HSS-ID.B.6.b

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The model $y = 2x + 1$ is used for the data in the table. Find the residuals (observed minus predicted) at $x = 1$ and at $x = 2$.

| x | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| y | 3.5 | 4.6 | 7.2 | 8.8 | 11.1 |

Residual at $x = 1$: _____

Residual at $x = 2$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • MAIN • Q1/5

A model predicts $y = 3x + 10$. Find the residuals (observed minus predicted) for the observed points (4, 20) and (6, 30).

Residual for (4, 20): _____

Residual for (6, 30): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

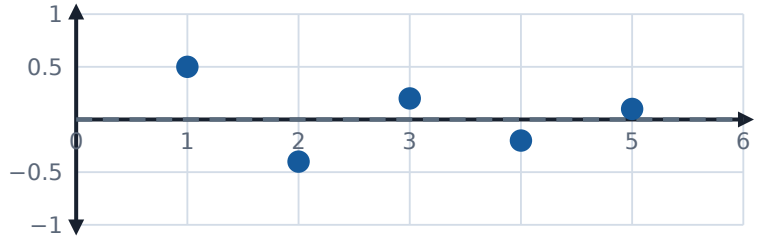
ID S16-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

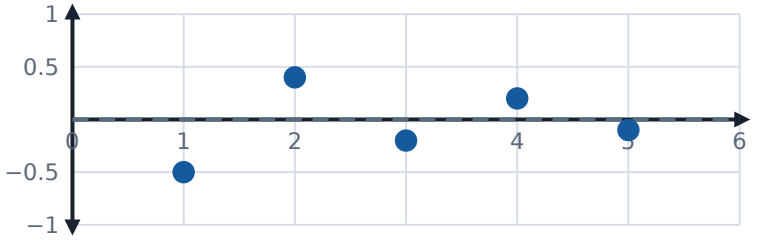
HSS-ID.B.6.b • Set 16 • MAIN • Q2/5

The model y = 2x + 1 is used for the data in the table (x = 1 to 5; y = 3.5, 4.6, 7.2, 8.8, 11.1). Which plot is the residual plot (observed minus predicted)?

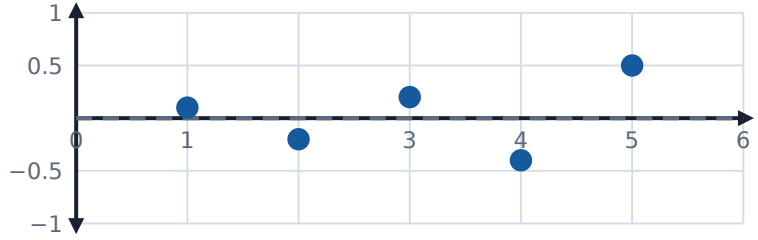
**A**



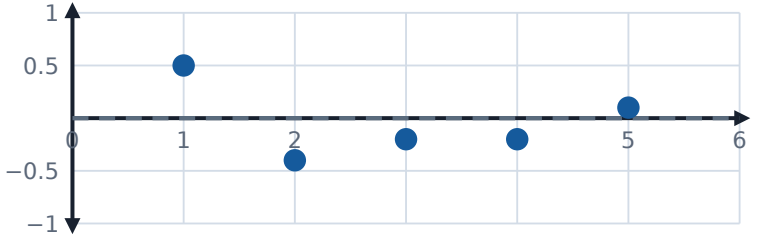

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 16 • MAIN • Q3/5**

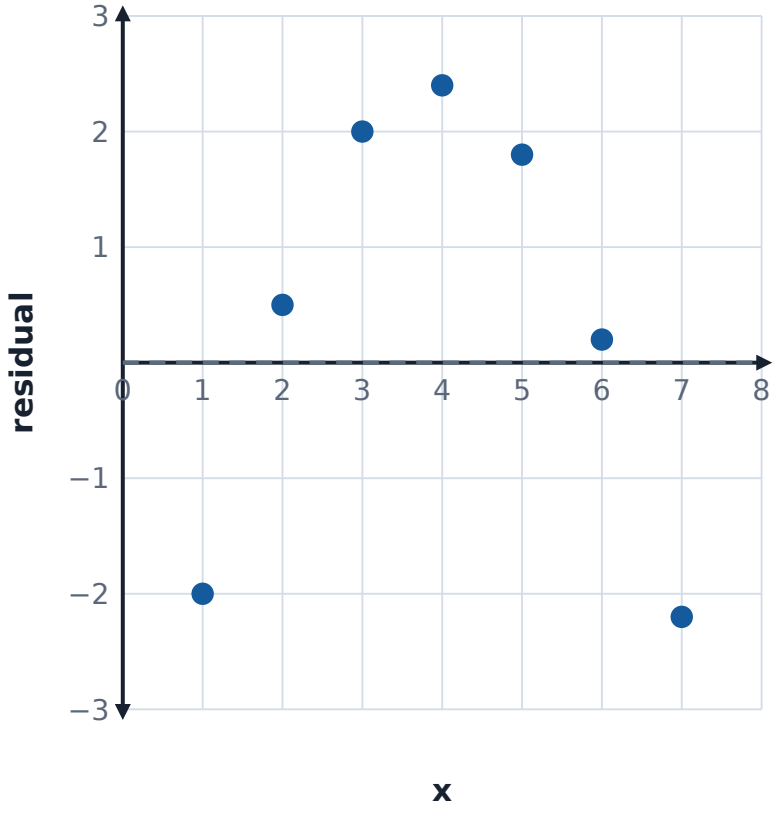
A line was fitted to a data set. The residual plot is shown. Which conclusion is correct?

**A.** The linear model is appropriate, because some residuals are near 0.

**B.** The linear model is appropriate, because residuals are both signs.

**C.** The data must contain an error, because residuals are not all 0.

**D.** A linear model is not appropriate; the curve suggests a curved model.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • MAIN • Q4/5

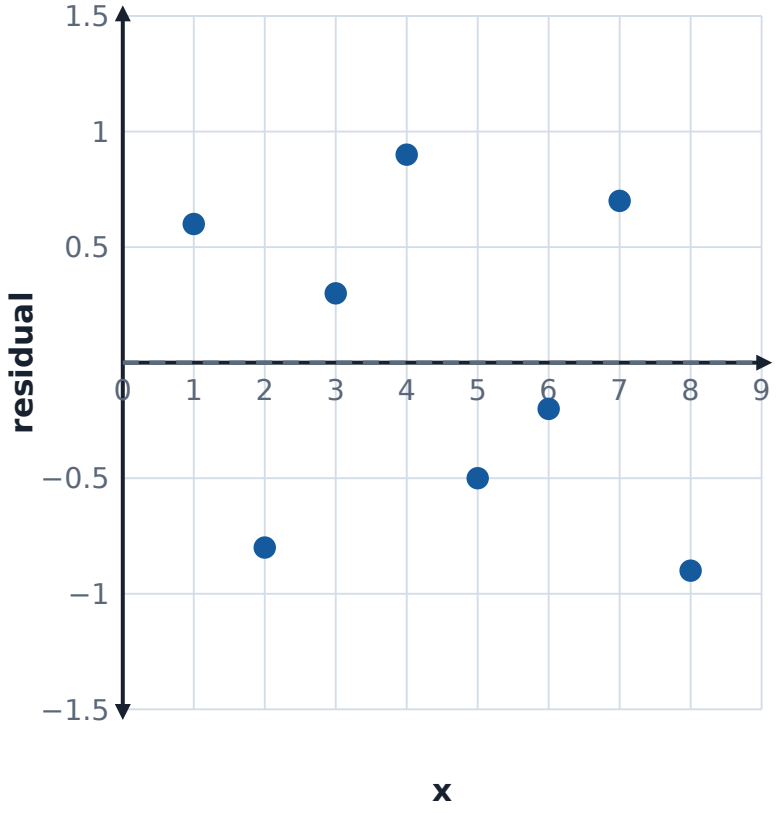
A line was fitted to a different data set, and its residual plot is shown. What does the plot show about the linear model?

**A.** The linear model is not appropriate, because residuals are not 0.

**B.** The linear model is appropriate: the residuals show no pattern.

**C.** A quadratic model is needed, because residuals change sign.

**D.** The model is perfect, because the residuals add to about 0.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • MAIN • Q5/5

SET 16 • HSS-ID.B.6.b

# BACKWARD 1

Subtract with signed decimals • Grade 7 • The Number System • 7.NS.A.1.c

Compute $3.5 - 3$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q1/5**

Compute $4.6 - 5$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q2/5**

**True or false?**

$20 - 22 = -2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q3/5**

Compute $7.2 - 7.5$.

**A.** $-0.3$

**B.** $-14.7$

**C.** $14.7$

**D.** $0.3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q4/5**

## True or false?

$11 - 11.4 = 0.4$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

**Grade 7 • The Number System • 7.NS.A.1.c • Set 16 • BACKWARD 1 • Q5/5**

SET 16 • HSS-ID.B.6.b

# BACKWARD 2

Find a predicted value from a model • High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2

For y = 2x + 1, find the predicted y at x = 3.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q1/5**

For y = 3x + 10, find the predicted y at x = 4.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q2/5**

## True or false?

For $y = -1.5x + 20$, the predicted y at $x = 4$ is 14.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q3/5

For y = 0.5x + 3, what is the predicted y at x = 8?

**A.** 8.5

**B.** 4

**C.** 7

**D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

**High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q4/5**

## True or false?

For $y = 4x - 2$, the predicted $y$ at $x = 0$ is 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

High School — Algebra I • Functions: Interpreting Functions • HSF-IF.A.2 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSS-ID.B.6.b

# BACKWARD 3

Compare an observed value with a prediction • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a

A model predicts y = 2.5x + 12 free throws for x hours of practice. A student practiced 8 hours and made 35. What is the predicted number, and how many more did the student make than predicted?

Predicted: _____

More by: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q1/5

## True or false?

If an observed value is less than the predicted value, the point lies below the fitted line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q2/5

The observed value is 50 and the predicted value is 46. What is observed minus predicted?

**A.** 4

**B.** 96

**C.** −4

**D.** 46

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-B3-Q3**

**High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q3/5**

## True or false?

In this collection, a residual is defined as predicted minus observed.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q4/5

The observed value is 18 and the predicted value is 20.5. What is observed minus predicted?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.a • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSS-ID.B.6.b

# FORWARD 1

Compute least-squares residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

The least-squares line for (1, 2), (2, 5), (3, 4), (4, 7) is $y = 1.4x + 1$. Find the residuals (observed minus predicted) at $x = 1$ and $x = 2$.

At $x = 1$: _____

At $x = 2$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q1/5

For $y = 1.4x + 1$ and the points (1, 2), (2, 5), (3, 4), (4, 7), what is the sum of all four residuals?

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q2/5

## True or false?

For a least-squares line, the residuals always add to 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q3/5

For $y = 1.4x + 1$ and the points (1, 2), (2, 5), (3, 4), (4, 7), which points have the largest residuals in absolute value?

**A.** (1, 2) only

**B.** (1, 2) and (4, 7)

**C.** (4, 7) only

**D.** (2, 5) and (3, 4)

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q4/5**

## True or false?

A residual of $-1.2$ means the point lies 1.2 units above the line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSS-ID.B.6.b

# FORWARD 2

Compare two models with residuals • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.b

Two models were fitted to the same data. Linear residuals: 3, −1, −3, −2, 1, 4. Exponential residuals: 0.2, −0.3, 0.1, 0.3, −0.2, −0.1. Which model fits better?

**A.** the exponential model

**B.** the linear model

**C.** neither model can be judged

**D.** they fit equally

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S16-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q1/5**

Find the sum of squared residuals for each model: linear 3, −1, −3, −2, 1, 4 and exponential 0.2, −0.3, 0.1, 0.3, −0.2, −0.1.

Linear: _____

Exponential: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q2/5

## True or false?

Of two models fitted to the same data, the one with the smaller sum of squared residuals fits those data more closely.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q3/5

A residual plot fans out: the residuals get larger as x increases. What does this show?

**A.** the slope is negative

**B.** prediction errors grow as x increases

**C.** the model is exact

**D.** the data have no outliers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q4/5

## True or false?

A residual plot with a clear U shape shows that the model is a good fit.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.b • Set 16 • FORWARD 2 • Q5/5

# Fit a linear function to a linear scatter plot

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.B.6.c** |
| **BACKWARD 1** | Fit a line by eye | **8.SP.A.2** |
| **BACKWARD 2** | Find slope and intercept from points | **8.F.B.4** |
| **FORWARD 1** | Use the properties of the least-squares line | **HSS-ID.B.6.c** |
| **FORWARD 2** | Interpret the fitted line's slope and intercept | **HSS-ID.C.7** |

SET 17 • HSS-ID.B.6.c

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Find the least-squares line $y = mx + b$ for hours studied ($x$) and score ($y$). Round $m$ and $b$ to two decimal places. A calculator or spreadsheet may be used.

| Hours | 1 | 2 | 2 | 3 | 4 | 5 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| Score | 52 | 58 | 61 | 63 | 70 | 72 | 76 | 80 | 83 | 90 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.c • Set 17 • MAIN • Q1/5

Find the least-squares line y = mx + b for car age (x) and value in $1000s (y). Round m and b to two decimal places. A calculator or spreadsheet may be used.

| Age (years) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Value ($1000s) | 24 | 21 | 22 | 17 | 18 | 13 | 14 | 9 | 11 | 6 |

m: _____

b: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

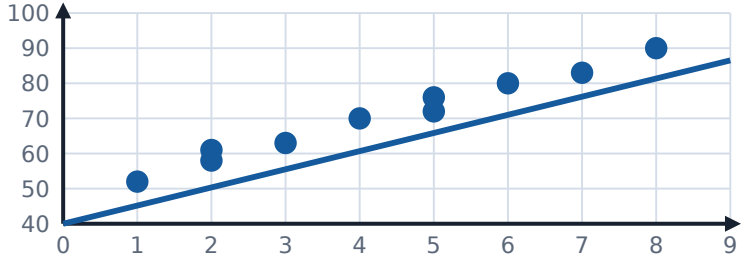
ID S17-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

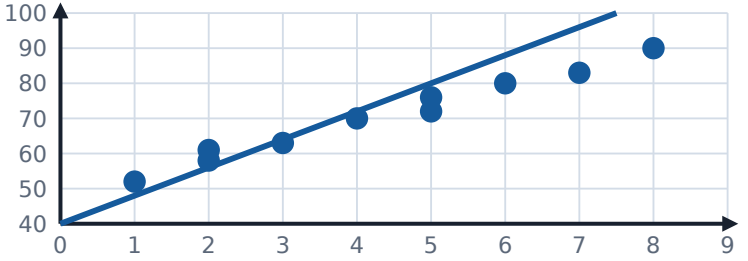
HSS-ID.B.6.c • Set 17 • MAIN • Q2/5

Which graph shows the line that best fits the study data (hours, score)?

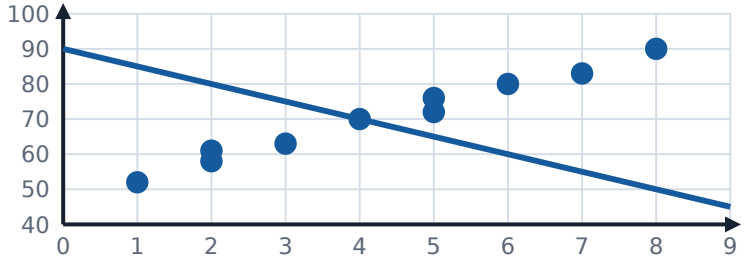
**A**



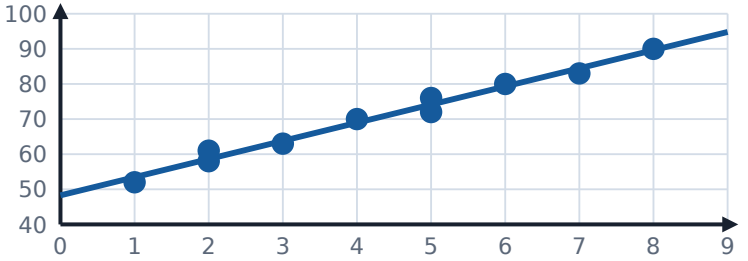

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.c • Set 17 • MAIN • Q3/5**

Using the least-squares line y = 5.17x + 48.28 for hours studied and score, predict the score for 3 hours.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • MAIN • Q4/5

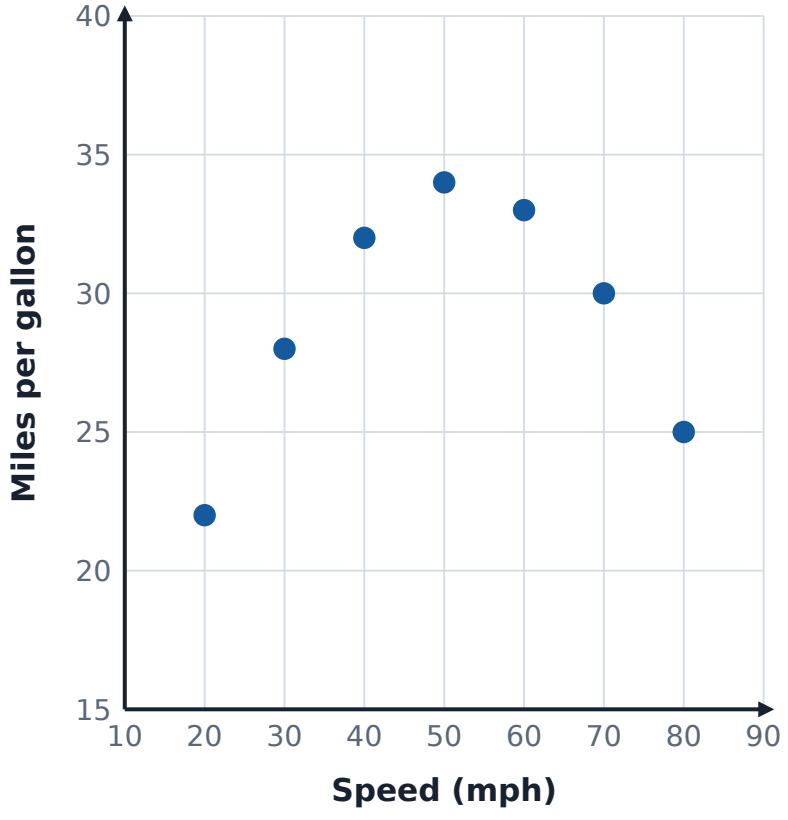
A student wants to fit a least-squares line to the data in this scatter plot. Which statement is correct?

**A.** A line is appropriate, because r can always be computed.

**B.** A line is appropriate, because the data have no outliers.

**C.** A line is not appropriate: the pattern is curved.

**D.** A line fits well, because the points rise at first.




**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.c • Set 17 • MAIN • Q5/5**

SET 17 • HSS-ID.B.6.c

# BACKWARD 1

Fit a line by eye • Grade 8 • Statistics & Probability • 8.SP.A.2

## True or false?

A good fitted line has about as many points above it as below it.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q1/5**

Which line passes through (0, 2), (2, 6) and (4, 10)?

**A.** $y = x + 2$

**B.** $y = 2x$

**C.** $y = 2x + 2$

**D.** $y = 4x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q2/5**

A line drawn by eye passes through (0, 50) and (5, 75). Find its slope and y-intercept.

Slope: _____

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q3/5**

## True or false?

A fitted line must pass through every data point.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q4/5**

Use the line $y = 5x + 50$ to predict y at $x = 4$.

y: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 17 • BACKWARD 1 • Q5/5**

SET 17 • HSS-ID.B.6.c

# BACKWARD 2

Find slope and intercept from points • Grade 8 • Functions • 8.F.B.4

Find the slope of the line through (2, 58) and (6, 80).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q1/5**

A line has slope 5 and passes through (2, 60). Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q2/5**

**True or false?**

The line through (0, 26) with slope −2 has y = 16 at x = 5.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q3/5**

Which equation is the line through (0, 10) and (4, 2)?

**A.** $y = 2x + 10$

**B.** $y = -2x + 2$

**C.** $y = -0.5x + 10$

**D.** $y = -2x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q4/5**

**True or false?**

The slope of the line through (1, 24) and (10, 6) is 2.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 17 • BACKWARD 2 • Q5/5**

SET 17 • HSS-ID.B.6.c

# FORWARD 1

Use the properties of the least-squares line • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6.c

Find the mean hours $\bar{x}$ and the mean score $\bar{y}$ for the study data.

| Hours | 1 | 2 | 2 | 3 | 4 | 5 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| Score | 52 | 58 | 61 | 63 | 70 | 72 | 76 | 80 | 83 | 90 |

$\bar{x}$: _____

$\bar{y}$: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q1/5

## True or false?

The least-squares line passes through the point $(\bar{x}, \bar{y})$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q2/5

The least-squares line is the line that makes which quantity as small as possible?

**A.** the number of points off the line

**B.** the largest single residual

**C.** the sum of the residuals

**D.** the sum of the squared residuals

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-F1-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q3/5**

A least-squares line has slope 2 and passes through $(\bar{x}, \bar{y}) = (5, 20)$. Find its y-intercept.

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q4/5

## True or false?

Adding one point far from the others can never change the least-squares line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.B.6.c • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSS-ID.B.6.c

# FORWARD 2

Interpret the fitted line's slope and intercept • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

For hours studied (x) and score (y), the least-squares line is $y = 5.17x + 48.28$. What does 5.17 mean?

**A.** A student who studied 0 hours scored exactly 5.17.

**B.** Each extra hour goes with about 5.17 more points, on average.

**C.** The score rises 48.28 points for each hour.

**D.** Each student scored 5.17 points per test.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 17 • FORWARD 2 • Q1/5**

Using $y = 5.17x + 48.28$, how much higher is the predicted score for a student who studies 2 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 17 • FORWARD 2 • Q2/5

For car age (x, years) and value (y, $1000s), $y = -1.88x + 25.87$. What does 25.87 mean?

**A.** The oldest car is worth about $25,870.

**B.** A car is predicted to be worth $0 after 25.87 years.

**C.** Cars lose about $25,870 each year.

**D.** A new car (age 0) has a predicted value of about $25,870.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S17-F2-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 17 • FORWARD 2 • Q3/5**

## True or false?

In $y = -1.88x + 25.87$, the slope means every car loses exactly $1,880 each year.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q4/5

Using $y = -1.88x + 25.87$, how much greater (in \$1000s) is the predicted value of a 3-year-old car than a 7-year-old car?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.7 • Set 17 • FORWARD 2 • Q5/5

# Interpret the slope and intercept of a linear model

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.7** |
| **BACKWARD 1** | Interpret the rate and the initial value of a linear function | **8.F.B.4** |
| **BACKWARD 2** | Use a linear model's equation to answer questions | **8.SP.A.3** |
| **FORWARD 1** | Interpret parameters with units and judge meaning | **HSS-ID.C.7** |
| **FORWARD 2** | Relate the slope to the correlation coefficient | **HSS-ID.C.8** |

SET 18 • HSS-ID.C.7

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A model for a family's savings is $y = 85x + 1200$, where $x$ is months and $y$ is dollars. What does 85 mean?

**A.** The family started with $85 in savings.

**B.** The family saves $1200 each month.

**C.** The savings will reach $85 after 1200 months.

**D.** The savings are predicted to grow by $85 each month.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • MAIN • Q1/5**

A model for weekly hours of exercise is $y = 6 - 0.2x$, where $x$ is weekly hours of TV. What does 6 mean?

**A.** A person who watches 0 hours of TV is predicted to exercise 6 hours.

**B.** Each hour of TV is predicted to add 6 hours of exercise.

**C.** Each hour of TV is predicted to cut exercise by 6 hours.

**D.** A person exercises at most 0.2 hours per week.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • MAIN • Q2/5**

A tree's height is modeled by $y = 0.4x + 1.5$, with $y$ in meters and $x$ in years since planting. What is the predicted growth per year, and what is the predicted height at planting?

Growth per year (m): _____

Height at planting (m): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.7 • Set 18 • MAIN • Q3/5

For adults with heights from 150 to 190 cm, arm span is modeled by $y = 1.1x - 15$ (both in cm). Which statement about the intercept $-15$ is correct?

**A.** It means a person 0 cm tall has an arm span of $-15$ cm.

**B.** It means arm spans are 15 cm less than heights.

**C.** It is not meaningful: a height of 0 cm is far outside the data.

**D.** It means the slope of the model is negative.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • MAIN • Q4/5**

Weekly gasoline use is modeled by $y = 0.35x + 4.2$ gallons, where $x$ is miles driven per day. By how many gallons do the predictions differ for two drivers whose daily miles differ by 20?

Gallons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • MAIN • Q5/5

SET 18 • HSS-ID.C.7

# BACKWARD 1

Interpret the rate and the initial value of a linear function • Grade 8 • Functions • 8.F.B.4

A plumber charges $y = 50x + 80$ dollars for $x$ hours. What does 80 mean?

**A.** the charge per hour

**B.** a fixed fee charged before any hours

**C.** the number of hours

**D.** the total for 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q1/5**

A phone plan costs y = 0.1x + 25 dollars for x minutes. What is the cost per minute, and what is the fixed cost?

Per minute ($): _____

Fixed ($): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q2/5**

**True or false?**

In $y = 3x + 12$, when $x$ increases by 1, $y$ increases by 12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q3/5**

A candle's height is $h = 20 - 1.5t$ centimeters after $t$ hours. What does 1.5 mean?

**A.** the candle burns 1.5 cm per hour

**B.** the candle starts 1.5 cm tall

**C.** the candle burns 20 cm per hour

**D.** the candle lasts 1.5 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q4/5**

**True or false?**

In y = 3x + 12, the value of y when x = 0 is 12.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

**Grade 8 • Functions • 8.F.B.4 • Set 18 • BACKWARD 1 • Q5/5**

SET 18 • HSS-ID.C.7

# BACKWARD 2

Use a linear model's equation to answer questions • Grade 8 • Statistics & Probability • 8.SP.A.3

Plant height is modeled by y = 1.5x + 5 (cm after x weeks). Predict the height at 6 weeks.

Height (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q1/5**

In the plant model $y = 1.5x + 5$, one more week of growth goes with

**A.** 1.5 more weeks

**B.** 5 cm more height

**C.** 1.5 cm more height

**D.** 6.5 cm more height

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q2/5**

## True or false?

In the plant model $y = 1.5x + 5$, the slope means the plants were 1.5 cm tall at the start.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q3/5**

In the plant model y = 1.5x + 5, how much greater is the predicted height at week 10 than at week 2?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q4/5**

## True or false?

A negative slope means the predicted y decreases as x increases.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 18 • BACKWARD 2 • Q5/5**

SET 18 • HSS-ID.C.7

# FORWARD 1

Interpret parameters with units and judge meaning • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

Using y = 5.17x + 48.28 (score versus hours studied), what is the predicted score increase for 3 more hours?

Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • FORWARD 1 • Q1/5

A shipping cost is modeled by $y = 0.02x + 3.5$ dollars for $x$ grams. What does 0.02 mean?

**A.** Each package costs $0.02 no matter what it weighs.

**B.** Each additional gram adds about $0.02 to the predicted cost.

**C.** The predicted cost rises by $3.50 for each gram.

**D.** A package weighing 0 grams is predicted to cost $0.02.

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.7 • Set 18 • FORWARD 1 • Q2/5

## True or false?

The intercept of a linear model always has a meaningful interpretation in context.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • FORWARD 1 • Q3/5

For ages 20 to 60, maximum heart rate is modeled by $y = 208 - 0.7x$ beats per minute, $x$ in years. Which statement about 208 is correct?

**A.** It is the predicted heart rate for a 20-year-old person.

**B.** It is the prediction at age 0, outside the data: not meaningful.

**C.** It is the decrease in maximum heart rate for each year of age.

**D.** It is the largest heart rate that any person can have.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-F1-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.7 • Set 18 • FORWARD 1 • Q4/5**

Using $y = 208 - 0.7x$, predict the maximum heart rate at age 40.

Beats per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.7 • Set 18 • FORWARD 1 • Q5/5

SET 18 • HSS-ID.C.7

# FORWARD 2

Relate the slope to the correlation coefficient • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

## True or false?

The sign of the correlation coefficient matches the sign of the slope of the least-squares line.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q1/5

The least-squares line is $y = -1.88x + 25.87$. Which could be the correlation coefficient?

**A.** 0.97

**B.** 1.88

**C.** $-1.88$

**D.** $-0.97$

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-F2-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 18 • FORWARD 2 • Q2/5**

## True or false?

A larger slope always means a stronger correlation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q3/5

The least-squares slope equals r·(s_y/s_x). If r = 0.8, s_y = 10 and s_x = 4, what is the slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 18 • FORWARD 2 • Q4/5

If the least-squares line has slope 0, what is r?

**A.** 1

**B.** $-1$

**C.** it cannot be found

**D.** 0

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S18-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.8 • Set 18 • FORWARD 2 • Q5/5**

# Compute the correlation coefficient with technology

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.8** |
| **BACKWARD 1** | Find means and deviations | **6.SP.B.5.c** |
| **BACKWARD 2** | Read the direction of an association | **HSS-ID.B.6** |
| **FORWARD 1** | Compute r from deviations | **HSS-ID.C.8** |
| **FORWARD 2** | See how changes to the data affect r | **HSS-ID.C.8** |

SET 19 • HSS-ID.C.8

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Find the correlation coefficient r for hours studied and score. Round to two decimal places. A calculator or spreadsheet may be used.

| Hours | 1 | 2 | 2 | 3 | 4 | 5 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|---|
| Score | 52 | 58 | 61 | 63 | 70 | 72 | 76 | 80 | 83 | 90 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q1/5

Find the correlation coefficient r for car age and value. Round to two decimal places. A calculator or spreadsheet may be used.

| Age (years) | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Value ($1000s) | 24 | 21 | 22 | 17 | 18 | 13 | 14 | 9 | 11 | 6 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q2/5

Find the correlation coefficient r for hours after sunrise (x) and ice thickness in cm (y). Round to two decimal places. A calculator or spreadsheet may be used.

| Hours | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Thickness (cm) | 48 | 44 | 41 | 35 | 33 | 28 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q3/5

Find r for the temperature and cones-sold data for all 7 days, and again without the 90-degree day. Round each to three decimal places. A calculator or spreadsheet may be used.

| Temperature (°F) | 60 | 65 | 70 | 75 | 80 | 85 | 90 |
|---|---|---|---|---|---|---|---|
| Cones sold | 120 | 140 | 155 | 175 | 190 | 210 | 60 |

All 7 days: _____

Without the 90-degree day: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.8 • Set 19 • MAIN • Q4/5

Find the correlation coefficient r for the spring data (mass and length). Round to two decimal places. A calculator or spreadsheet may be used.

| Mass (g) | 2 | 4 | 5 | 7 | 8 | 10 |
|---|---|---|---|---|---|---|
| Length (cm) | 15 | 19 | 24 | 26 | 31 | 35 |

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • MAIN • Q5/5

SET 19 • HSS-ID.C.8

# BACKWARD 1

Find means and deviations • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean of 1, 2, 3, 4, 5.

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q1/5**

For the values 2, 5, 4, 7 (mean 4.5), find the deviations of 2 and 7 from the mean.

2: _____

7: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q2/5**

## True or false?

The deviations of a data set from its mean always add to 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q3/5**

What is the mean of 48, 44, 41, 35, 33, 28, rounded to two decimal places?

**A.** 35

**B.** 38.17

**C.** 41

**D.** 38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q4/5**

## True or false?

A value below the mean has a positive deviation.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

**Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 19 • BACKWARD 1 • Q5/5**

SET 19 • HSS-ID.C.8

# BACKWARD 2

Read the direction of an association • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

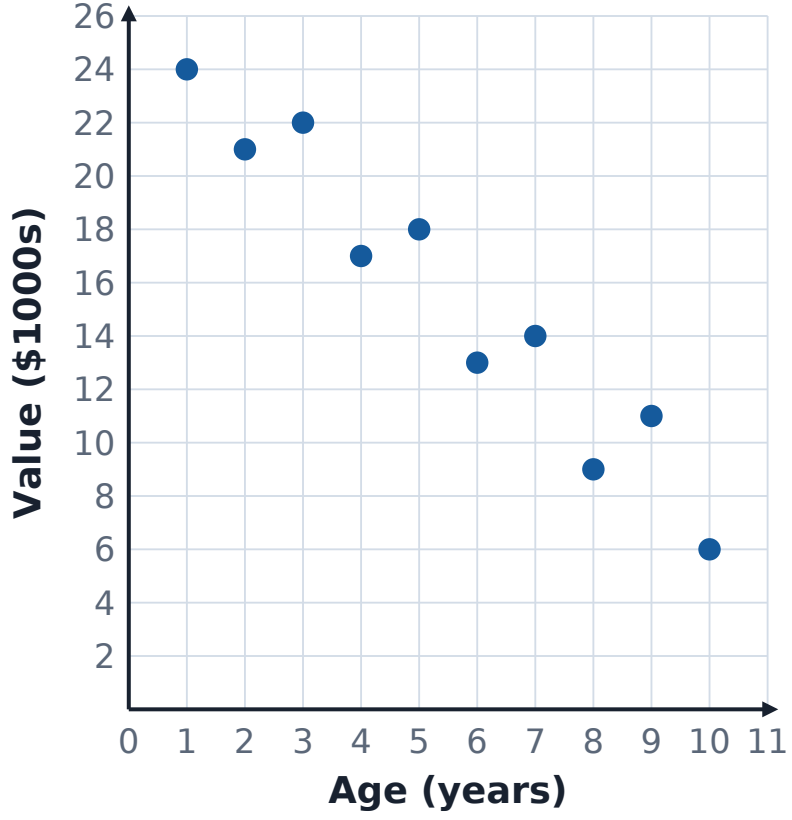
The scatter plot shows car age and value. What is the sign of r?

**A.** positive

**B.** cannot be found

**C.** zero

**D.** negative




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

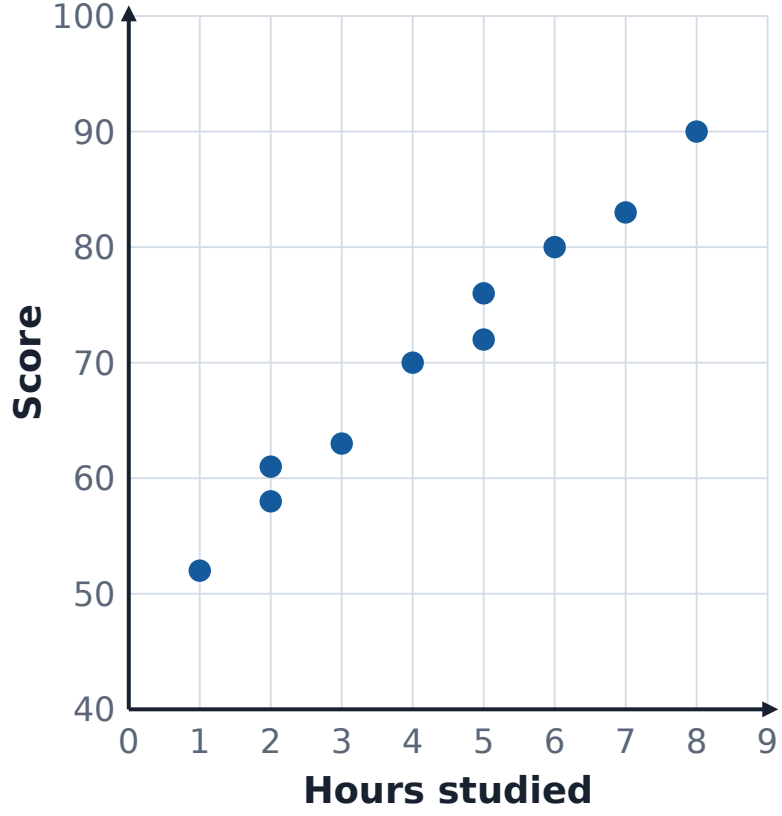
High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q1/5

## True or false?

For the hours-studied scatter plot, r is positive.

**A.** True

**B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q2/5

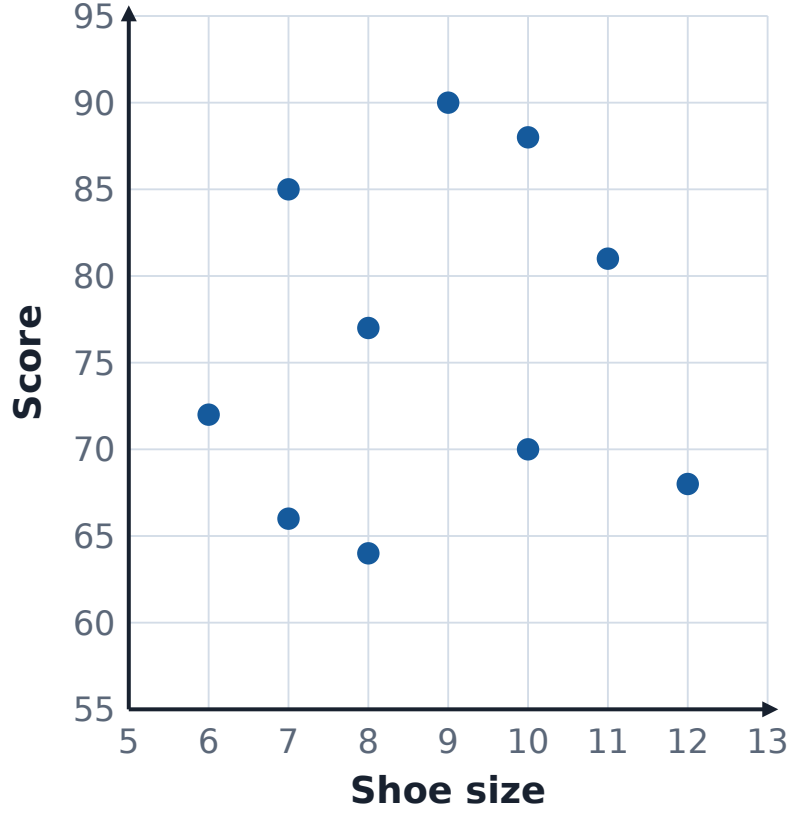
For the shoe-size scatter plot, $r$ is closest to

**A.** 1

**B.** 0

**C.** $-1$

**D.** 0.9




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q3/5

## True or false?

For a cloud of points with no pattern, r is close to 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q4/5

Which scatter plot would have r close to −1?

**A.** points along a U-shaped curve

**B.** a shapeless cloud

**C.** points falling closely along a line

**D.** points rising closely along a line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSS-ID.C.8

# FORWARD 1

Compute r from deviations • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

For (1, 2), (2, 5), (3, 4), (4, 7): $\Sigma(x - \bar{x})(y - \bar{y}) = 7$, $\Sigma(x - \bar{x})^2 = 5$ and $\Sigma(y - \bar{y})^2 = 13$. Compute $r = 7/\sqrt{(5 \cdot 13)}$, rounded to two decimal places.

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 1 • Q1/5

## True or false?

If all points lie exactly on a line with positive slope, r = 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q2/5

For the points (1, 2), (2, 4), (3, 6), what is r?

r: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 1 • Q3/5

A calculator reports r = 1.4. What should you conclude?

**A.** The association is very strong and positive.

**B.** The data have one outlier.

**C.** A mistake was made: r is always between −1 and 1.

**D.** The slope of the line is 1.4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 1 • Q4/5

## True or false?

For the points (1, 6), (2, 4), (3, 2), r = 1.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSS-ID.C.8

# FORWARD 2

See how changes to the data affect r • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

## True or false?

Changing x from hours to minutes does not change r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q1/5

## True or false?

Swapping the roles of x and y changes the value of r.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data HSS-ID.C.8 • Set 19 • FORWARD 2 • Q2/5

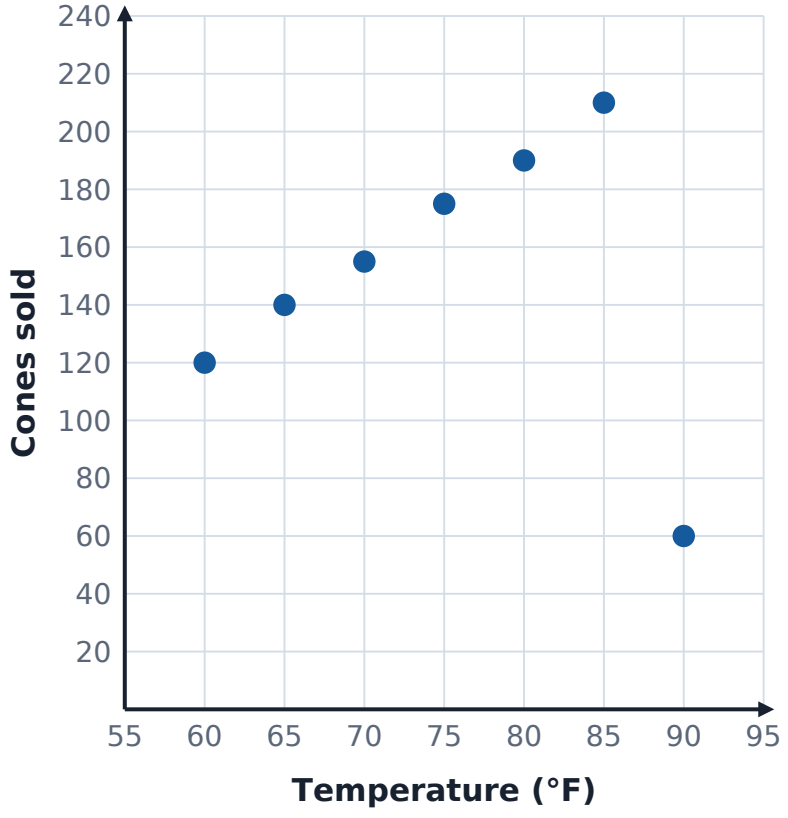
For the cones data, r = 0.999 for six days and −0.008 with the 90-degree day included. What does this show?

**A.** Temperature causes cone sales to fall.

**B.** r is not affected by outliers.

**C.** The six days have no association.

**D.** One outlier can change r dramatically.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.8 • Set 19 • FORWARD 2 • Q3/5

Find r for (1, 3), (2, 5), (3, 7), and for the same points with every y increased by 10. Give both.

Original: _____

After adding 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q4/5

Which change to a data set could change r?

**A.** adding 5 to every x-value

**B.** converting y from meters to centimeters

**C.** adding a new point far from the others

**D.** doubling every y-value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 19 • FORWARD 2 • Q5/5

# Interpret the correlation coefficient

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.8** |
| **BACKWARD 1** | Describe strength and direction from a plot | **HSS-ID.B.6** |
| **BACKWARD 2** | Compare numbers on a scale from -1 to 1 | **6.NS.C.7.c** |
| **FORWARD 1** | Match correlation coefficients to scatter plots | **HSS-ID.C.8** |
| **FORWARD 2** | Correlation is not causation | **HSS-ID.C.9** |

SET 20 • HSS-ID.C.8

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

For car age and value, $r = -0.97$. Which interpretation is correct?

**A.** weak negative linear association

**B.** strong positive linear association

**C.** strong negative linear association

**D.** no association

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q1/5

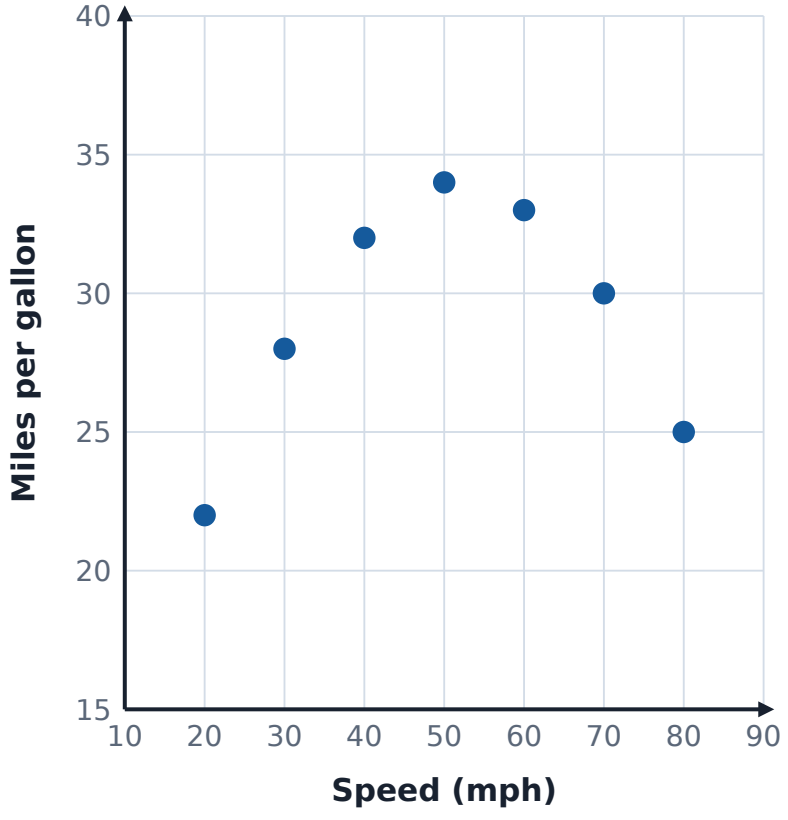
For speed and fuel efficiency, r = 0.25 but the scatter plot shows a clear curve. Which statement is correct?

**A.** r must have been computed incorrectly.

**B.** There is a weak linear relationship and nothing more.

**C.** r measures only linear association, so it misses the curve.

**D.** There is no relationship between the variables.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q2/5

Which correlation coefficient shows the strongest linear association?

**A.** $-0.91$

**B.** $0.85$

**C.** $-0.40$

**D.** $0.10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q3/5

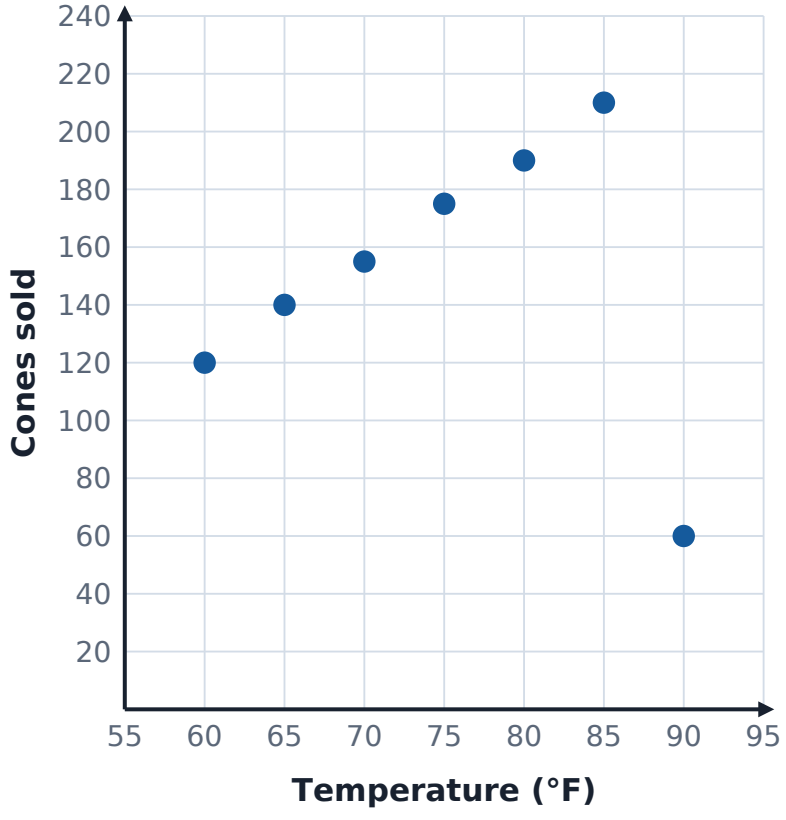
For the cones data, r = −0.008 for all 7 days but 0.999 without the 90-degree day. Which interpretation is best?

**A.** The six typical days show a strong positive linear association.

**B.** The 90-degree day is the most typical day.

**C.** Temperature and cone sales have no relationship on any day.

**D.** Hotter days have fewer cones sold.




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data

HSS-ID.C.8 • Set 20 • MAIN • Q4/5

Which statement about r is correct?

**A.** r changes when x is converted from hours to minutes.

**B.** r is negative whenever the data contain an outlier.

**C.** r can be as large as 5 for very strong data.

**D.** r has no units and stays the same if the units change.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • MAIN • Q5/5

SET 20 • HSS-ID.C.8

# BACKWARD 1

Describe strength and direction from a plot • High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6

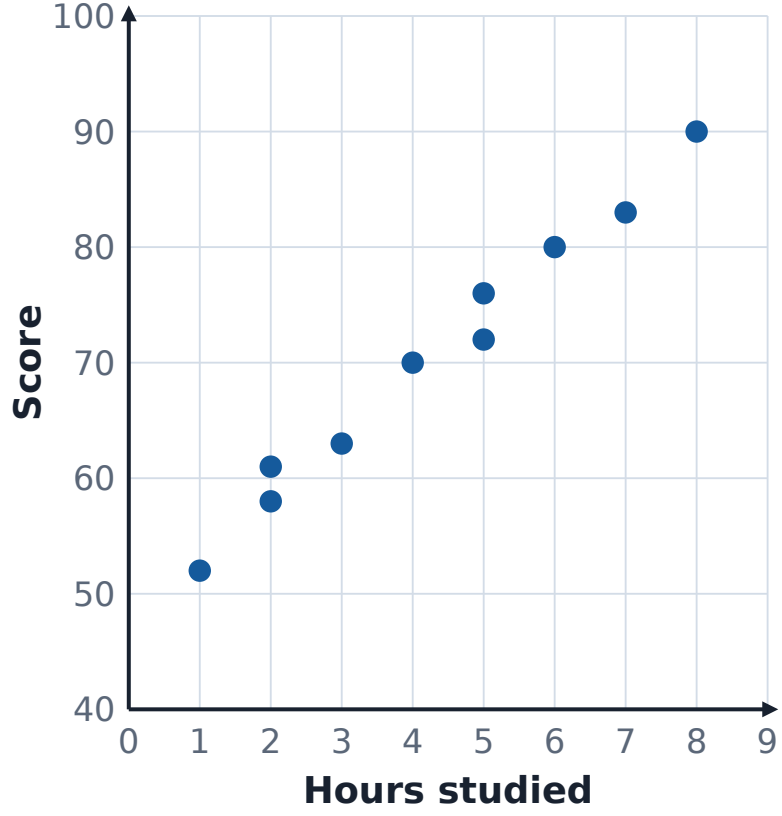
The scatter plot shows hours studied and score. How strong is the linear association?

**A.** strong

**B.** weak

**C.** cannot be judged

**D.** none




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q1/5

## True or false?

In a strong association, the points lie close to a line or curve.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q2/5

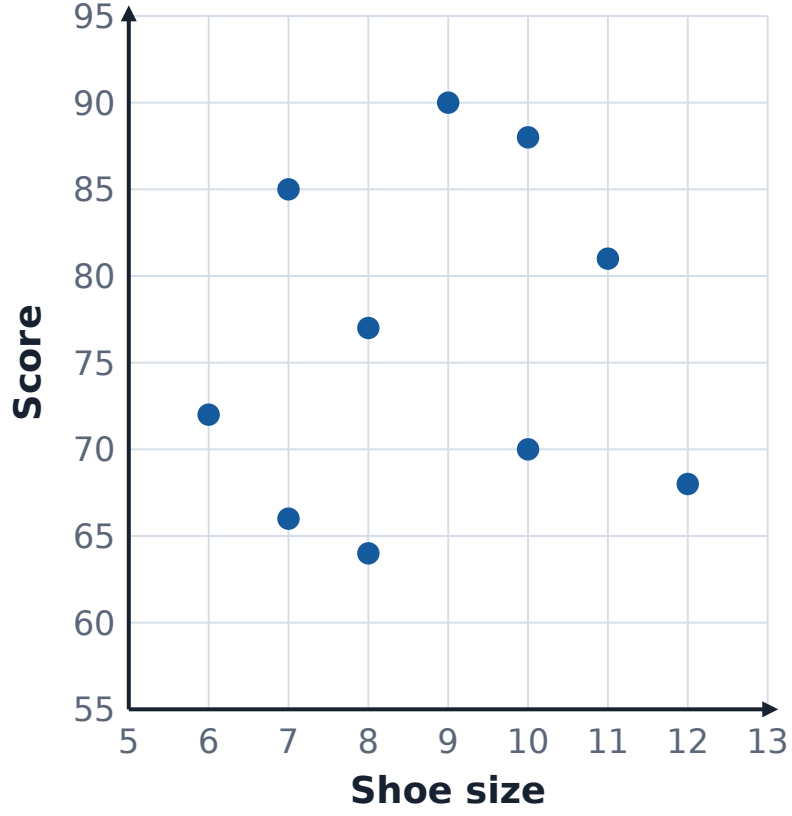
The scatter plot shows shoe size and quiz score. How strong is the association?

**A.** very weak or none

**B.** strong negative

**C.** strong positive

**D.** perfect




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q3/5

## True or false?

A scatter plot that falls from left to right shows a positive association.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q4/5

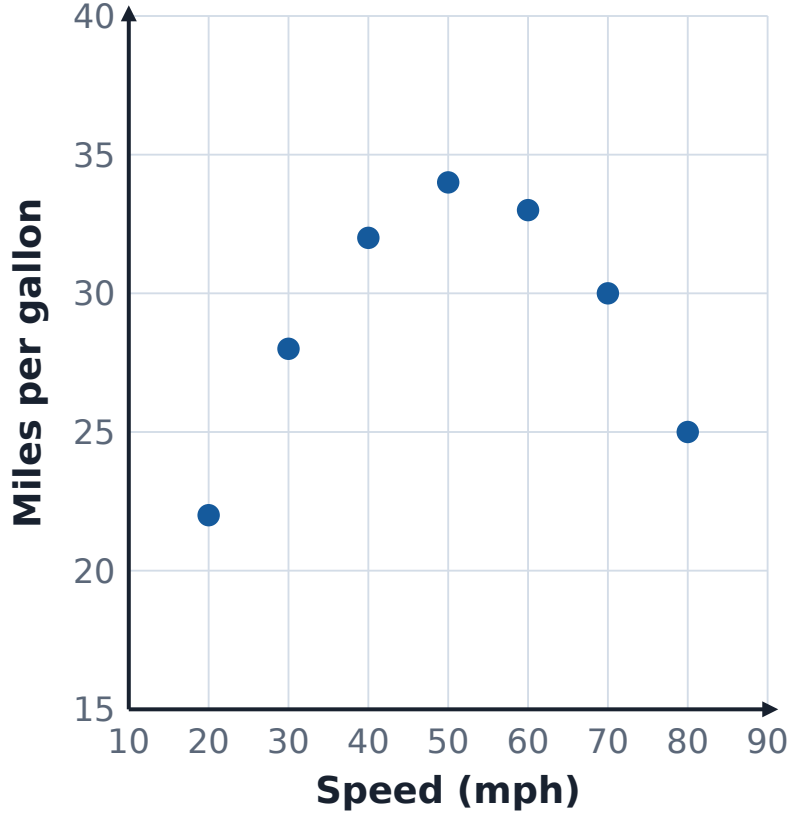
The scatter plot of speed and fuel efficiency has what form?

**A.** exponential growth

**B.** linear

**C.** curved

**D.** no form




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

High School — Probability & Statistics • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.B.6 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

# BACKWARD 2

Compare numbers on a scale from −1 to 1 • Grade 6 • The Number System • 6.NS.C.7.c

What is $|-0.91|$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q1/5**

## True or false?

$|-0.8| > |0.6|$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q2/5**

Which is closest to 0?

**A.** $-0.1$

**B.** $-0.6$

**C.** 0.5

**D.** 0.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q3/5**

**True or false?**

$-0.9$ is greater than $0.2$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q4/5**

Which has the greater absolute value, $-0.75$ or $0.7$? Enter the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

**Grade 6 • The Number System • 6.NS.C.7.c • Set 20 • BACKWARD 2 • Q5/5**

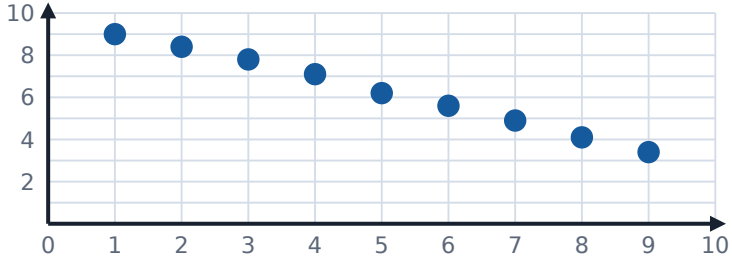
SET 20 • HSS-ID.C.8

# FORWARD 1

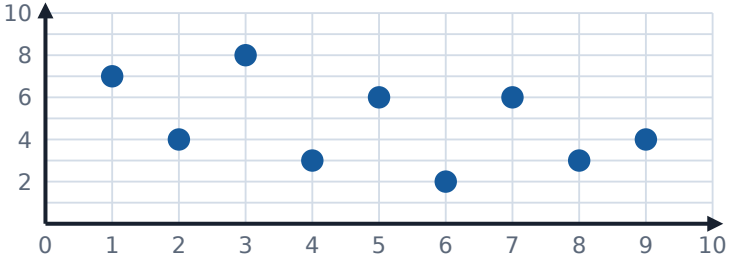
Match correlation coefficients to scatter plots • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

Which scatter plot has $r \approx -0.45$?

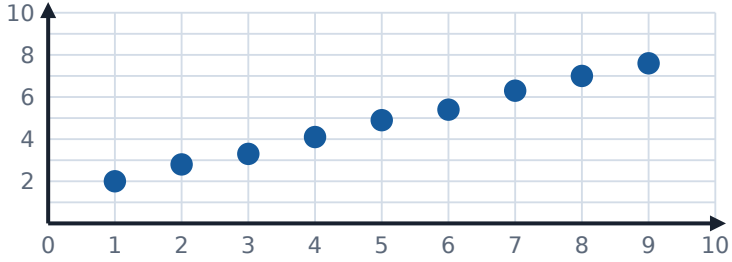
**A**



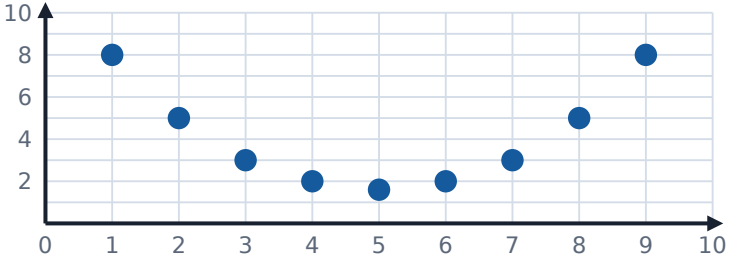

**B**




**C**




**D**




**A.** Graph A

**B.** Graph B

**C.** Graph C

**D.** Graph D

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F1-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.8 • Set 20 • FORWARD 1 • Q1/5**

A U-shaped scatter plot that is perfectly symmetric has r equal to

**A.** 0.5

**B.** 0

**C.** $-1$

**D.** 1

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F1-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.8 • Set 20 • FORWARD 1 • Q2/5**

## True or false?

$r = -0.45$ indicates a stronger linear association than $r = 0.30$.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q3/5

Which value of r goes with points falling steadily in a narrow band?

**A.** 0.99

**B.** −0.99

**C.** 0

**D.** −0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q4/5

## True or false?

Two data sets with the same r must have scatter plots of the same shape.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.8 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSS-ID.C.8

# FORWARD 2

Correlation is not causation • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Across towns, the number of libraries and the number of crimes have r = 0.9. Which is the best explanation?

**A.** Libraries cause crime.

**B.** r = 0.9 proves a cause in one direction.

**C.** Crime causes towns to build libraries.

**D.** Larger towns tend to have more of both.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F2-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 20 • FORWARD 2 • Q1/5**

## True or false?

A correlation of r = 0.95 shows that changes in x cause changes in y.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q2/5

Which kind of study can show that x causes y?

**A.** a scatter plot of past data

**B.** a study with r close to 1

**C.** a randomized experiment

**D.** a survey with many people

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q3/5

## True or false?

A lurking variable can produce a strong correlation between two variables that do not affect each other.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data
HSS-ID.C.9 • Set 20 • FORWARD 2 • Q4/5

Students who sleep more have higher grades ($r = 0.6$) in an observational study. Which conclusion is justified?

**A.** Higher grades cause these students to sleep more.

**B.** Sleep and grades are positively associated in this group.

**C.** Sleep and grades are unrelated for these students.

**D.** More sleep causes higher grades for these students.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S20-F2-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 20 • FORWARD 2 • Q5/5**

# Distinguish between correlation and causation

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-ID.C.9** |
| **BACKWARD 1** | Recognize a strong correlation | **HSS-ID.C.8** |
| **BACKWARD 2** | Describe the association in a scatter plot | **8.SP.A.1** |
| **FORWARD 1** | Tell experiments from observational studies | **HSS-IC.B.3** |
| **FORWARD 2** | Judge causal claims from experiment results | **HSS-IC.B.5** |

SET 21 • HSS-ID.C.9

# MAIN

Probability & Statistics main question • High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.9

Over several years, a city's sunscreen sales and its number of pool visits are strongly correlated. What best explains this?

**A.** Swimming causes people to buy sunscreen.

**B.** Sunny, hot weather raises both.

**C.** Sunscreen causes people to swim.

**D.** The data must be wrong.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q1**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q1/5**

Countries with more doctors per person have longer life spans. Which conclusion is justified?

**A.** Longer life spans cause countries to train more doctors.

**B.** Adding more doctors in a country will lengthen life spans.

**C.** Doctors per person and life span are positively associated.

**D.** Doctors per person and life span are unrelated.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q2**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q2/5**

Researchers want to know whether a new study app raises test scores. Which design can show cause?

**A.** Find the correlation of app use and scores.

**B.** Survey students who already chose to use the app.

**C.** Compare the scores of last year's users and nonusers.

**D.** Randomly assign students to use the app or not.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q3**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q3/5**

Children with larger shoe sizes have higher reading scores. What is the most likely lurking variable?

**A.** age

**B.** reading-book color

**C.** school lunch menu

**D.** shoe brand

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q4**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q4/5**

Which statement is correct?

**A.** Association implies causation when r is above 0.9.

**B.** If two variables are not correlated, neither can cause anything.

**C.** Causation implies association, but association does not imply causation.

**D.** Causation can only be shown with a scatter plot.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-M-Q5**

**High School — Probability & Statistics, revisits an Algebra I standard • Statistics & Probability: Interpreting Categorical & Quantitative Data**
**HSS-ID.C.9 • Set 21 • MAIN • Q5/5**

SET 21 • HSS-ID.C.9

# BACKWARD 1

Recognize a strong correlation • High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8

## True or false?

$r = -0.92$ shows a strong linear association.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q1/5

Which r shows the weakest linear association?

**A.** $-0.7$

**B.** $0.05$

**C.** $0.6$

**D.** $-0.95$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q2/5

## True or false?

A strong correlation means the two variables must be causally related.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q3/5

A data set has r = 0.88. Is the linear association positive or negative? Enter positive or negative.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q4/5

$r = 0.88$ for study time and grades in a survey. What does it describe?

**A.** a curved association

**B.** a weak association

**C.** a strong positive linear association

**D.** proof that study time causes grades

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Algebra I • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSS-ID.C.9

# BACKWARD 2

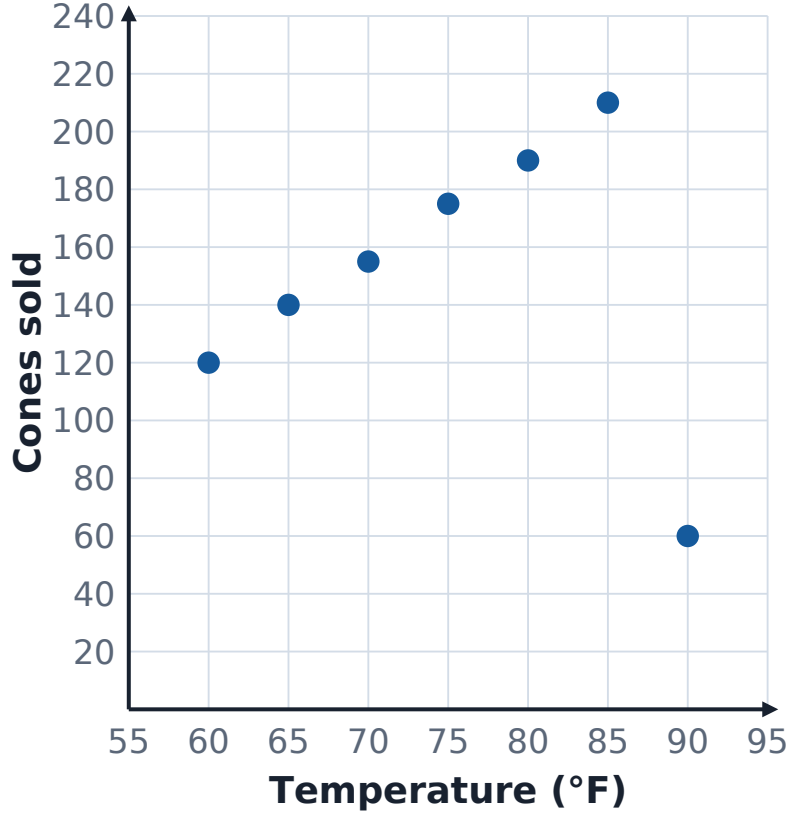
Describe the association in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

## True or false?

In the ice-cream scatter plot, the six typical days show a positive association.

**A.** True

**B.** False




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q1/5**

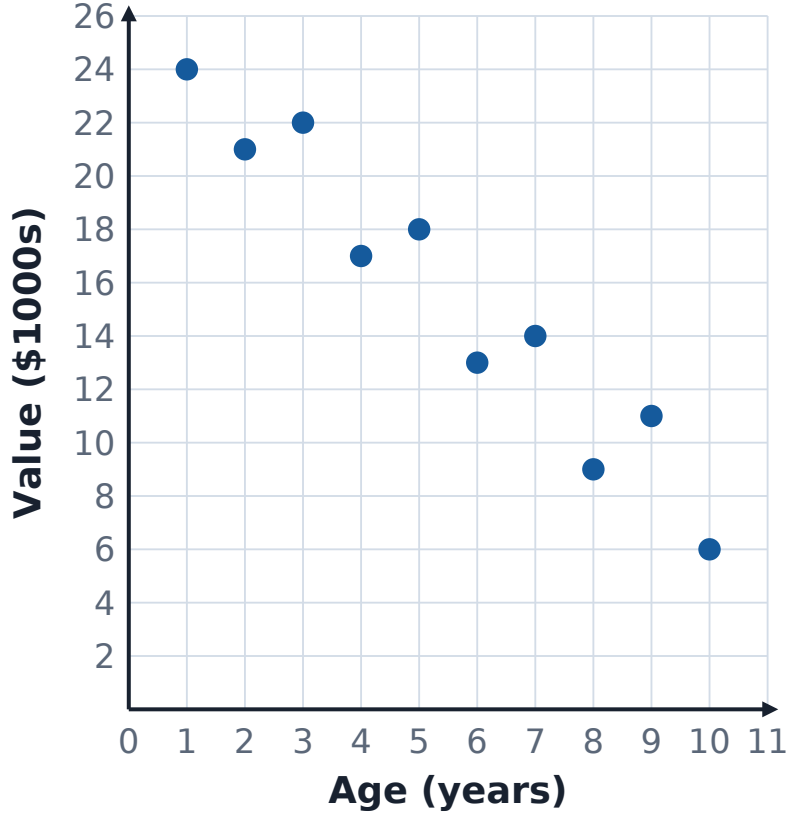
The scatter plot shows car age and value. Which describes it?

**A.** curved association

**B.** positive linear association

**C.** no association

**D.** negative linear association




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q2/5**

## True or false?

A scatter plot can show an association between two variables that do not cause each other.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q3/5**

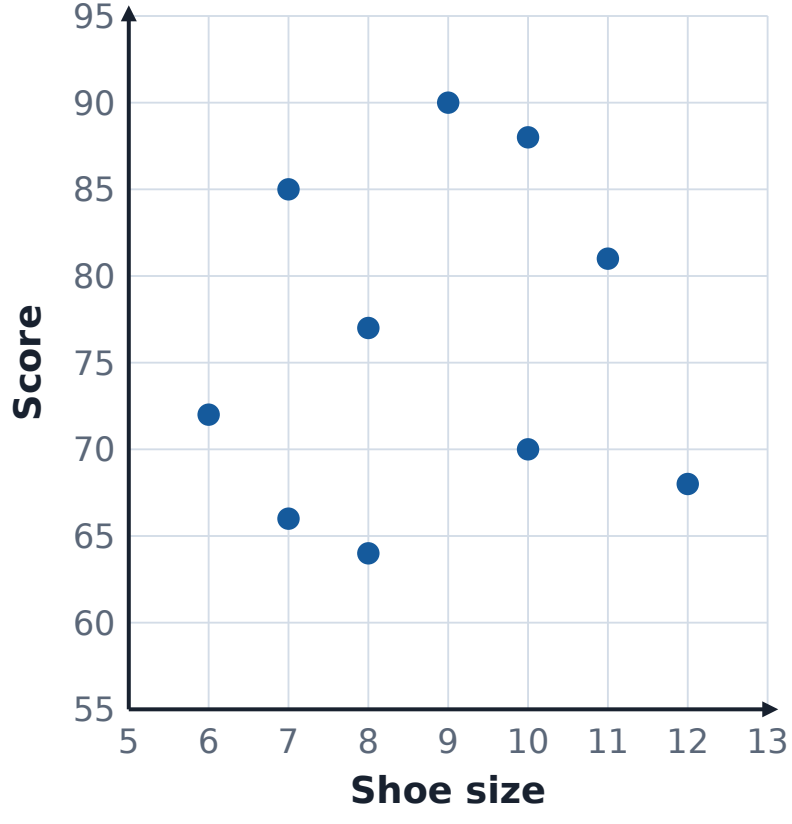
The scatter plot shows shoe size and quiz score of adults. Which describes it?

**A.** curved

**B.** no clear association

**C.** strong negative

**D.** strong positive




REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q4/5**

## True or false?

An association seen in a scatter plot shows which variable is the cause.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

**Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 21 • BACKWARD 2 • Q5/5**

SET 21 • HSS-ID.C.9

# FORWARD 1

Tell experiments from observational studies • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3

Researchers record the coffee intake and blood pressure of 500 adults without changing anything. What type of study is this?

**A.** randomized experiment

**B.** simulation

**C.** sample survey of opinions

**D.** observational study

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q1/5**

## True or false?

In a randomized experiment, the researcher assigns the treatments by chance.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q2/5

Which study is a randomized experiment?

**A.** Researchers compare towns that chose different school lunches.

**B.** Volunteers are randomly assigned to a new diet or the usual diet.

**C.** A survey asks people which diet they prefer.

**D.** Doctors record the diets their patients already follow.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F1-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q3/5**

## True or false?

An observational study can establish cause as well as a randomized experiment can.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q4/5

Why does random assignment help support causal conclusions?

**A.** It removes all variability from the measurements.

**B.** It guarantees that r will be close to 1.

**C.** It makes the sample larger than other designs do.

**D.** It tends to balance other variables between the groups.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F1-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.3 • Set 21 • FORWARD 1 • Q5/5**

SET 21 • HSS-ID.C.9

# FORWARD 2

Judge causal claims from experiment results • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5

In a randomized experiment, 40 plants got fertilizer and grew a mean of 18 cm; 40 got none and grew a mean of 12 cm. What is the difference in means (fertilizer minus none)?

Difference (cm): _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q1/5

Re-randomizing the plants many times gave differences of 6 cm or more only 1% of the time. What can be concluded?

**A.** Fertilizer is unrelated to growth.

**B.** The fertilizer likely caused more growth.

**C.** The difference is due to chance alone.

**D.** No conclusion is possible without r.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q2/5**

## True or false?

In a randomized experiment, a difference much larger than chance alone would produce supports a causal conclusion.

**A.** True

**B.** False

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F2-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q3/5**

An observational study found a large difference between coffee drinkers and others. Why can't it prove cause?

**A.** Observational studies do not collect any data.

**B.** Coffee drinkers may differ in other ways that matter.

**C.** Coffee intake cannot be measured accurately.

**D.** The difference was too large to be believable.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S21-F2-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q4/5**

## True or false?

If an experiment's difference could easily happen by chance, the experiment proves the treatment has no effect.

**A.** True

**B.** False

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.5 • Set 21 • FORWARD 2 • Q5/5

# Statistics as a process of inference from samples

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.A.1** |
| **BACKWARD 1** | Recognize random and representative samples | **7.SP.A.1** |
| **BACKWARD 2** | Use a sample to estimate a population count | **7.SP.A.2** |
| **FORWARD 1** | See how sample size affects sampling variability | **HSS-IC.A.1** |
| **FORWARD 2** | Attach a margin of error to an estimate | **HSS-IC.B.4** |

SET 22 • HSS-IC.A.1

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the sample proportion who bike? Give a decimal.

Sample proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q1/5

A random sample of 200 of a school's 1,500 students found that 62 bike to school. What is the parameter of interest?

**A.** the 200 students who were in the sample

**B.** the sample proportion 0.31 from the survey

**C.** the proportion of all 1,500 students who bike

**D.** the 62 sampled students who said they bike

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q2/5**

In a random sample of 200 of a school's 1,500 students, the proportion who bike is 0.31. Use it to estimate how many of the 1,500 students bike.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q3/5

Which sample of a school's students allows an inference about all of the students?

**A.** the first 200 students to arrive at school one morning

**B.** a random sample of 200 students chosen from the school roster

**C.** all 200 students who belong to the school bike club

**D.** the 200 students who chose to answer an online poll

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-M-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q4/5**

Two random samples of 200 students from the same school give sample proportions 0.31 and 0.27. Why do they differ?

**A.** One of the two samples must have been chosen incorrectly.

**B.** Sample proportions can never be used to estimate a parameter.

**C.** Sample statistics vary from one random sample to another by chance.

**D.** The population proportion changed between the two samples.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-M-Q5**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • MAIN • Q5/5**

SET 22 • HSS-IC.A.1

# BACKWARD 1

Recognize random and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

## True or false?

A random sample tends to be representative of its population.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q1/5**

A town wants residents' opinion on a new park. Which method gives a random sample?

**A.** ask the first 100 people at the park

**B.** ask the members of the parks committee

**C.** post an online survey and use the first replies

**D.** pick 100 names by lottery from the voter list

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q2/5**

## True or false?

A sample of volunteers who choose to respond is a random sample.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q3/5**

A survey of shoppers at a sporting-goods store is used to estimate how much all town residents exercise. What is the main problem?

**A.** the sample is not representative

**B.** the sample is a census

**C.** the sample is too random

**D.** the sample has no data

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q4/5**

## True or false?

The whole group a study wants to learn about is called the population.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 22 • BACKWARD 1 • Q5/5**

SET 22 • HSS-IC.A.1

# BACKWARD 2

Use a sample to estimate a population count • Grade 7 • Statistics & Probability • 7.SP.A.2

In a random sample of 50 students, 12 play an instrument. Estimate how many of 600 students play.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q1/5**

In a random sample of 40 light bulbs, 2 are defective. Estimate the number defective in 2,000 bulbs.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q2/5**

**True or false?**

If 30% of a random sample prefer pizza, about 30% of the population is estimated to prefer pizza.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q3/5**

In a random sample of 80 voters, 44 support a measure. What is the estimated percent of all voters who support it?

**A.** 55%

**B.** 36%

**C.** 44%

**D.** 80%

---

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q4/5**

## True or false?

Estimates from a random sample are always exactly equal to the population value.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 22 • BACKWARD 2 • Q5/5**

SET 22 • HSS-IC.A.1

# FORWARD 1

See how sample size affects sampling variability • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1

Many random samples of size 25 and many of size 400 are taken from the same population. Which is true?

**A.** Larger samples are always biased.

**B.** Proportions from samples of 25 vary less.

**C.** Both sizes vary the same amount.

**D.** Proportions from samples of 400 vary less.

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q1/5**

## True or false?

Increasing the size of a random sample reduces bias caused by a poor sampling method.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q2/5

A random sample of 400 voters has 228 who favor a law. What is the sample proportion? Give a decimal.

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q3/5

Which estimate of a population proportion is likely to be closest to the true value?

**A.** one from 1,000 volunteers who called in

**B.** one from a random sample of 50

**C.** one from 50 friends of the researcher

**D.** one from a random sample of 1,000

---

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q4/5**

## True or false?

Sample proportions from repeated random samples center around the population proportion.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.1 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSS-IC.A.1

# FORWARD 2

Attach a margin of error to an estimate • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4

A random sample gives a sample proportion of 0.57 with a margin of error of 0.05. Give the interval of plausible values for the population proportion.

From: _____

To: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q1/5

A poll reports 48% with a margin of error of 3 percentage points. Which value is plausible for the population percent?

**A.** 52%

**B.** 55%

**C.** 44%

**D.** 50%

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S22-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q2/5**

## True or false?

A larger random sample usually gives a smaller margin of error.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q3/5

Simulated sample proportions have standard deviation 0.025. Using margin of error = 2 × standard deviation, what is the margin of error?

Margin of error: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q4/5

## True or false?

A margin of error accounts for bias from a sample of volunteers.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.B.4 • Set 22 • FORWARD 2 • Q5/5

# Simulate a probability model with random digits

| | | |
|---|---|---|
| **MAIN** | Probability & Statistics main question | **HSS-IC.A.2** |
| **BACKWARD 1** | Design a simulation for a compound event | **7.SP.C.8.c** |
| **BACKWARD 2** | Find a probability as a relative frequency | **7.SP.C.6** |
| **FORWARD 1** | Use two-digit groups for probabilities in hundredths | **HSS-IC.A.2** |
| **FORWARD 2** | Use simulation results to judge a model | **HSS-IC.A.2** |

SET 23 • HSS-IC.A.2

# MAIN

Probability & Statistics main question • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

A player makes 70% of free throws. To simulate, digits 0-6 mean "make" and 7-9 mean "miss". Use the first 10 digits of line 1 of the table as 10 shots. How many shots are made?

| Line | Digits | Line | Digits |
|---|---|---|---|
| 1 | 15881 39986 98797 30911 | 5 | 17784 86488 76634 40029 |
| 2 | 41707 53645 56815 18844 | 6 | 78753 47787 68045 71784 |
| 3 | 72940 55761 69871 68976 | 7 | 72671 47513 62701 07981 |
| 4 | 84697 88093 20819 67273 | 8 | 93319 08452 05997 28379 |

Shots made: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q1/5**

A player makes 70% of free throws (digits 0-6 = make, 7-9 = miss). Each line's first 10 digits are one trial of 10 shots, giving 8 trials. In how many trials does the player make 8 or more shots?

| Line | Digits | Line | Digits |
|---|---|---|---|
| 1 | 15881 39986 98797 30911 | 5 | 17784 86488 76634 40029 |
| 2 | 41707 53645 56815 18844 | 6 | 78753 47787 68045 71784 |
| 3 | 72940 55761 69871 68976 | 7 | 72671 47513 62701 07981 |
| 4 | 84697 88093 20819 67273 | 8 | 93319 08452 05997 28379 |

Trials: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q2/5**

An event has probability 0.4. Which assignment of single random digits simulates it?

**A.** 4 = success; all other digits = failure

**B.** 0-3 and 9 = success; 4-8 = failure

**C.** 0-4 = success; 5-9 = failure

**D.** 0-3 = success; 4-9 = failure

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-M-Q3**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q3/5**

A spinner lands on red with probability 0.25. Read line 5 of the table as ten two-digit numbers; 00-24 means red. How many of the ten spins are red?

| Line | Digits | Line | Digits |
|---|---|---|---|
| 1 | 15881 39986 98797 30911 | 5 | 17784 86488 76634 40029 |
| 2 | 41707 53645 56815 18844 | 6 | 78753 47787 68045 71784 |
| 3 | 72940 55761 69871 68976 | 7 | 72671 47513 62701 07981 |
| 4 | 84697 88093 20819 67273 | 8 | 93319 08452 05997 28379 |

Red spins: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q4/5

A simulation of 50 families of 4 children (each birth a girl with probability 0.5) gave the results in the table. What proportion of the simulated families had 3 or more girls? Give a decimal.

| Girls in 4 births | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Number of simulated trials | 3 | 12 | 19 | 13 | 3 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • MAIN • Q5/5

SET 23 • HSS-IC.A.2

# BACKWARD 1

Design a simulation for a compound event • Grade 7 • Statistics & Probability • 7.SP.C.8.c

To simulate a fair coin with random digits, which assignment works?

**A.** 0-4 = heads; 4-9 = tails

**B.** 0 = heads; 1-9 = tails

**C.** even digits = heads; odd digits = tails

**D.** 0-6 = heads; 7-9 = tails

In the digits 4 1 7 0 7 5 3 6 4 5, digits 0-4 mean heads. How many heads?

Heads: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q2/5**

**True or false?**

A trial of a simulation for "number of heads in 10 flips" uses 10 random digits.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q3/5**

To simulate rolling a die with digits, what should be done with 0, 7, 8 and 9?

**A.** skip them and read the next digit

**B.** stop the simulation

**C.** count them as a 6

**D.** count them as a 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q4/5**

## True or false?

Running more trials of a simulation makes the estimated probability less reliable.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.8.c • Set 23 • BACKWARD 1 • Q5/5**

SET 23 • HSS-IC.A.2

# BACKWARD 2

Find a probability as a relative frequency • Grade 7 • Statistics & Probability • 7.SP.C.6

In 50 simulated trials, an event happened 16 times. What is its relative frequency? Give a decimal.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q1/5**

An event happened in 9 of 200 trials. What is its relative frequency as a percent?

Percent: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q2/5**

## True or false?

A relative frequency of 0.32 means the event happened in 32% of the trials.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q3/5**

An event happened in 12 of 60 trials. What is its relative frequency?

**A.** 0.2

**B.** 0.6

**C.** 5

**D.** 0.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q4

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q4/5**

## True or false?

The relative frequency from 10 trials is always exactly equal to the true probability.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q5

**Grade 7 • Statistics & Probability • 7.SP.C.6 • Set 23 • BACKWARD 2 • Q5/5**

SET 23 • HSS-IC.A.2

# FORWARD 1

Use two-digit groups for probabilities in hundredths • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

An event has probability 0.35. Which assignment of two-digit numbers simulates it?

**A.** 35 = success; others = failure

**B.** 00-35 = success; 36-99 = failure

**C.** 00-34 = success; 35-99 = failure

**D.** 01-34 = success; others = failure

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**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q1**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q1/5**

Read line 3 as ten two-digit numbers; 00-34 means success. How many successes?

| Line | Digits | Line | Digits |
|---|---|---|---|
| 1 | 15881 39986 98797 30911 | 5 | 17784 86488 76634 40029 |
| 2 | 41707 53645 56815 18844 | 6 | 78753 47787 68045 71784 |
| 3 | 72940 55761 69871 68976 | 7 | 72671 47513 62701 07981 |
| 4 | 84697 88093 20819 67273 | 8 | 93319 08452 05997 28379 |

Successes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q2/5

## True or false?

Two-digit groups 00-99 give 100 equally likely outcomes.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q3/5

A fair coin model uses 0-4 = heads. Using the first 10 digits of each of the 8 lines as 8 trials of 10 flips, in how many trials are there 3 or fewer heads?

| Line | Digits | Line | Digits |
|---|---|---|---|
| 1 | 15881 39986 98797 30911 | 5 | 17784 86488 76634 40029 |
| 2 | 41707 53645 56815 18844 | 6 | 78753 47787 68045 71784 |
| 3 | 72940 55761 69871 68976 | 7 | 72671 47513 62701 07981 |
| 4 | 84697 88093 20819 67273 | 8 | 93319 08452 05997 28379 |

Trials: _____

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F1-Q4**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q4/5**

## True or false?

To simulate a probability of 1/3 with single digits, use 1-3 = success, 4-9 = failure and skip 0.

**A.** True

**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 1 • Q5/5

SET 23 • HSS-IC.A.2

# FORWARD 2

Use simulation results to judge a model • High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2

In 100 simulated sets of 10 fair-coin flips (table), in what proportion of the sets were there 9 or more heads? Give a decimal.

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

Proportion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q1/5

A coin claimed to be fair gave 9 heads in 10 flips. Use the table of 100 simulated sets. Decision rule: if results at least as extreme as the observed result happen in fewer than 5% of the simulated trials, the model is not consistent with the data; otherwise it is consistent. Which decision is correct?

| Heads in 10 flips | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Number of simulated trials | 0 | 1 | 4 | 12 | 20 | 25 | 20 | 12 | 4 | 2 | 0 |

**A.** Not consistent: 9 or more heads happened in only 2% of the trials.

**B.** Not consistent: a fair coin must give exactly 5 heads.

**C.** Consistent: a fair coin can land heads 9 times.

**D.** Consistent: 9 or more heads happened in 2% of the trials.

**REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION** **ID S23-F2-Q2**

**High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q2/5**

## True or false?

A result that happens in 25% of the simulated trials is consistent with the model under the 5% rule.

**A.** True

**B.** False

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REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

High School — Probability & Statistics • Statistics & Probability: Making Inferences & Justifying Conclusions • HSS-IC.A.2 • Set 23 • FORWARD 2 • Q3/5